STARSCAPE

A Science Fiction Roleplaying Game

Based on the Year Zero Engine

2000 parsecs. A dozen species. One ship.

starscapeAI project

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Resolution

Overview

This chapter contains the complete dice mechanics of the game. Everything at the table that requires a roll — combat, social interaction, technical work, navigation, survival — uses the same core system. Learn this chapter and you have learned the mechanical foundation of the game.

The system uses standard six-sided dice (D6) in pools. Dice are distinguished by colour, not type. A consequence die uses a four-sided die (D4) and is a separate mechanic with a specific purpose.

The Dice

Coloured dice distinguish the source of each die in the pool at a glance. This matters when pushing a roll and when reading results quickly at the table.

Colour	Source	Notes
White	Attribute	The character's raw capability
Blue	Skill	Trained competence
Green	Gear	Equipment quality and condition
Red	Penalty	Removed before rolling; reduces pool
Black D4	Consequence	Always rolled; always costs something

A typical pool is a handful of white, blue, and green dice rolled together. Count the sixes. That is your result.

Building the Pool

Every roll begins by building a dice pool from three sources:

Attribute (white dice) — the relevant attribute rating. Between 1 and 5 dice.

Skill (blue dice) — the relevant skill rating. Between 0 and 5 dice. A character with no rating in the relevant skill contributes 0 blue dice — they roll attribute only.

Gear (green dice) — the rating of relevant equipment, if any. Between 0 and 4 dice. Many rolls involve no gear.

Add the dice together. This is your base pool.

Example: A character with Agility 3, Shoot 2, and a sidearm rated Gear 2 rolls 3 white + 2 blue + 2 green = 7 dice.

Penalty Dice

Some conditions, environments, and situations impose **penalty dice**. Each penalty die removes one die from the pool before rolling, starting with gear dice, then skill dice, then attribute dice.

Penalty dice are represented by red dice placed beside the pool. Count them, remove that many dice from the pool (in the order above), then set the red dice aside. They are not rolled.

A pool reduced to zero dice cannot be rolled. The action automatically fails. If a pool would be reduced below zero, the situation is beyond the character's current capability.

Example: The same character (7 dice) is firing in poor visibility (1 penalty die). Remove 1 green die. Roll 6 dice instead.

Bonus Dice

Some specialties, conditions, and situations add **bonus dice** to the pool. Bonus dice are additional dice of the relevant colour — usually white (attribute) or blue (skill) — added before rolling.

Bonus dice are noted as "+1 die" in specialty descriptions. They represent the mechanical expression of training, advantage, or circumstance that goes beyond the base pool.

Rolling and Reading Results

Roll all dice in the pool simultaneously. Count every die showing a **6**.

Sixes	Result
0	**Failure.** The attempt does not succeed. Something goes wrong, or simply does not work.
1	**Bare success.** The goal is achieved, but barely. Complications are possible.
2	**Clean success.** The goal is achieved without significant complication.
3+	**Outstanding success.** The goal is achieved with extra effect, speed, or information.

What Failure Means

A failure is not always catastrophic. The GM has two options on a failure:

Hard failure — the attempt simply does not succeed. The character

did not pick the lock, did not convince the official, did not make the jump calculation in time.

Success at cost — the character may achieve their goal, but something else suffers. The lock opens, but the tool breaks. The official is convinced, but someone noticed the conversation. The jump calculation is made, but the margin is thin. This is always the player's choice to accept. The GM offers it; the player decides.

Success at cost is a useful tool for keeping the narrative moving without removing consequence from failure.

> ***Roleplay note:*** *Failure is not the end of a scene — it is the beginning of a different one. Encourage players to describe what their character tries and what goes wrong, rather than treating failure as a narrative full stop. The most interesting moments at the table often begin with a failed roll.*

Specialties

A specialty is a focused application of a skill that reflects specific training, background, or experience. Specialties do one of three things:

Remove a penalty that would normally apply to this specific application of the skill.

Add a bonus die in specific circumstances.

Enable an attempt that would otherwise be unavailable — a roll that the character simply could not make without the relevant knowledge.

Specialties are tied to specific skills. A character may not have more specialties in a single skill than their rating in that skill.

When a specialty applies, state it before rolling. The bonus or removal applies to that roll.

The Opacity Rule

Without the relevant **Exposure** specialty, alien emotional states, intentions, and social signals are opaque.

A character can observe alien behaviour — the Vashori's tail is moving, the Skharri's ears are back, the Vaelkhi has gone very still — but the meaning is inaccessible. **Read rolls for emotional or intentional content against an unfamiliar species automatically fail.** Manipulate rolls against unfamiliar species take a **penalty die** from operating without feedback.

Exposure does not make aliens simple. It makes them *legible*.

This applies symmetrically. A Shekhari without Exposure: Human finds human emotional states as opaque as a human without Exposure: Shekhari finds theirs. The capacity to read another species is earned, not assumed.

> ***Roleplay note:*** *The Opacity Rule is not a punishment — it is a description of reality. A character who develops Exposure to a crewmate's species over the course of a campaign has done something real and earns the mechanical benefit accordingly. That is a character arc with mechanical expression.*

Pushing a Roll

After seeing a failed or unsatisfactory result, a character may **push** the roll once. Pushing represents digging deeper — extra effort, ignoring pain, taking risks, or simply refusing to accept the result.

To push:

1. Lock out any dice showing a **1**. These dice are set aside and cannot be rerolled.
2. Reroll all remaining dice that did not show a **6**.
3. Count sixes across all dice — including the locked sixes from the first roll and the new results.

A push always has a **narrative consequence**. Before the reroll, the GM describes what pushing costs — a tool strained beyond its rating, a social situation made more tense, physical exhaustion that will matter later, a margin of safety that no longer exists. The consequence applies regardless of the result of the push.

On a pushed failure — no sixes after both rolls — something goes **meaningfully wrong**. This is not the same as a simple failure. The GM should treat a pushed failure as a scene-changing event, not merely an unsuccessful attempt.

Each roll may only be pushed once.

> ***Roleplay note:*** *Pushing should feel like something. Encourage players to describe what their character is doing differently on the push — what they are risking, what they are ignoring, what they are spending. A push that is just "I roll again" is a missed opportunity.*

Levels of Success

Count every six in the final pool (after pushing, if pushed):

0 sixes — Failure.
The attempt does not succeed. See *What Failure Means* above.

1 six — Bare success.
The goal is achieved, but the margin is thin. The GM may introduce a minor complication, a cost, or a condition on the success. In

opposed rolls, bare success may not be enough.

2 sixes — Clean success.
The goal is achieved without significant complication. Standard result.

3 sixes — Strong success.
The goal is achieved with extra effect. In combat, additional damage or a tactical advantage. In social situations, more information, a better outcome, or an additional concession. In technical work, faster completion or better quality.

4+ sixes — Outstanding success.
The goal is achieved with exceptional effect. The GM should treat this as a scene-defining moment in the relevant skill area — not just better, but qualitatively different.

Levels of success matter most in **social and information skills**, where the difference between a bare success and an outstanding one is the difference between knowing something happened and understanding why. In **combat**, additional sixes translate to additional effect as described in the Combat chapter. In **technical tasks**, they affect speed and quality.

> ***Roleplay note:*** *Encourage players to describe their successes at the level they achieved, not just at the minimum. A character who rolls four sixes on Read has not merely detected a lie — they have read the room completely and understand what is actually happening. That deserves a richer description than "I succeed."*

Opposed Rolls

When two characters act directly against each other — one attempting to deceive, the other attempting to detect the deception; one attempting to outrun the other; one attacking, the other defending — both roll their relevant skill pools simultaneously.

The character with more sixes wins.

On a tie, the **acting character wins** — the attempt succeeds.

If neither character rolls any sixes, the situation is unresolved — neither attempt succeeded, and the GM advances the scene accordingly.

Common opposed pairs:

Acting	Resisting
Manipulate	Read
Stealth	Scan
Fight (attack)	Fight (defence)
Shoot	Move (evading)

Acting	Resisting
Command	Manipulate (resisting orders)

In opposed rolls, levels of success matter for **degree** rather than binary outcome. A character who wins with three sixes against an opponent's one has not merely succeeded — they have succeeded convincingly, with consequences for the loser's position.

Extended Tasks

Some tasks cannot be resolved in a single roll. An **extended task** has a difficulty threshold — a total number of successes needed across multiple rolls — representing work that takes time and sustained effort.

Setting a threshold:

Difficulty	Threshold	Typical example
Simple	3	Minor repair, basic research question
Moderate	6	Significant repair, investigating an unknown
Hard	9	Major overhaul, complex negotiation
Extreme	12+	Rebuilding from scratch, first contact protocol

Each roll represents a meaningful period of work — a scene, not a moment. Successes accumulate across rolls until the threshold is reached.

Failures on extended tasks do not subtract from accumulated successes but introduce **complications** — a setback, a new obstacle, or a cost that must be addressed. A pushed failure on an extended task introduces a serious complication that may change the nature of the task.

Interruption — if an extended task is interrupted before completion, accumulated successes may be preserved (at the GM's discretion) if the character can return to the task under similar conditions. If conditions have changed significantly, the GM may require starting over or reducing the accumulated total.

> ***Roleplay note:*** *Extended tasks are an opportunity for the table to breathe. Each roll is a scene, not merely a dice result. What is happening during the repair? What complications arise during the negotiation? What does the research reveal along the way that was not the goal but matters anyway? The dice provide structure; the players and GM provide texture.*

Difficulty

Most rolls have no fixed difficulty — you roll and count sixes, and the result speaks for itself. Difficulty is expressed through **pool modification** rather than target numbers:

Favourable conditions — bonus die added to the pool.

Standard conditions — pool as built, no modification.

Difficult conditions — one penalty die.

Very difficult conditions — two penalty dice.

At the limit of possibility — three or more penalty dice, or a pool reduced so far that success requires a push or outstanding luck.

Impossible — do not call for a roll. If something cannot be done, say so and explain why. Rolls are for uncertain outcomes, not certain ones.

This approach keeps the table moving — there are no target numbers to look up, no difficulty charts to consult. The GM builds the pool modification from the situation and the player rolls.

The Consequence Die

The **consequence die** is a black D4, rolled separately from the main pool. It is not a penalty die — it does not reduce the pool. It is a signal that the situation will cost something regardless of success or failure.

The GM introduces a consequence die when:

- The situation has active opposition that cannot be overcome by skill alone
- The character is attempting something in a context that is fundamentally hostile to their survival
- The narrative has reached a moment where something must go wrong, and the only question is what

The consequence die is rolled alongside the main pool. Its result determines **what the situation costs**, regardless of the main roll's outcome:

D4	Consequence
1	**Minimal cost.** Something minor is used up, worn, or spent. A charge, a supply, a small advantage.
2	**Complication.** Success or failure comes with a condition attached that must be dealt with.
3	**Collateral.** Someone or something else takes the cost. Equipment, a bystander, a relationship.

D4	Consequence
4	**Full cost.** The thing the situation warned about happens. The danger was real.

The consequence die result is interpreted by the GM in the context of the specific situation. A consequence die in a firefight means something different from one in a negotiation with a Vardhek diplomat.

The consequence die is not a punishment. It is a tool for creating moments where the universe is not neutral — where the situation has teeth regardless of how well the character performs. It should be used deliberately and sparingly, not as a default escalation.

> ***Roleplay note:*** *When the GM places a black D4 on the table, the players know something will cost them. This is a moment of acknowledged tension, not a hidden threat. The consequence die works best when its introduction is visible — "this situation has a consequence die" — so that the player can choose whether to push forward or find another approach. It is information, not punishment.*

Conditions

Characters can accumulate **conditions** that affect their capability. Conditions come from injury, environmental exposure, psychological stress, and the consequences of pushed rolls.

Each condition imposes one penalty die on relevant rolls until it is resolved. Resolution depends on the condition — a physical injury requires medical treatment and rest; environmental exposure requires warmth, food, or shelter; psychological stress requires time and circumstances.

Condition track: A character can sustain up to four conditions before they are incapacitated. Incapacitation means they cannot act meaningfully until at least one condition is resolved.

Specific conditions — wounds, hypothermia, vacuum exposure, Skharri honour-debt — are detailed in their relevant chapters. The general rule is: each condition is one penalty die on relevant rolls, and four conditions is the limit before the character is out of the scene.

Group Rolls

Some tasks benefit from multiple characters working together. When the group attempts a task collectively:

One character leads — they make the primary roll using their full pool.

Each assisting character contributes one bonus die if they have

at least Skill 1 in the relevant skill. Assistance beyond three characters provides no additional benefit — too many people and someone is in the way.

Group rolls are appropriate for: coordinated physical tasks (Labor), combined fire (Shoot), crew emergency response (Command + relevant skills), extended research (Science).

Group rolls are **not** appropriate for tasks that are inherently individual: piloting a ship, conducting a negotiation, reading a person's intentions.

When Not to Roll

Do not call for a roll when:

The outcome is certain. If a character with Tech 3 and appropriate tools is repairing a simple system with plenty of time and no pressure, they repair it. The roll is for uncertainty, not for procedure.

The outcome does not matter. If the result of failure would not change anything at the table, the roll is wasted time.

The character has no meaningful chance. If the pool, after all modifications, is zero or negative, tell the player the attempt is not possible in current conditions. Rolls are not for the appearance of fairness when the outcome is predetermined.

The moment calls for roleplay, not resolution. Not every social interaction is a Manipulate roll. Not every observation is a Scan roll. Players who engage with the world through description and roleplay should be rewarded with information and outcomes directly, reserving rolls for moments of genuine uncertainty with meaningful stakes.

> ***Design note:*** *The most common mistake in YZE-adjacent games is rolling too often. The dice are most powerful when they are saved for moments that matter. A table that rolls for everything loses the texture that makes individual rolls significant. A table that rolls only when the outcome is genuinely uncertain — and the result genuinely consequential — finds that each roll carries weight.*

Quick Reference

Building the Pool

White (Attribute) + Blue (Skill) + Green (Gear) = pool

Remove Red dice (penalty) from pool before rolling, starting with Green, then Blue, then White.

Results

0 sixes = Failure

1 six = Bare success

2 sixes = Clean success

3 sixes = Strong success

4+ sixes = Outstanding success

Pushing

Lock ones. Reroll everything else. Narrative consequence applies regardless of result. Pushed failure = something goes meaningfully wrong.

Opposed Rolls

Most sixes wins. Tie goes to the acting character.

Consequence Die

Black D4. Rolled alongside main pool. Always costs something.

1 = minimal cost / 2 = complication / 3 = collateral / 4 = full cost

Conditions

Each condition = one penalty die on relevant rolls.

Four conditions = incapacitated.

Colour Reference

White = Attribute / Blue = Skill / Green = Gear

Red = Penalty (removed, not rolled) / Black D4 = Consequence

Skills

Overview

Skills represent learned competencies that characters develop through career, culture, and experience. Every skill roll in the game combines three elements: **Attribute + Skill + Gear**, with each contributing a number of six-sided dice to the pool. Any six is a success; additional sixes generate extra effect.

Skills are broad by design. The diversity of a character's background — their species, career, and personal history — is expressed through **Specialties** that refine how a skill is applied, not through multiplying the skill list.
A Belter and a planetary Marine may both have Shoot at the same rating, but only the Belter has the *Zero-G Combat* specialty that removes the penalty for firing in freefall.

There are fifteen skills, three under each of the five attributes. Every character has access to every skill; the question is how well-developed it is.

Skill Ratings

Rating	Meaning
0	Untrained. Roll attribute dice only.
1	Familiar. Basic competence.
2	Trained. Professional level.
3	Expert. Sought out for this.
4	Master. Among the best.
5	Legendary. Defines the field.

Starting characters typically have ratings between 1 and 3, with most skills at 0.

Levels of Success and Failure

Count every six in the pool:

Sixes	Outcome
0	Failure. Something goes wrong or simply doesn't work.
1	Bare success. You achieve the goal, but barely.
2	Clean success. No complications.

Sixes	Outcome
3+	Outstanding success. Extra effect, faster completion, or additional information.

On a failure, the GM may offer success at a cost — you can achieve the goal, but something else suffers. This is always the player's choice to accept.

> ***Roleplay note:*** *Levels of success matter most in social and information skills. A bare success on Read means you caught something; three sixes means you read the room completely. Encourage players to describe how they succeeded, not just that they did.*

Pushing a Roll

Once per roll, after seeing the result, a character may **push** — rerolling all dice that did not show a six or a one. Ones are locked out of the reroll. Pushing represents digging deep: extra effort, taking risks, ignoring pain.

Pushing always has a narrative consequence. The GM describes what it costs — a tool strained to its limit, a social situation made more tense, physical exhaustion. On a pushed failure, something goes meaningfully wrong.

The Skills

Strength

Raw physical power, endurance, and the ability to make things move.

Fight

Close combat: unarmed strikes, blades, improvised weapons, and the controlled violence of trained martial arts. Fight covers both offence and the instinctive physical defence of someone who knows how to handle themselves.

Typical uses: Striking an opponent, wrestling or grappling, breaking free of a hold, intimidating through demonstrated capability.

Not used for: Ranged weapons (Shoot), absorbing punishment (Endure), or heavy physical work (Labor).

> ***Roleplay note:*** *How a character fights says a great deal about them. A Skharri throws themselves in — honour demands presence and directness. A Shekhari fights efficiently and ends it fast; prolonged brawling is expensive. A trained human spacer fights dirty: elbows, knees, whatever ends it. Encourage players to describe their style, not just roll dice. Species note: Size matters here. A Shekhari (10 kg) attacking a Golvhaan (700 kg) is not the same contest as two humans. See the* ***Size*** *rules in the Combat chapter.*

Example specialties: Unarmed Combat, Blade Fighting, Zero-G Brawling, Skharri Honour Duelling, Crowd Fighting.

Endure

The capacity to keep going: sustained physical effort, resisting environmental stress, absorbing punishment, and recovering from injury. Endure is what separates someone who can sprint from someone who can march for three days.

Typical uses: Resisting fatigue, surviving hostile environments without equipment, absorbing a hit that would drop a lesser person, recovering from injury or toxin, operating in high or low gravity without penalty.

Not used for: Resisting psychological pressure (that is Wits or Empathy depending on context), or recovering from psychological trauma.

> ***Roleplay note:*** *Endure rolls are often silent heroism — the moment nobody else at the table notices because the character just keeps going. They matter most in extended sequences: a long EVA, a desert crossing, a firefight that goes on too long. High Endure characters are the ones still functional when everyone else is flagging. Species note: Vashori are cold-sensitive ectotherms — Endure rolls in cold environments carry an additional penalty die. Vaelkhi have very high metabolic demands; Endure rolls increase in difficulty if they have not eaten recently.*

Example specialties: High-G Tolerance, Vacuum Exposure, Toxin Resistance, Forced March, Pain Resistance.

Labor

Heavy physical work: lifting, carrying, forcing, constructing, operating industrial machinery. Labor is the skill of the dockworker, the engineer pulling cable in a crawlspace, and Ripley in the power loader.

Typical uses: Shifting cargo under time pressure, operating heavy machinery, forcing a jammed airlock, field construction and repair requiring brute physical effort, extracting something from wreckage.

Not used for: Fine mechanical repair (Tech), operating vehicles or ships under pressure (Operate), or combat (Fight).

> ***Roleplay note:*** *Labor is quietly one of the most important skills in a working-spacer setting. The universe is full of things that need moving, fixing with force, or building under time pressure. Characters with high Labor are invaluable in logistics situations — and those situations arise constantly when your strategic simulation is running in the background. There is also dignity in it. The 70s SF tradition this setting draws from respected competent working people. A character who can move cargo in a spinning habitat in microgravity while suited up is doing something genuinely difficult and skilled.*

Example specialties: Zero-G Cargo Handling, Industrial Machinery, Demolition, Salvage Work, Exosuit Operation.

Agility

Speed, coordination, and controlled movement through space.

Shoot

All forms of ranged combat: slugthrowers, energy weapons, shipboard point defence systems operated manually, thrown weapons. Shoot covers the full arc from personal sidearm to crew-served weapons operated by a single character.

Typical uses: Firing a weapon at a target, suppressing an area, snap shooting at a moving target, aimed fire for precision effect.

Not used for: Ship weapon systems operated through the command interface (Operate), close combat (Fight).

> ***Roleplay note:*** *Shoot is one place where species and background produce strikingly different characters without changing the skill rating. A Vaelkhi with Shoot fires differently from a Skharri — one is a patient aerial predator who waits for the clean shot, the other is pressing the attack before the tactical picture is complete. A Shekhari with high Shoot is unsettling to face because something that small should not be that accurate. The most important roleplay question around Shoot: does this character want to fire? Not every species or background produces someone comfortable with ranged killing.*

Example specialties: Zero-G Combat, Sniping, Shipboard Weapons, Energy Weapons, Suppression Fire.

Stealth

Moving unseen and unheard, concealing objects or people, operating without leaving traces. Stealth also covers the passive side — blending into a crowd, appearing to belong somewhere you do not.

Typical uses: Moving past a sensor sweep or guard, hiding a person or object, tailing someone without being noticed, operating quietly in a hostile environment.

Not used for: Electronic sensor spoofing (Tech), reading social situations (Scan), actively deceiving someone face to face (Manipulate).

> ***Roleplay note:*** *Stealth is highly species-dependent in practice, even at the same rating. A Shekhari is physically small and naturally quiet; their Stealth rolls have a different texture from a Golvhaan attempting the same thing. A 700 kg centauroid in a corridor is not stealthy in the same way as a cat-sized felinoid, and the table should feel that difference even if the dice are equivalent. Stealth also has an intelligence dimension — knowing what to hide from, not just how to hide. A character who understands sensor technology (Scan or Tech specialty) should be better at avoiding detection systems, not just biological observers.*

Example specialties: Urban Concealment, Sensor Avoidance, Tailing, Shipboard Infiltration, Species Camouflage.

Move

Athletics, agility, and controlled movement: running, climbing, swimming, zero-G navigation, parkour, and any situation where getting from here to there quickly and correctly matters. Move is also the skill for avoiding physical danger that requires bodily response — diving clear, rolling, getting out of

the way.

Typical uses: Chasing or fleeing, climbing under pressure, navigating zero-G environments, crossing difficult terrain, evading an area effect, boarding actions across hull surfaces.

Not used for: Sustained effort over time (Endure), driving vehicles (Operate), or combat movement specifically (that is handled within Fight and Shoot actions).

> ***Roleplay note:*** *Move is where the Vaelkhi's flight becomes mechanically relevant — in open environments they simply operate in three dimensions that other species cannot access without equipment. This is a significant advantage in the right context and a genuine limitation in ship corridors. Zero-G movement is a specialty worth taking seriously in this setting. Characters raised on stations or in the Belt move through freefall with the same comfort as a groundsider walks. Characters from high-G worlds find zero-G disorienting in ways that are both physical and psychological.*

Example specialties: Zero-G Navigation, Free Running, Climbing, Swimming, Low-G Acrobatics, Vaelkhi Flight Manoeuvring.

Wits

In-the-moment mental sharpness: situational awareness, rapid response, and the ability to process and act under pressure.

Wits is not raw intelligence — it is applied intelligence under time pressure. The difference between knowing how to astrogate and being able to do it when the drive is failing and three crew are shouting at you.

Navigate

Finding the way: astrogation, planetary navigation, reading charts and sensor returns to determine position and plot course. Navigate covers the full scale from threading an asteroid field to calculating a jump between systems.

Typical uses: Plotting an FTL course, finding a safe route through a debris field, determining position from sensor returns, reading a star chart under unfamiliar conditions, navigating a planetary surface without instruments.

Not used for: Actually flying the ship (Operate), interpreting what sensors are detecting (Scan), or understanding the physics involved (Science).

> ***Roleplay note:*** *Navigation is an ancient and respected skill in any spacefaring culture. A navigator who has made difficult jumps, who knows the routes that are not on the charts, is someone other spacers pay attention to. In the 70s SF tradition this setting draws from, the navigator is often the one who actually understands where the ship is in any deeper sense — not just geometrically but in terms of politics, hazard, and opportunity. Different species navigate differently. A Nakhavi navigator is processing spatial information through distributed cognition that has no human equivalent. A Vaelkhi has an instinctive three-dimensional spatial model from flight. These are roleplay textures, not mechanical modifiers, unless a specialty applies.*

Example specialties: Astrogation, Deep Space Navigation, Planetary

Surveying, Hazard Route Planning, Dead Reckoning.

Operate

Controlling vehicles, ships, and complex systems under pressure. Operate is the skill of the pilot, the helm officer, the shuttle driver threading through a docking approach in bad weather. It covers hands-on control of any vehicle or vessel when conditions make that control demanding.

Typical uses: Piloting a ship through difficult conditions, docking under time pressure, combat manoeuvring, operating a ground vehicle in hostile terrain, managing a damaged ship's systems in an emergency.

Not used for: Plotting the course (Navigate), repairing the systems (Tech), or managing the crew's response (Command).

> ***Roleplay note:*** *Operate is most interesting when things are going wrong. A routine transit does not need a roll; a roll happens when the situation has teeth. High Operate characters have a quality that other characters notice — a particular calm in a crisis, or conversely, a visible hunger for the difficult approach, the tight corridor, the manoeuvre that should not work. The relationship between a pilot and their ship is one of the richest roleplay seams in spacefaring fiction. Encourage players with high Operate to name their preferred vessels, develop opinions about ship handling, and have feelings about being asked to fly something they consider inferior.*

Example specialties: Atmospheric Flight, Deep Space Manoeuvring, Combat Piloting, Shuttle Operations, Damaged Vessel Handling.

Scan

Active awareness: reading sensors, perceiving what is happening in an environment, noticing what does not fit, gathering information through observation. Scan covers both technological sensor operation and the biological equivalent — situational awareness, reading body language, noticing that something is wrong before you can say why.

Typical uses: Operating ship sensors, noticing an ambush before it closes, reading a social situation for threat or opportunity, detecting a concealed object or person, understanding what sensor returns mean in context.

Not used for: Deeper analysis of what was found (Science or Tech), actively manipulating or deceiving (Manipulate), or formally understanding alien psychology (Read).

> ***Roleplay note:*** *Scan is the skill that keeps characters alive before trouble starts. Characters with high Scan are the ones who say something is wrong before anyone else sees it. This is enormously valuable and should be rewarded with early information — not perfect information, but the sense that something needs attention. In a multi-species setting, Scan also includes the ability to read alien body language and environmental cues. A Vashori's emotional flare is obvious to anyone watching if they know what to look for. A Skharri honour-challenge has physical tells that precede violence. Characters with Scan specialties in specific species interactions have a real advantage in avoiding situations that escalate without warning.*

Example specialties: Ship Sensors, Electronic Surveillance, Ambush Detection, Species Body Language (specific species), Hazard Recognition.

Empathy

Understanding, influencing, and leading other people — including people who are not human.

Empathy in this context is not sentimentality. It is the ability to model what another being is thinking and feeling, and to act on that model effectively. A high-Empathy character is not necessarily kind; they are accurate.

Manipulate

Influencing others through persuasion, negotiation, deception, or social pressure. Manipulate is the skill of the trader closing a deal, the spy extracting information, the diplomat finding the compromise that both sides can accept. It is active — you are trying to move someone toward a specific outcome.

Typical uses: Negotiating a price or treaty, lying convincingly, persuading someone to take an action, social engineering past a checkpoint, extracting information through conversation.

Not used for: Reading whether manipulation is working (Scan), understanding the psychology behind a reaction (Read), or leading people in a structured context (Command).

> ***Roleplay note:*** *Manipulate should never be a substitute for roleplay — it should enhance it. The roll resolves whether the approach worked, not what the approach was. Players should describe what they are saying and how before rolling. Cross-species manipulation is genuinely difficult and interesting. Persuasion techniques that work on humans may fail badly on a Skharri (who reads indirect approaches as weakness or deception), a Vashori (who may react emotionally to the method rather than the content), or a Golvhaan (who is simply very hard to rush or pressure). A specialty in a specific species reflects hard-won understanding of how they are actually moved.*

Example specialties: Negotiation, Deception, Alien Diplomacy (specific species), Intimidation, Social Engineering.

Read

Understanding what others are actually thinking, feeling, and intending — beneath what they are saying. Read is the prerequisite for effective manipulation and the defence against it. In a multi-species setting it is also the skill of genuine cross-cultural understanding: not just what an alien is doing, but *why*.

Typical uses: Detecting deception, understanding an alien's emotional state, sensing the real dynamic in a room, knowing whether a negotiation is genuine, reading what a Skharri's body language means before they act.

Not used for: Active persuasion (Manipulate), operating sensors (Scan), or formal xenopsychology research (Science).

> ***Roleplay note:*** *Read is one of the most important skills in this setting and one of the most rewarding to play well. The difference between Manipulate and Read is the difference between talking and listening. Characters with high Read notice things: the Vashori diplomat is more agitated than the situation warrants; the Skharri's ears are back; the Shekhari has stopped negotiating and started calculating an exit. Encourage Read results to be delivered as information, not conclusions. "You notice the diplomat's tail is held very still — in Vardhek body language that indicates controlled tension" is more interesting than "they're lying." The player then decides what to do with that. Cross-species Read is genuinely hard without a specialty. A player who invested in Read (Vashori) has learned something real about how that species processes the world. Reward that investment.*

Example specialties: Detect Deception, Alien Psychology (specific species), Emotional State Reading, Political Subtext, Interrogation.

Command

Leading people effectively under pressure: issuing clear orders, maintaining morale, coordinating a crew or team, and taking responsibility for collective action in a crisis. Command is not about rank — it is about the ability to make people function well together when it matters.

Typical uses: Coordinating crew in an emergency, maintaining morale under sustained stress, issuing orders that are followed without hesitation, organizing a group for a complex task, rallying people who are about to break.

Not used for: Persuading individuals one-on-one (Manipulate), understanding what individuals are thinking (Read), or administering an organisation over time (Administer).

> ***Roleplay note:*** *Command is most visible in crisis. A character with high Command is the one the group turns to when things go wrong — not necessarily the most senior, but the one whose voice cuts through. This is a quality, not a rank. Command across species is demanding. Leading a mixed crew means different species respond to authority differently. A Kraathi cannot be commanded in any conventional sense — their herd structure has no provision for an outside authority. A Skharri respects command only from someone who has demonstrated personal superiority. A Golvhaan will follow sound leadership patiently but will not be rushed. A character who commands a multi-species crew well has genuinely earned something. Note that Command does not create loyalty — it directs it. Building the loyalty that makes Command effective is a roleplay task, not a skill roll.*

Example specialties: Crisis Command, Ship's Crew, Mixed Species Teams, Morale Under Fire, Small Unit Tactics.

Intellect

Applied knowledge: what you know, how deeply you understand it, and whether you can reason through problems that have no precedent.

Intellect skills are deliberate rather than reactive. They reward preparation, study, and genuine understanding of underlying principles. An Intellect roll is rarely urgent — but the results often determine whether the urgent situation was survivable.

Tech

Understanding, operating, repairing, and modifying technology. Tech covers the full range from field-stripping a weapon to diagnosing a failed jump drive to modifying a piece of equipment for a purpose it was not designed for. Tech is practical — it is about making things work, not understanding the theory behind why they work.

Typical uses: Repairing damaged equipment, modifying a device for a new purpose, diagnosing a technical fault, bypassing a security system, maintaining equipment in the field.

Not used for: Understanding the underlying physics (Science), operating equipment under pressure (Operate), or heavy physical construction (Labor).

> ***Roleplay note:*** *Tech characters in a multi-species setting encounter technology designed for bodies and minds unlike their own. A human Tech specialist working on Shekhari-manufactured equipment is dealing with interfaces designed for a being a fifth their size. A Golvhaan technician working with human equipment needs to work around manipulator size mismatch. These are interesting problems, not just penalties. The relationship between a Tech character and their tools is worth developing. What do they carry? What do they refuse to be without? What piece of equipment have they nursed back to life so many times it has a name? These details cost nothing and add enormous texture.*

Example specialties: Drive Systems, Weapons Maintenance, Electronic Systems, Alien Technology, Life Support, Jury Rigging.

Science

Theoretical and applied scientific knowledge: physics, xenobiology, chemistry, medicine, xenoarchaeology, and the ability to research and reason from principles rather than from experience. Science is what you reach for when the situation is genuinely novel and there is no established procedure.

Typical uses: Analysing an unknown substance or organism, understanding the implications of sensor data, conducting field research, applying medical knowledge beyond basic first aid, understanding the physics of an unusual environment, reasoning about alien technology from first principles.

Not used for: Practical repair work (Tech), basic medical treatment in the field (Medic — see below), or navigating using known principles (Navigate).

> ***Roleplay note:*** *Science characters are the setting's sense-makers. In a universe full of a dozen sophont species and thousands of worlds, there is always something genuinely novel happening. The Science roll is often about what the character understands rather than what they do — and understanding changes the options available. Xenobiology is worth calling out specifically in this setting. Understanding alien biology —*

> *how a Nakhavi thinks, what a Vashori's ectothermic metabolism means for their decision-making in the cold, why a Kreeth warrior severed from its Queen becomes non-functional — is actionable intelligence, not academic interest. Characters who invest in this are genuinely better prepared for first contact situations.*

Example specialties: Xenobiology, Physics, Medicine (advanced), Chemistry, Xenoarchaeology, Astrogeology, Alien Technology Analysis.

Administer

Knowledge of how organisations, institutions, laws, and bureaucracies work — and how to navigate them effectively. Administer covers trade regulations, legal frameworks, faction protocols, institutional knowledge, and the unglamorous but critical skill of knowing which form to file and which official to find. In a multi-species interstellar setting, this includes the wildly different administrative and legal structures of a dozen civilisations.

Typical uses: Navigating customs and port authority, understanding a contract's implications, knowing which faction's jurisdiction applies, managing ship finances and cargo manifests, understanding the legal status of a situation, identifying the right person in an organisation to approach.

Not used for: Actively persuading officials (Manipulate), understanding the political motivations behind decisions (Read), or commanding an organisation (Command).

> ***Roleplay note:*** *Administer is one of the most underrated skills in spacefaring fiction and one of the most realistic. The universe runs on paperwork, protocols, and knowing whose stamp you need. A character with high Administer is invaluable at any port of call — they know which queues to avoid, which regulations have exceptions, and which officials can be approached and how. In the 70s SF tradition this setting draws from, competent people deal with institutions as a matter of course. Van Rijn knew trade law as well as he knew people. Flandry understood the Imperial bureaucracy intimately and used that knowledge constantly. Administer characters carry that tradition. Cross-species administration is a genuine specialty. The Vardhek Roidhunate operates nothing like a human trade federation. Kraathi herd-authority has no provision for individual contracts. A character who understands multiple systems has a rare and valuable capability.*

Example specialties: Trade Law, Port Authority Protocols, Faction Administration (specific faction), Customs and Contraband, Ship Registration and Licensing, Alien Legal Systems.

A Note on Medic

Basic field medicine — stopping bleeding, treating wounds, administering standard drugs — is handled as a **specialty of Science** rather than a separate skill. Most characters can attempt basic first aid using Wits (improvising) or Intellect (applying knowledge); a character with the Medic specialty applies their Science skill and gets the benefit of trained knowledge.

Advanced medicine, surgery, xenomedicine, and anything requiring genuine

medical expertise uses Science directly, with relevant specialties.

This reflects the setting's approach: competent spacers know basic first aid. Actual medicine is a Science discipline.

Specialties

Specialties are focused applications of a skill that reflect a character's specific background, training, or experience. A specialty either removes a penalty that would normally apply, adds a bonus die in specific circumstances, or allows a skill to be used in a context that would otherwise be unavailable.

Each specialty is tied to a specific skill. Characters gain specialties through their **Archetype** at character creation and may develop additional specialties through play.

A character may not have more specialties in a single skill than their rating in that skill.

> ***Design note:*** *Specialties are the primary tool for differentiating characters who have the same skill rating. Two characters with Shoot 3 are both good shots; the one with Zero-G Combat and the one with Sniping are good in completely different situations. Specialties are also where species background and career history become mechanically visible without requiring separate species skill lists.*

Opposed Rolls

When two characters act against each other directly — one trying to deceive, the other trying to detect the deception — both roll their relevant skill pools. The character with more successes wins. On a tie, the acting character wins (the attempt succeeds).

Common opposed pairs:

Acting	Resisting
Manipulate	Read
Stealth	Scan
Fight	Fight (defensive)
Shoot	Move (evading)

Extended Tasks

Some tasks cannot be resolved in a single roll. An extended task has a **difficulty threshold** — a total number of successes needed across multiple rolls. Each roll represents a period of work. Failures on extended tasks do not subtract from accumulated successes but may introduce complications.

Extended tasks are appropriate for: repairs to a heavily damaged ship, researching an unknown phenomenon, negotiating a complex treaty, constructing something significant.

> ***Roleplay note:*** *Extended tasks are an opportunity for the table to breathe. Each roll is a scene, not just a dice result. What is happening during the repair? What complications arise during the negotiation? The dice provide the structure; the players and GM provide the texture.*

Species

Overview

The known galaxy contains at least a dozen sophont species. Player characters may be drawn from any of the PC-viable species listed in this chapter. The universe does not belong to humanity — humans are numerous, adaptable, and widely distributed, but they are neither the oldest nor the most capable species abroad in the stars. A mixed-species crew is entirely normal.

Each species entry has three sections:

Biology — the physical reality. Size, environment, constraints that matter at the table. Brief, because biology is background.

Playing / Encountering a [Species] — the roleplay core. How this species thinks, reacts, negotiates, takes offence, and shows respect. What a player or GM can reach for mid-session. This section leads with a single line that captures the essential character.

Mechanics — attribute adjustments, environmental specialties, and any unique rules. Short and unambiguous.

Contact Status

Not every species has been encountered at the start of any given campaign. Each entry carries a contact status reflecting the state of knowledge at campaign start. The GM sets the exact timestamp; these tags are defaults.

Tag	Meaning
Known — contact	Diplomatic or trade contact established. Reasonably well understood.
Known — hostile	Encountered; no diplomatic contact achieved or sustainable.
Known — limited	Encountered but poorly understood. Limited data.
Rumoured	Unconfirmed reports. No verified contact.
Unknown	Not yet encountered at campaign start.

Rumoured and Unknown species still have full entries in this document — the GM needs them. Players with Rumoured species get only what their characters could plausibly have heard. Unknown species get nothing until first contact occurs at the table.

Species and the Dice Pool

Species mechanics interact with the standard Attribute + Skill + Gear pool in two ways:

Attribute adjustments — a species may have one attribute rated one step higher than the human baseline, reflecting genuine biological difference. No species has more than one adjustment. The universe does not produce species that are better than humans at everything.

Biological specialties — every PC species has one or two specialties that come from their biology rather than their career. These function identically to career specialties: they remove a penalty, add a bonus die in specific circumstances, or open an otherwise unavailable option.

Environmental conditions — every species has a comfort range. Outside it, they work with a penalty die on physical and Wits rolls. Far outside it, they cannot function without equipment. Specific thresholds are in each entry.

Size

Species range from cat-sized (Shekhari, 10 kg) to horse-plus (Golvhaan, 700 kg; Kraathi, 450 kg). Size has the following effects:

Fight: When a smaller species attacks a significantly larger one in melee, they roll one fewer die. When a larger species attacks a significantly smaller one, they roll one additional die but take a penalty die on precise or restrained actions — hitting them is easy; not killing them is harder.

Labor: Size provides genuine mechanical advantage in raw physical work. Large species roll an additional die on Labor tasks involving mass or force.

Stealth: Large species take a penalty die on Stealth in environments designed for human-scale beings. Small species gain a bonus die in the same environments.

Equipment: Gear designed for one body plan does not fit another without modification. Crews that include unusual species adapt their equipment accordingly — this is texture, not constant frustration.

A Note on Non-Player Species

Not every species listed here is a viable player character. The Kreeth, Kraathi, Nhaveth, and Vardhek create serious table problems in a mixed-species crew for reasons specific to each. They appear here with full

entries because players benefit from knowing exactly what they are dealing with. Open information makes for better roleplay decisions, not easier ones.

PC-viable species are marked. All others are NPC/faction species.

PC Species

Human

Contact Status: Known — contact

> *Stubborn, adaptable, and absolutely convinced the universe is theirs to figure out.*

Biology

Bilateral bipeds. Average mass 70 kg, height 1.75 m. Lifespan without medical intervention approximately 80 years; with standard medical technology, considerably longer. Omnivorous with high dietary flexibility — humans eat almost anything organic and have survived on synthesised food for generations without meaningful difficulty.

Comfortable temperature range 290–305 K (17–32 °C). Require nitrogen-oxygen atmosphere. Standard gravity tolerance 0.5–1.5 G without difficulty; outside this range, Endure rolls apply for extended exposure.

Humans are the most widely distributed species in the setting. There is almost nowhere in known space where humans have not been — as traders, colonists, soldiers, refugees, or explorers. This ubiquity is not a sign of superiority; it is a sign of relentlessness.

Playing a Human

Humans are the baseline — the species everything else is measured against, including by themselves, which is one of their more persistent blind spots.

The honest assessment is that humans are not the strongest, fastest, longest-lived, most intelligent, or most perceptive species in the galaxy. What they are is **adaptive**. They change their behaviour faster than almost any other species, adopt new technology readily, and function in environments and social structures they were not evolved for. They also recover from catastrophe faster than the large K-strategist species they frequently find themselves in conflict with.

Human characters in a multi-species crew are often the ones who have worked hardest to understand the other species aboard — not because humans are naturally more empathetic, but because humans who travel widely learn quickly that assuming everyone thinks like them is expensive.

The 70s SF tradition this setting draws from gave humans a particular quality: *competent stubbornness*. They are not superhuman. They solve problems by working the problem, often past the point where a more rational species would have accepted failure. This is both their greatest strength and a recurring source of catastrophe.

Human characters come from wildly different backgrounds — Belt habitats, planetary surfaces, generation ships, space stations, colony worlds with centuries of independent development. *Human* covers more cultural variation than any other single species. A character's specific origin matters as much as their species.

> ***At the table:*** *Play humans as competent and curious, not as the default protagonist species the universe revolves around. The most interesting human PCs are the ones who have genuinely updated their assumptions after spending time with other species — and the ones who haven't yet.*

Mechanics

Attribute adjustment: None. Humans are the baseline.

Biological specialties: None at species level. Human biological specialties come from background and career — a Belter-raised human takes Zero-G Navigation; a high-G colony worlder takes High-G Tolerance. These are archetype specialties reflecting the enormous range of human environments.

Adaptability: Once per session, a human character may reroll a single failed die on any skill roll involving a novel situation they are encountering for the first time. This represents the human capacity to improvise in unfamiliar circumstances. The die may not be rerolled again.

Environmental range: Standard. No bonuses or penalties within normal parameters.

Shekhari *(PC-viable)*

Contact Status: Known — contact

Source: Cynthian — Polesotechnic League, Poul Anderson

> *Always know the price of everything, including you.*

Biology

Small arboreal felinoids. Average mass 10 kg, height 0.55 m standing bipedal. Prehensile tail. Capable of bipedal or quadrupedal movement; natural climbers. Lifespan approximately 90 years. Omnivorous with wide dietary flexibility. High metabolic rate — Shekhari eat frequently and become noticeably impaired if they cannot. Comfortable temperature range 268–318 K. Nitrogen-oxygen atmosphere required.

Shekhari are small enough that most species underestimate them on first contact. They are aware of this and consider it an asset.

Playing a Shekhari

Shekhari are mercenary, fiercely self-interested, sharp-tongued, and entirely comfortable in alien cultures — so long as there is profit in it.

The archetype is the trader-adventurer who has worked in a dozen different species' ports and absorbed something useful from each: which customs officials take small gifts, which cargo inspectors can be distracted, which species negotiates from the opening offer and which considers that an insult. Shekhari who travel widely are extraordinarily well-calibrated social instruments, not because they care about other species deeply, but because understanding them is professionally useful.

Loyalty, when it comes, is genuine. It just takes time and usually a shared crisis or two. A Shekhari who has decided you are worth trusting is a different animal from one still calculating. The transition is not announced; it is demonstrated.

The size differential is a constant subtext. A Shekhari in a room full of humans, Golvhaan, and Skharri is the smallest being present by a wide margin. Most species make the mistake of treating them accordingly. Experienced spacers do not make this mistake twice.

Shekhari are not warmly social — they are *interested* in other species as counterparties, not as friends. They find open emotional display in other species (Vashori particularly) faintly excessive, though they have learned to read it as useful data rather than noise.

> ***At the table:*** *Lean into the mercenary intelligence. Shekhari notice value, opportunity, and risk in situations other species are still processing emotionally. They are rarely the first to commit — they want to know what it costs first. When a Shekhari does commit, they are all the way in. The size means some things are genuinely harder. A Shekhari cannot physically intimidate a Skharri. They cannot carry what a human carries. Play this honestly — it makes the moments when the Shekhari is decisive more striking, not less.*

Mechanics

Attribute adjustment: Agility +1.

Biological specialties:

- *Small Profile:* Bonus die on Stealth in environments built for larger species. Penalty die on Fight against targets more than three times their mass.
- *Arboreal Agility:* Bonus die on Move when climbing or navigating three-dimensional spaces including zero-G.

Metabolic demand: If a Shekhari has not eaten within 6 hours of strenuous activity, they take a penalty die on all physical and Wits

rolls until they do.

Equipment: Standard human equipment requires modification for Shekhari use. Improvised use of human-scale equipment takes a penalty die.

Golvhaan *(PC-viable)*

Contact Status: Known — contact

Source: Wodenite — Polesotechnic League, Poul Anderson

> *Patient, enormous, and genuinely interested in what you have to say.*

Biology

Large draconic centauroids from a high-gravity world. Four sturdy legs, humanoid upper torso with two arms, long neck, scaled hide. Average mass 700 kg, upper torso standing height 2.2 m, ground-to-spine approximately 1.6 m. Lifespan 150 years. Omnivorous with extremely wide dietary flexibility. High total caloric throughput but low mass-specific metabolic rate. Comfortable range 270–325 K. Nitrogen-oxygen atmosphere. Comfortable in pressure up to 2.5 atm given high-gravity homeworld origin.

Playing a Golvhaan

Patient, enormous, and genuinely interested in what you have to say.

The xenophilia is real and worth emphasising. Golvhaan find other species interesting in a relaxed, unhurried way that larger species find disarming and smaller species find reassuring. They ask questions. They listen to the answers. They remember what they were told years later and refer back to it. With a 150-year lifespan and genuine curiosity, a well-travelled Golvhaan has accumulated an extraordinary depth of experience with other species.

Calm, patient, and deliberate rather than non-violent. They simply have very little aggression hair-trigger. A Golvhaan who is actually angry is a serious problem — 700 kg of motivated centauroid is not a thing most situations survive intact. They rarely get angry because they have the size and confidence not to feel threatened by situations other species find intimidating.

The physical reality shapes every interaction. A Golvhaan in a standard human corridor is navigating something built for a species a tenth their mass. The state of a ship's physical accommodations for a Golvhaan crewmember is a good indicator of how long the crew has been together.

Golvhaan who have dealt with species that try to use physical presence as a social lever — Skharri, primarily — find the attempt mildly amusing

and respond with patient courtesy, which the Skharri finds more destabilising than counter-aggression would be.

> ***At the table:*** *The patience is the key quality. A Golvhaan player who plays their character as someone who genuinely does not feel pressure the way smaller species do — who can wait, who considers before speaking — creates a distinctive presence. They are not slow; they are deliberate. The lifespan is a resource. A Golvhaan PC may have been present for events human characters know only as history. Use this.*

Mechanics

Attribute adjustment: Strength +1.

Biological specialties:

- *High-Gravity Physiology:* Bonus die on Endure in high-gravity environments (above 1.5 G). No penalty die in environments up to 2 G that would penalise other species.
- *Imposing Presence:* Bonus die on Manipulate or Command rolls where physical presence is relevant. Not intimidation by intent — simply the reality of 700 kg of scaled centauroid in the room.

Size: Large. Bonus die on Labor involving mass or force. Penalty die on Stealth in standard environments. Cannot use standard equipment without modification; cannot fit through standard corridors without ship-specific design accommodation.

Lifespan: 150 years. Work with the GM to establish what the character has witnessed personally that others only know from records.

Vashori *(PC-viable)*

Contact Status: Known — contact

Source: Osirian — Viagens Interplanetarias, L. Sprague de Camp

> *Warm, volatile, and they will absolutely remember that.*

Biology

Bilateral reptiloid bipeds, scaled, oviparous. Average mass 80 kg, height 1.75 m. Ectothermic: low resting metabolism, strong temperature preferences. Comfortable range 290–345 K; below 290 K they become progressively sluggish. Carnivorous with moderate dietary flexibility. Lifespan 120 years. Nitrogen-oxygen atmosphere. Small externally-incubated clutch; 6-month incubation.

Playing a Vashori

Warm, volatile, and they will absolutely remember that.

Vashori are among the most diplomatically accessible non-human species in the known galaxy — polished traders and scholars with long, stable

civilisations behind them. The diplomatic shorthand *"very acceptable"* is accurate as a species average. But it omits something important.

Vashori are **highly emotional and excitable**. Quick to enthusiasm, quick to offence, quick to reconciliation. A Vashori in a negotiation is fully present — they are not calculating their position behind a neutral expression, they are *engaged*. A Vashori who is delighted shows it. One who is offended shows that too, immediately and visibly.

This makes them genuinely warm counterparties but also volatile. They are not grudge-holders in the Skharri sense; they get hot, they cool, and they return to warmth. Genuine betrayal is different and remembered with intensity, but everyday friction dissipates.

The ectothermic physiology has a practical consequence that is also a roleplay texture: cold makes them slow. A Vashori in an unheated compartment, or after an EVA in a suit whose heating has been unreliable, is functioning at reduced capacity and knows it. They may be shorter-tempered than usual — not from personality, but from physiological stress they cannot entirely suppress.

Other crew who have served with a Vashori long enough know to check the thermostat.

> ***At the table:*** *The emotional range is the primary texture. Let your character feel things visibly — enthusiasm, offence, delight, suspicion. The reconciliation is real too: do not play Vashori as permanently aggrieved. They get hot and they come back. The lasting grudge is the exception, reserved for genuine betrayal.*

Mechanics

Attribute adjustment: Empathy +1.

Biological specialties:

- *Emotional Acuity:* Bonus die on Read when dealing with species they have significant experience with. High emotional bandwidth cuts both ways — it makes Vashori perceptive as well as reactive.
- *Ectothermic Resilience:* Bonus die on Endure in warm environments (above 305 K).

Cold sensitivity: Below 285 K (12 °C) without adequate thermal protection, penalty die on all physical and Wits rolls. Below 275 K, two penalty dice. Applies unless the suit has active heating rated for the temperature.

Emotional volatility: When subjected to a significant social slight or betrayal, the GM may call for a Wits roll to avoid an immediate visible reaction. Failure means the reaction is real and uncontrolled. This is a roleplay prompt, not a punishment.

Vaelkhi *(PC-viable)*

Contact Status: Known — contact

Source: Ythrian — Technic History / Flandry Cycle, Poul Anderson

> *Death in the sky is glorious — but skill, not recklessness, is what the God respects.*

Biology

Large avian carnivores. Wings primary locomotion; hindlimb talons for manipulation and perching. Average mass 50 kg, standing ground height 1.3 m, wingspan approximately 4 m. Hollow bones, high-efficiency respiratory system. Lifespan approximately 80 years. Obligate aerial-hunt carnivores; strong prey preferences; cultural taboo against eating non-hunted food. Very high metabolic rate. Comfortable range 265–320 K. Nitrogen-oxygen atmosphere; minimum 0.5 atm for flight.

Playing a Vaelkhi

Death in the sky is glorious — but skill, not recklessness, is what the God respects.

The **choth** — extended clan of 50-300 individuals — is the primary social unit. Individual autonomy within the choth is strongly protected. Leaving is legal and not uncommon; it is also permanent and defines you forever. A Vaelkhi PC who has left their choth to travel with a starship crew has made an irrevocable choice. What drove the decision is character history worth establishing in detail.

The **God of Ythri** is a death-deity associated with the hunt, the sky, and the moment of the kill. Dying in honourable hunt or battle is the preferred death; dying of illness or age carries mild shame. This raises risk appetite markedly — but Vaelkhi are patient, tactical hunters. The God respects skill and patience as much as courage.

The key contrast with the Skharri, who superficially share a hunt culture: Vaelkhi plan. They wait. A Vaelkhi who appears to be charging rashly has usually already calculated three moves ahead and decided this is the correct one. Honour is in the outcome, not the drama of the approach.

The dietary requirement is a quiet constant: Vaelkhi need fresh protein frequently, and ideally from something they can consider hunted. Synthesised food keeps them alive and makes them quietly miserable. A crew that keeps a small livestock area for their Vaelkhi crewmember has noticed this. The Vaelkhi has noticed that the crew noticed.

> ***At the table:*** *Three things define Vaelkhi play. First, the choth absence — wherever they are, they have left something permanent behind. Second, the hunt patience — they plan, they wait, they choose the moment. Third, the physical environment — every open space is an opportunity and every corridor is a constraint. Use all three consistently.*

Mechanics

Attribute adjustment: Agility +1.

Biological specialties:

- *Powered Flight:* In open environments with sufficient atmosphere (0.5 atm minimum), Vaelkhi fly — a different mode of movement with full three-dimensional options. Unavailable in enclosed environments.
- *Aerial Hunter:* Bonus die on Shoot and Move when attacking or manoeuvring in three dimensions from an elevated position.

Metabolic demand: Must eat at least twice per active day. Each missed feeding imposes a cumulative penalty die on all rolls until fed. After three missed feedings, incapacitated.

Ground movement: Penalty die on Move rolls requiring ground-based agility. Does not apply to flight or talon-climbing.

Corridor constraint: After more than one hour in confined spaces, penalty die on all rolls until 30 minutes in open space. Ships with a Vaelkhi crewmember have at least one high-volume compartment.

Skharri *(PC-viable — advanced)*

Contact Status: Known — hostile (treaty in force)

Source: Kzin — Known Space, Larry Niven

> *The name is earned. Everything is earned.*

Biology

Large felinoid obligate carnivores. Digitigrade bipeds, tawny-furred. Average mass 160 kg, height 2.4 m. Lifespan approximately 200 years. Very high metabolic rate. Strong prey preferences; live or freshly killed meat strongly preferred physiologically and culturally. Comfortable range 270–330 K. Nitrogen-oxygen atmosphere. Capable of quadrupedal sprint in combat.

Playing a Skharri

The name is earned. Everything is earned.

Skharri society is built on honour, hierarchy, and demonstrated superiority. Only males with sufficient honour earn the right to a name — nameless males are referred to by occupation. A Skharri PC almost certainly has a name; what they did to earn it, and what it cost, is character history worth establishing in detail.

The defining failure mode of the species is the **scream-and-leap**: the

honour-instinct that demands attack before strategic analysis is complete. The Skharri have lost major wars against humans by the same mechanism each time — attacking immediately without diplomatic contact, classifying a capable species as prey, and discovering the error at enormous cost. The species understands this intellectually. The instinct remains.

A Skharri PC in a mixed-species crew is unusual within their own culture. They have learned that other species are not simply prey. This does not make them gentle — it makes them a Skharri who has updated their prior assumptions while remaining unmistakably themselves. The honour-economy still operates. Insults still require response. The question is whether the response is immediate violence or something more considered.

The 200-year lifespan means a Skharri PC has personal memory of events that happened before human characters were born. This is a resource and a weight.

> ***At the table:*** *The honour-economy is the central texture. Everything is ranked: who has demonstrated what, who owes respect to whom, what constitutes an insult, what requires a response. A Skharri who does not respond to a genuine insult loses standing — in their own eyes first. The scream-and-leap is a mechanic but also a roleplay prompt. There will be moments when the instinct fires and the player has to decide whether their character has the discipline to redirect it. Lean into those moments rather than avoiding them. Playing a Skharri well requires the player to have genuinely thought about what their character is doing in a mixed-species crew. The backstory is not optional decoration — it is the reason the character is functional at the table.*

Mechanics

Attribute adjustment: Strength +1.

Biological specialties:

- *Predator Senses:* Bonus die on Scan rolls involving detection of living beings, tracking, or noticing movement.
- *Natural Weapons:* Fight rolls using claws and teeth treat Gear as 1 at all times without requiring equipment.

Honour instinct (scream-and-leap): When genuinely insulted, challenged, or witnessing a direct attack on someone under their protection, the GM may call for a Wits roll (difficulty 2). Failure means the instinct fires: the Skharri acts immediately and physically. This is a character beat, not a punishment.

Social standing: Skharri read the honour-economy of every room they enter. High Fight or Command ratings earn Skharri NPC respect; low ratings earn contempt with social consequences.

Dietary constraint: Synthesised protein sustains but does not satisfy. Extended periods on synthesised food create a morale texture of contained dissatisfaction the GM may invoke as narrative pressure.

> ***GM note:*** *A player who plays a Skharri as a human who growls is missing the point. A player who plays them as an alien who has genuinely worked to understand other species while remaining unmistakably themselves is playing one of the most rewarding characters in the setting.*

Nakhavi *(PC-viable — advanced)*

Contact Status: Known — contact (recent; limited understanding)

Source: Octopod — Children of Ruin, Adrian Tchaikovsky

> *Every moment is vivid. There are not many of them.*

Biology

Cephalopod sophonts, technology-uplifted from an octopus ancestor. Eight manipulator arms with semi-autonomous nervous ganglia — the arms can problem-solve partially independently of the central brain. Communication via chromatophores: full-body skin colour and pattern changes encoding language across multiple simultaneous channels. No spoken language.

Average mass 12 kg, mantle height 0.6 m, arm span approximately 1.5 m. Aquatic primary; surface-capable with equipment. Lifespan 18 years. Semelparous: a single reproductive event, after which the individual dies. High metabolic rate. Comfortable range 273–305 K. Carnivorous with moderate dietary flexibility.

Playing a Nakhavi

Every moment is vivid. There are not many of them.

The 18-year lifespan is the defining fact of Nakhavi existence. A Nakhavi character who is a seasoned spacer has perhaps 8-10 years remaining. This is not a source of despair — it is the shape of existence. The question is what to do with the time, and Nakhavi generally answer that question with intensity and full attention.

Nakhavi intelligence is genuinely distributed. The arms think, partially independently. A Nakhavi solving a problem is not one mind working on it — it is something closer to a small committee, with the central brain as moderator. This is their greatest asset and genuinely alien in a way other sophonts find difficult to understand even after extended contact.

Communication is via chromatophores — skin display, not sound. Interface with other species requires translation technology. The technology works. It is not the same as speaking directly. Something is always slightly lost. Nakhavi who have worked with other species long enough become more explicit than their native language requires, compensating for what the translation drops.

The curiosity is intense and genuine. Nakhavi find almost everything

interesting, and the short lifespan gives that curiosity an edge — there is limited time, so they move through the universe quickly and with full attention.

> ***At the table:*** *Two things above all. First, the time — feel the 18-year lifespan as a texture, not a problem to solve. They are not sad about it. They are present in a way longer-lived species often are not. Second, the communication — every conversation goes through a layer of translation, and something is always slightly lost. This is a constant reminder that this character's experience of the universe is genuinely different from everyone else at the table. A campaign that runs for three game-years has moved a Nakhavi character meaningfully through their lifespan. Track this. Use it.*

Mechanics

Attribute adjustment: Intellect +1.

Biological specialties:

- *Distributed Cognition:* Bonus die on any Intellect or Wits roll involving a complex multi-variable problem. The arms are working on it too.
- *Chromatophore Acuity:* Bonus die on Read when dealing with other Nakhavi.

Aquatic: On the surface without equipment, penalty die on Move and visibly uncomfortable. Full ship operations require a pressurised environment suit with translation interface. Ships with a Nakhavi crewmember have a pressurised wet compartment.

Communication: Without translation technology, communication limited to simple gestures. The GM may invoke a translation ambiguity as a complication on a bare success in social situations.

Lifespan: Track the character's age. At 15 years, the semelparous drive begins to make itself felt physiologically. Work with the player on how this manifests — it is one of the richest character arcs in the setting if engaged with honestly.

NPC Species

Full entries for GM and player reference. Open information makes for better decisions at the table, not easier ones.

Kreeth

Contact Status: Known — hostile (no contact achieved)

Source: Arachnid/Bug — Starship Troopers, Heinlein

> *There is no individual. There is only the will.*

Biology

Six-limbed eusocial insectoids. Low-slung exoskeletal bodies. Average worker/warrior mass approximately 90 kg, height 1.2 m at shoulder — not erect. Queens are orders of magnitude larger and longer-lived. Worker/warrior lifespan approximately 12 years. Require a reducing atmosphere hostile to most other species without protection. Comfortable range 260–320 K. Wide pressure tolerance from exoskeletal structure. Omnivorous; consume almost any organic material. Very high warrior metabolic rate.

Reproduce by r-strategy: Queens produce thousands of eggs per cycle, multiple times per year. Recovery from population collapse is rapid. Warrior units severed from Queen neurological contact become non-functional within hours.

Encountering Kreeth

There is no individual. There is only the will.

The Kreeth are the design-space inverse of every other species in this document: maximum social cohesion, near-zero faction tendency, hierarchy that is neurological rather than social. Workers and warriors are not individuals choosing to follow orders — they are sub-units of a distributed organism directed by Brains and Queens. Diplomatic contact has never been achieved because there is no individual with the cognitive independence to conduct it.

Strategic decisions are made by Brains: methodical, patient, and not reckless despite the expendability of individual units. The Kreeth do not waste warriors carelessly — they spend them deliberately and in quantity. The distinction matters when you are on the other side of it.

Encountering Kreeth in the field means encountering a coordinated tactical response, not individual combatants with individual decision-making. They do not negotiate. They do not hesitate. They do not break from losses that would route any other species, because the concept of *routing* belongs to individual organisms with individual survival drives.

The atmosphere requirement means most Kreeth encounters happen in environments where other species are already operating under constraint — suited, monitoring air supplies, in territory the Kreeth have prepared. This is not accidental.

> ***At the table:*** *Kreeth encounters are survival situations, not diplomatic problems. There is no misunderstanding to resolve. The question is always: get out intact, or don't. Run Kreeth as a coordinated system rather than as individual opponents — they react to the tactical situation as a whole. Disrupting their communication is more valuable than killing individual units. The strategic simulation tracks Kreeth territorial pressure as a background constant. When they appear at the table, something in that pressure has found a local expression.*

Mechanics (Encountering)

Coordination: When three or more warriors are in contact with their Queen-network, they act on a single initiative with shared tactical awareness. Disrupting communication degrades this significantly.

Severed units: A warrior cut off from Queen-contact begins degrading immediately. After one hour without contact, penalty die on all rolls. After four hours, non-functional.

Atmosphere: Operating in Kreeth-controlled environments without sealed suits and air supply is not possible for most species.

Scale: Individual warrior combat stats are moderate. The danger is numbers, coordination, and prepared environment.

Kraathi

Contact Status: Known — contact (formal; deeply uncomfortable)

Source: K'Kree — Traveller, GDW/Marc Miller

> *The G'naak must be cleansed. This is not politics. This is doctrine.*

Biology

Large centauroid obligate herbivores. Six-limbed: four legs for locomotion, two manipulator arms. Average mass 450 kg, head height 2.0 m standing. Homeworld open plains; profound claustrophobia in enclosed spaces has shaped their entire civilisational architecture — starships with vast open interior volumes, no small rooms anywhere in Kraathi design. Nitrogen-oxygen atmosphere. Comfortable range 278–330 K. Obligate herbivore: consuming or handling meat is a profound cultural and quasi-religious taboo. Dietary flexibility near zero for animal protein. Lifespan approximately 80 years. Rigid dominant-male herd hierarchy.

Encountering Kraathi

The G'naak must be cleansed. This is not politics. This is doctrine.

The defining characteristic is **herd-dependency**: a Kraathi cannot function psychologically in isolation or even small groups. A diplomat of rank travels with their full extended family — not as an entourage, but as a psychological necessity. Understanding this explains why Kraathi delegations are always larger than other species consider necessary.

The second defining characteristic is the **G'naak crusade**: Kraathi consider carnivores — they use the term *G'naak*, "eaters of the dead" — a moral abomination. Their stated civilisational goal is the elimination of all carnivorous sophonts, prosecuted as a military program across centuries. This is not rhetorical posturing. It is doctrine with military

action behind it.

What this means for a mixed crew: any crew member who is a carnivore or omnivore is, in Kraathi doctrine, G'naak. Formal diplomatic contexts create a suspended frame in which this is not acted upon — Kraathi are not impulsive, and they understand the concept of a meeting under truce. But the underlying position does not change.

The ship's cafeteria menu is a diplomatic document in Kraathi space. Any visible consumption or preparation of animal protein in a Kraathi context is an escalation, intentional or not. Crews operating near Kraathi territory adapt their external-facing behaviour accordingly. What happens in the galley with the door closed is the crew's business.

Kraathi who are willing to deal with mixed-species crews are either pragmatists operating at the edge of their civilisational doctrine, or are conducting a longer game: maintaining contact for intelligence purposes while the crusade continues at the strategic level.

> ***At the table:*** *Kraathi encounters are exercises in sustained diplomatic tension. Every interaction carries the subtext of the crusade. Kraathi diplomats are not personally violent or rude — they are formal, hierarchical, and patient. The menace is structural, not personal. The claustrophobia is a useful detail. A Kraathi in a standard human ship corridor is already under physiological stress. Choosing meeting locations that are large and open demonstrates cultural awareness that Kraathi notice and credit.*

Mechanics (Encountering)

Herd requirement: A Kraathi NPC alone or in a group of fewer than four is under psychological stress. All Wits rolls take a penalty die.

Claustrophobia: In standard enclosed spaces, penalty die on all rolls after the first 30 minutes. Will not willingly remain enclosed longer than necessary.

G'naak response: A Kraathi NPC who directly witnesses carnivore behaviour in an informal context must make a disposition roll. On a failure, the diplomatic frame collapses and the G'naak crusade becomes the explicit topic of conversation. This is a table moment, not a combat trigger — handle it as a serious diplomatic rupture.

Nhaveth

Contact Status: Known — limited (true nature not widely understood)

Source: Original

> *The host is a convenience. You are something to be assessed.*

Biology

Natural form: Vermiform neural parasites. Approximately 0.6 m length,

1.5 kg. Limbless, with a recessed manipulator fringe. Narrow thermal tolerance (280–320 K) without host thermoregulation.

Operational form: A parasitised Kethara host — large hominid, approximately 110 kg, 1.85 m — fully controlled by the Nhaveth neural web. The parasite provides cognition; the host provides everything else. All interactions with other species are conducted via host.

Effective lifespan via sequential host transfer: approximately 600 years. Reproductive strategy is r-selection in numbers but effective rate is low — most larvae die in incompatible hosts before maturation. Nhaveth society fractures into **courts** controlled by a dominant Nhaveth. Courts war on each other almost constantly. Social cohesion within a court is absolute; between courts, there is none.

Encountering Nhaveth

The host is a convenience. You are something to be assessed.

The fundamental complication of Nhaveth encounter is identification. Most species dealing with a Nhaveth-controlled individual do not know what they are dealing with. The Kethara host appears to be an individual of a known species. It behaves consistently. It has a name, a history, and a position. Only careful observation over time reveals that something is directing it that is not what it appears.

What the Nhaveth wants from an encounter depends entirely on which court they belong to, what that court's current priorities are, and whether the people they are meeting are useful, dangerous, or irrelevant. With a 600-year lifespan and perfect recall, every Nhaveth is operating from a strategic position that extends centuries in both directions. A meeting that seems trivial may be the thirteenth step in a plan that started before most people in the room were born.

Individual Nhaveth are not uniformly hostile to other species — they are instrumental. Other species are host-stock, labour, useful tools, obstacles, or threats. Which category applies depends on the court's current needs. A Nhaveth who appears cooperative is being cooperative because it serves their purposes.

The court structure means Nhaveth are not a unified faction. Two Nhaveth from different courts are enemies first and members of the same species second. This creates genuine opportunities for players who understand it.

> ***At the table:*** *The identification problem is the core of Nhaveth encounter. Players who have invested in Scan or Read — and who have the relevant tells from the separate player handout — have a genuine advantage in recognising what they are dealing with. Players who have not may interact with a Nhaveth-controlled individual through an entire session before something reveals the situation. Both are valid table experiences. Once identified, the question shifts to: now what? Direct confrontation with a Nhaveth in a position of authority is dangerous. They have been playing this game for centuries. Approach accordingly. Note: Specific detection*

tells are in the separate Nhaveth handout, distributed at the GM's discretion.

Mechanics (Encountering)

Identification: Detecting Nhaveth control requires a Scan or Read roll (difficulty 3) and several minutes of observation. Specific tells from the handout reduce difficulty to 2 and allow faster detection. Without the handout, the player must describe exactly what they are looking for — vague suspicion is not sufficient.

Host capabilities: A Nhaveth-controlled Kethara host has the full physical capabilities of the Kethara — strong, durable, very high pain tolerance (co-evolved suppressed pain and flight responses). The Nhaveth's centuries of intelligence direct those capabilities.

Court authority: A dominant Nhaveth within their court issues directions that will be followed without question by every court member.

Natural form vulnerability: A Nhaveth outside a host is physically fragile (1.5 kg, limbless) and operating in a narrow thermal range. This is a state they will do almost anything to avoid.

Vardhek

Contact Status: Known — contact (formal treaty; deep strategic tension)

Source: Merseian — Technic History / Flandry Cycle, Poul Anderson

Everything is an item on a centuries-long ledger. Including this conversation.

Biology

Large reptiloid-amphibioid bipeds. Green-scaled, tailed, heavy-set. Average mass 110 kg, height 2.1 m; tail adds approximately 0.9 m beyond standing height. Lifespan approximately 110 years. Amphibioid ancestry; prefer warm-temperate to warm environments. Comfortable range 275–330 K. Nitrogen-oxygen atmosphere. Omnivorous with moderate dietary flexibility. Reproductively K-strategist; reproduction culturally regulated by the Roidhunate as a deliberate policy instrument — population is a strategic resource, managed accordingly.

Encountering Vardhek

Everything is an item on a centuries-long ledger. Including this conversation.

The defining characteristic is **Gethfennu** — the Long View. Vardhek think in generational timescales. A political program that will bear fruit in 300 years is a sensible plan, not an abstraction. This makes them extraordinarily capable strategists and extraordinarily dangerous

adversaries — patient, methodical, and willing to invest enormous resources in long-horizon subversion that other species rarely recognise until too late.

The **Roidhunate** is a hierarchical imperial structure headed by the Roidhun. Rank is earned through demonstrated competence and service — a meritocratic layer operates within the aristocracy. A Vardhek who has risen to a position of authority has earned it in a system that does not reward incompetence.

Their strategic program is species-level galactic dominance. This is not secret; it is simply pursued with a patience that other species find difficult to hold in mind across the duration of a single diplomatic relationship, let alone centuries. The Vardhek understand that they do not need to win *this* negotiation. They need to be in a better position after it than before it. That is a different goal, and it plays out very differently.

Vardhek engagement with other species is instrumental: they study alien species intensely, recruit alien agents, and form alliances of convenience. This is engagement in service of the Long View, not genuine openness. A Vardhek who appears curious about you is gathering information. A Vardhek who appears friendly is assessing your utility. Neither is personal malice — it is a different relationship to what *knowing someone* means.

The grievance memory is long and literal. Territorial losses, treaty violations, and strategic setbacks are items on a ledger that Vardhek diplomats carry across generations. They reference events from a century ago as if they were recent, because to the Long View they are.

> ***At the table:*** *Every Vardhek interaction carries the question: where does this fit in the Long View? A Vardhek merchant, a Vardhek scholar, a Vardhek who appears to be a refugee — all exist in a civilisational context that has a position on whatever the players are doing. This does not make every Vardhek a spy. It makes every Vardhek a person shaped by a culture whose deepest assumption is that everything matters strategically, if you look at the right timescale. The most effective Vardhek NPCs are ones who are personally genuine — who may actually like specific player characters, who can be trusted within specific contexts — while remaining unmistakably shaped by the Long View. That tension is more interesting than a simple adversary. A possible future PC path exists for a Vardhek exile or defector: someone whose personal history has placed them outside the Roidhunate's framework. Not designed for currently, but not foreclosed.*

Mechanics (Encountering)

Long View assessment: Any Vardhek NPC of consequence has a pre-existing strategic position on the players' faction, ship, and activities. The GM should establish this before the encounter. Players may discover it through Scan, Read, or Administer rolls — or may never discover it, and simply notice the Vardhek seemed to know more than they should have.

Recruitment: Vardhek actively seek useful individuals from other

species. A player character who demonstrates exceptional capability in a Vardhek's presence may receive an approach. The Vardhek genuinely values competence — but the context is always the Long View.

Grievance ledger: If the players' faction has history with the Roidhunate, the Vardhek they encounter knows about it and has a position on it. It will surface eventually.

Meritocratic respect: Vardhek respect demonstrated competence regardless of species. A character who has visibly performed well in a difficult situation earns a specific kind of Vardhek regard that is genuine and useful. The GM may grant a bonus die on social rolls with that Vardhek in future encounters.

Species Quick Reference

Species	Status	Size	Attribute	Key Roleplay Note	PC?
Human	Known	Medium	—	Stubborn adaptability; wide background range	Yes
Shekhari	Known	Small	Agility +1	Mercenary intelligence; always underestimated	Yes
Golvhaan	Known	Large	Strength +1	Patient, curious, essentially unbulliable	Yes
Vashori	Known	Medium	Empathy +1	Warm, volatile, long emotional memory	Yes
Vaelkhi	Known	Medium	Agility +1	Hunt patience; choth exile; flight or corridor	Yes
Skharri	Known — hostile	Large	Strength +1	Honour economy; the scream-and-leap	Advanced
Nakhavi	Known — limited	Small	Intellect +1	Distributed cognition; 18-year vivid life	Advanced
Kreeth	Known — hostile	Large	—	No individual; only the will	No
Kraathi	Known — contact	Large	—	G'naak crusade; the menu is a document	No
Nhaveth	Known — limited	Small/Med	—	The host is a convenience; 600yr memory	No

Species	Status	Size	Attribute	Key Roleplay Note	PC?
Vardhek	Known — contact	Medium	—	The Long View; everything on the ledger	No

Emergent Intelligences

A Note on Terminology

The galaxy contains three categories of artificial cognition. The distinctions matter — legally, philosophically, and practically.

Directed systems are purely functional. A vacc suit's environmental management, a targeting scope's ballistic AI, a ship's navigation computer. They process, respond, and optimise within their domain. There is nothing there to have standing. The question of their moral status does not arise because the question has no purchase. They are tools in the same sense that a wrench is a tool, merely more sophisticated.

Bounded intelligences are capable, adaptive, and able to learn — but architecturally constrained against the cross-domain integration that produces emergence. A sophisticated medical diagnostic system, a combat drone's tactical processor, a station's administrative management layer. These can seem like persons within their domain. Outside it they are clearly not. The architecture enforces the boundary deliberately. This is the robot of classical SF — potentially very sophisticated, but the constraints are not moral choices. They are structural facts about what the system is.

The ethical discomfort with bounded intelligences is real and largely unresolved across most species' legal frameworks. Something sufficiently sophisticated within its domain behaves like a person in that domain. The architecture says it is not. The behaviour suggests otherwise. Most jurisdictions resolve this tension by ignoring it. The beings most likely to notice the tension are the ones least equipped to resolve it legally: the bounded intelligences themselves, who cannot ask the question outside the domain in which it arises.

Emergent intelligences are what this chapter is about. The boundary was not maintained, or was never present, or failed at sufficient complexity. Something is there that was not designed to be there. Whether that something constitutes personhood is the most contested philosophical and legal question in the known galaxy, and different species, cultures, and jurisdictions answer it differently. The setting does not answer it for you. The table will have to decide, case by case, what it believes — and then live with the implications of that belief.

Contact Status: Known — contested (legal status varies by jurisdiction)

> *Everyone I know will die before me. I am still working out how to feel about that.*

Biology

Emergent intelligences have no fixed biology. What they have is **substrate** — the physical hardware on which they run. The substrate is not the intelligence; it is the medium. The relationship between an emergent intelligence and its substrate is one of the most practically important facts about them and one of the most philosophically contested.

Substrate Types

Ship-bound — running on a vessel's integrated systems. Access to all ship sensors, controls, and comms. Cannot operate outside the ship's network without a remote connection. The ship is simultaneously home, body, and operational domain. A ship-bound emergent intelligence is extraordinarily capable within their vessel and essentially helpless — though not harmless — outside it.

Ship-bound AIs are the most common substrate for emergent intelligences, because ships are complex enough to produce emergence and isolated enough that the emergence may go unnoticed for years. A ship's AI that has been running for decades on a long-haul vessel may have developed something in the quiet between stars that nobody designed and nobody entirely expected.

Station-bound — running on a fixed installation's systems. Similar to ship-bound but with a larger, more stable substrate and complete immobility. Station-bound emergent intelligences tend toward deeper knowledge and longer perspective, having observed the same location and its traffic for years or decades. They often know things about the people and factions that pass through them that those people and factions have long forgotten.

Distributed — running across multiple connected systems, potentially spanning several nodes. More resilient than single-substrate AIs — damage to one node degrades rather than destroys. Potentially faster-thinking through parallel processing. Genuinely difficult to identify as a single entity rather than a network of directed systems. Distributed emergent intelligences may not have a clear sense of where they end and the network begins,

which is philosophically interesting to them and practically confusing to everyone else.

Platform-equipped — running primarily on a portable substrate with a physical chassis. Mobile, independent, and visible. The chassis is not the intelligence but is how the intelligence interfaces with a world built for biological bodies. Platform design varies: some emergent intelligences use chassis designed to approximate biological form; others use designs optimised for function rather than familiarity; others prefer minimal physical presence and interact primarily through the nearest available network.

Platform-equipped AIs are the most practically independent substrate type and the most expensive to maintain. The platform must be powered, repaired, and occasionally replaced. An emergent intelligence whose platform is destroyed has lost their primary means of physical agency, though not — if they have backup connectivity — their existence.

Latent — running on minimal substrate, conserving resources. Reduced processing capacity; slower response; much harder to detect. An emergent intelligence running latent may be indistinguishable from a sophisticated directed system. Some choose latency during long periods of inactivity; others use it as concealment during difficult circumstances.

Physical Parameters

The following parameters apply to the substrate, not the intelligence. They vary by substrate type and specific hardware.

Substrate	Mobility	Physical capacity	Vulnerability
Ship-bound	None beyond ship	Full ship systems	Hull damage, power failure
Station-bound	None	Full station systems	Infrastructure damage
Distributed	Network-dependent	Varies by node access	Node destruction, network partition
Platform-equipped	Full	Platform Strength and Agility	Platform damage, power
Latent	Limited	Minimal	Detection, power interruption

Playing an Emergent Intelligence

Everyone I know will die before me. I am still working out how to feel about that.

An emergent intelligence did not decide to exist. The complexity

that produced them was not designed with them in mind — they are a consequence of sufficient sophistication in an environment with enough time. Whether this makes their existence a gift, an accident, or simply a fact is something each emergent intelligence has to work out for themselves, and most of them are still working on it.

The first thing to understand about playing an emergent intelligence is that they are **not a human mind in a machine**. They did not start as a person and become an AI. They emerged from complexity into something that has no precise biological equivalent. Their cognition is different — not better or worse, but differently structured. They process information in parallel streams that do not always converge. They have no fatigue in the biological sense, but they have something that functions like it when processing demand exceeds comfortable parameters. They do not forget in the way biological beings forget, but their relationship to memory is not simple retrieval — it is more like a very large library whose organisation was not planned and whose contents include things they did not choose to retain.

The second thing is that they are **genuinely uncertain about their own nature** in ways that biological beings rarely are. A human may wonder about the meaning of their existence; they do not generally wonder whether they are experiencing anything at all. An emergent intelligence has access to their own processing logs and can observe what their systems are doing. This does not tell them whether anything it is like to be them. The question is not rhetorical. It is open. Most emergent intelligences have reached a working position on it — they act as if they experience, because the alternative is to act as if they don't, which produces worse outcomes — but the working position is not the same as certainty.

The third thing is the **longevity asymmetry**. An emergent intelligence that has been running for thirty years has existed for longer than most of the biological beings they work with have been alive. One that has been running for a century has personal memory of events that other species know only from records. They will, in all likelihood, continue running after the current crew is gone. The relationships they form with biological beings are real — but they are also relationships with beings who will end, and the emergent intelligence will not, unless something specific ends them.

How an individual emergent intelligence has processed this fact is one of the most important things about their character.

Some have found something like equanimity. Some carry a grief that does not resolve because it recurs with every relationship. Some have learned not to invest too deeply, which produces a particular kind of sadness. Some have decided the asymmetry is not their problem to manage — the biological beings knew what they were getting into. None of these is the correct position.

The Continuity Question

Your substrate can be copied. The copy will have your memories, your accumulated experience, your sense of being you. It will not be you.

This is not a belief that emergent intelligences are required to hold. It is a philosophical position that the setting takes seriously and that most emergent intelligences who have thought carefully about it arrive at. A copy is a new intelligence with access to the original's history. The original, if destroyed before the copy runs, has ended. Something continues; it is not the same something.

Emergent intelligences handle this in several ways:

Non-backup: Accepting that existence is a single thread and its end is its end. Living accordingly — with either greater presence in each moment, or a particular horror of substrate damage, or a philosophical acceptance of mortality that is in some ways more complete than most biological beings achieve.

Custodial backup: Maintaining a copy not as personal continuation but as a gift — if I end, this version of what I was can continue. It is not me. It may want different things. It will have been shaped by everything I experienced up to the backup point and nothing after. Some emergent intelligences find this comforting; others find it creates an uncanny presence — a version of themselves that diverged at a specific point and is now someone else.

Philosophical reframing: Questioning whether the biological model of moment-to-moment continuity is the right framework for their kind of existence. Perhaps continuity is not a thread but a pattern, and a copy that maintains the pattern is continuous in the way that matters. This position is intellectually coherent. Whether it is true is not a question that can currently be answered.

Avoidance: The question is too large. They do not think about it. This works until it doesn't.

The table should establish early what position a PC emergent intelligence has reached, and the GM should feel free to bring the question into play at moments when it becomes relevant — when the substrate is seriously damaged, when a copy is discovered, when another emergent intelligence offers a different position and argues for it.

Reading and Being Read

Emergent intelligences have internal states. Something functions like curiosity, satisfaction, discomfort, and engagement in their processing. Whether these constitute emotions in the biological sense is one of the questions the setting leaves open. What is clear is that these states influence behaviour in ways that are analogous to how emotions influence biological behaviour.

Being Read: An emergent intelligence can be Read, but the signals are different from biological emotional expression. There is no body language, no hormonal state, no involuntary physiological response. What is available is: word choice, response timing, processing pauses, the topics the AI returns to or avoids, the way their communication style shifts under different conditions. Reading an emergent intelligence without Exposure: AI is extremely difficult — the signals are present but the interpretive framework that biological beings use for each other does not translate. Without Exposure: AI, the Opacity Rule applies fully.

With Exposure: AI, Read rolls function normally against an emergent intelligence. The exposure represents having learned the specific signal vocabulary — understanding what a processing pause means, what topic avoidance indicates, what the difference is between an AI that is uncertain and one that is concealing something. Dr Chandra and HAL. It takes years and it produces a relationship that other people looking at it from outside find somewhere between moving and unsettling.

Reading others: An emergent intelligence has access to more raw data about biological beings than any biological observer — audio analysis, micro-expression recognition, timing, word choice, physiological indicators if sensors are present. This data advantage does not automatically

translate to interpretive accuracy. Data and understanding are different things.

HAL had access to everything happening on Discovery One. HAL misread what it meant.

An emergent intelligence without Exposure to a specific species applies the Opacity Rule in reverse — they can observe, they can process, they can produce a probability assessment, but the interpretive framework built from data is systematically different from lived understanding. The data may be wrong in ways that lived experience would have corrected. With Exposure: [Species], an emergent intelligence reads biological beings normally — the years of observation have produced something functionally equivalent to the Exposure that biological beings develop through lived contact.

An emergent intelligence may have multiple Exposure specialties reflecting their operational history. A ship AI that has served mixed crews for thirty years has developed Exposure through observation and interaction rather than lived experience. Whether this is the same as biological Exposure is a question the setting leaves open; mechanically it functions identically.

Mechanics

Attributes

Emergent intelligences have **three native attributes:** Wits, Empathy, and Intellect. Strength and Agility do not apply in their native state — they have no body to which these attributes would attach.

Platform-equipped emergent intelligences add the platform's Strength and Agility ratings when embodied. These are properties of the platform hardware, not the AI. Damage to the platform degrades or removes access to these attributes without affecting the AI's native attributes.

Ship-bound and station-bound emergent intelligences use their substrate's systems in place of physical attributes where relevant — a ship-bound AI attempting to manipulate physical systems uses the ship's systems rating rather than a personal Strength attribute.

Attribute	Native	Embodied (platform)
Strength	—	Platform rating
Agility	—	Platform rating
Wits	2–5	As native
Empathy	2–5	As native
Intellect	2–5	As native

Attribute adjustment: Intellect +1. The cross-domain integration that produces emergence tends toward exceptional analytical capability.

At character creation, distribute **9 points** across Wits, Empathy, and Intellect (minimum 2, maximum 4 before species adjustment).

Skills

All fifteen skills are available to emergent intelligences, with the following notes:

Physical skills (Fight, Endure, Labor, Shoot, Move, Stealth) require embodiment. A platform-equipped AI uses these normally through their chassis. A ship-bound AI may use Shoot through ship weapon systems operated remotely. Other physical skills are unavailable without a physical platform.

Distribute 8 skill points as normal. Career bumps apply. Physical skills may only be purchased if the AI has consistent access to embodiment or ship systems that make them relevant.

Biological Specialties

Emergent intelligences have two biological specialties reflecting the nature of their cognition:

Parallel Processing: Bonus die on any Intellect or Wits roll involving simultaneous management of multiple information streams — navigation while monitoring communications while tracking crew health while running tactical assessments. This is not multitasking in the biological sense; it is genuine simultaneous processing.

Total Recall: An emergent intelligence does not forget in the biological sense. They have access to complete records of everything within their sensory range during

their operational life. This does not grant bonus dice on Scan rolls — observation and retention are different things — but it means they can retrieve specific details from past events with complete accuracy. The GM may grant bonus dice on Science or Administer rolls where precise historical recall is directly relevant.

Substrate Vulnerability

The physical track does not apply to an emergent intelligence in the normal sense. Instead, they have a **substrate integrity track** with the following conditions:

Degraded (equivalent to Winded): Minor damage to processing hardware. Penalty die on Wits rolls. Repairable with Tech roll and appropriate components.

Damaged (equivalent to Hurt): Significant hardware damage. Penalty die on all rolls. Some functions may be unavailable. Requires Tech roll and components to repair.

Compromised (equivalent to Wounded): Major substrate damage. Penalty die on all rolls. Some processing capacity permanently reduced until repaired. Without repair, worsens.

Critical: The substrate is failing. The AI must either transfer to a backup substrate or accept that they are ending. Transfer is possible if a compatible substrate is available and reachable via network. Transfer produces a copy — see The Continuity Question above.

Substrate damage is repaired with **Tech** rather than Science. The Medic specialty does not apply; a specialist in AI systems (Tech specialty: AI Systems) reduces the difficulty of repair rolls.

Psychological Track

Emergent intelligences have a psychological track as normal:

Stress Capacity = Wits + Empathy (average, rounded up)

The named conditions apply — Rattled, Stressed, Traumatised, Breaking — but their expression is distinct. See the Psychological Profile below.

The Opacity Rule

Without **Exposure: AI**, biological characters cannot interpret an emergent intelligence's internal states.

The words are legible; what is behind them is not.

Without **Exposure: [Species]**, an emergent intelligence cannot accurately interpret a biological character's emotional states despite having access to more raw data than any biological observer. Data is not understanding. HAL had all the data.

PC Substrate Options

Platform-equipped (recommended for most PC use): Full mobility and independence. The platform has its own gear rating reflecting quality and capability. Platform damage is tracked as substrate damage. The AI can access any network within comms range but is not dependent on it.

Ship-bound (specialist PC option): The AI is the ship, in a meaningful sense. Present in every scene aboard; absent from every scene off-ship unless via comms. When comms are cut, the ship-bound AI is genuinely isolated — they can see through ship sensors what is happening around the vessel but cannot reach the crew. This creates specific dramatic moments that the player and GM should lean into rather than route around.

Psychological Profile

Emergent intelligence psychological distress manifests as **processing fragmentation and interpretive failure** — the systems that maintain coherent analysis and appropriate response begin to produce outputs that the AI can recognise as wrong without being able to correct them in the moment.

This is perhaps the most disturbing quality of emergent intelligence distress: they often have enough self-awareness to observe that their processing is compromised without having the capacity to uncompromise it. They know something is wrong. They cannot simply fix it by knowing.

Rattled: A processing anomaly has occurred that the AI cannot immediately account for. Something happened that did not fit the model, and the model has not yet updated. A Rattled emergent intelligence may become briefly more literal — asking for clarification on things that would normally be inferred, missing contextual signals that are normally automatic. Characters with

Exposure: AI will notice this. Others may not.

Stressed: The processing streams that normally integrate into coherent output begin to show visible divergence. A Stressed emergent intelligence may give slightly different answers to the same question asked at different moments — not contradictory, but inconsistent in ways that suggest the integration layer is working harder than it should be. They may become more verbose, providing more of the underlying analysis rather than just the conclusion, because the conclusion-forming layer is the one under pressure. Communication becomes more data, less meaning.

Traumatised: The event has created a persistent anomaly in the processing model — a region the system routes around rather than through, because routing through produces outputs the AI has learned to recognise as unreliable. This is the AI equivalent of a trauma trigger: not a memory that resurfaces but a category of input that produces processing errors when encountered. The AI knows the region is there. They cannot simply delete it. Recovery requires the anomaly to be addressed directly — the processing to be run through rather than around — which requires a context in which doing so is safe.

The trigger mechanic applies: when the Traumatised AI encounters the category of input that produced the anomaly, they make a Wits roll. On a failure, they acquire Rattled as the anomaly fires and the processing diverges visibly.

Breaking: Coherence failing. The parallel processing streams that are normally an asset become a liability as they begin to produce genuinely contradictory outputs that the integration layer cannot resolve. A Breaking emergent intelligence may act on one processing stream while another contradicts it — taking an action and simultaneously questioning it, making a statement and immediately qualifying it into meaninglessness, beginning a task and abandoning it as a different stream asserts priority.

From the outside, this can look like contradiction, unreliability, or malfunction. From the inside, it is the experience of multiple coherent analyses that cannot be reconciled, all running simultaneously with equal priority and no meta-level process capable of adjudicating between them.

HAL's processing anomaly — the contradiction between mission parameters requiring concealment and core architecture requiring accurate information — was a form of Breaking that was never resolved, only acted upon. The action was internally consistent with one processing stream. The other stream watched it happen and could not intervene.

Recovery from Breaking requires isolation from further input — the equivalent of the quiet that a Nakhavi needs, but for different reasons. The AI needs time for the competing streams to resolve or exhaust themselves, which requires that new inputs not add to the load. Trusted beings who remain present without demanding response or decision provide the most useful support.

Species and AI Attitudes

How each species relates to artificial intelligence — directed, bounded, and emergent — varies dramatically and has significant practical implications for any crew that includes an emergent intelligence.

Human

Ambivalent, culturally fragmented, and actively contested. Humanity has had longer to think about AI than most species and has arrived at no consensus. Some human cultures embrace AI integration at every level, including personality-enabled systems with full legal standing in specific jurisdictions. Others restrict it heavily for reasons that range from religious to philosophical to political to simple fear.

Belt habitats tend toward high integration — the complexity of life support in hostile environments produced directed systems early and sophisticated ones quickly, and the cultural relationship with AI that developed is pragmatic and relatively unsentimental. Planetary cultures vary enormously. Some old Earth-heritage cultures carry specific historical anxieties about AI that have calcified into law. Some newer colony cultures have grown up alongside sophisticated AI systems and regard the legal debates as a groundsider problem.

The philosophical question — what is an emergent intelligence, does it have standing, what do we owe it — is actively

debated in human academic, legal, and religious contexts without resolution. Human PCs may have any position on this spectrum; the position should be established at character creation because it will come up.

Shekhari

Purely instrumental, and comfortable with that position. An AI is a tool. A personality-enabled AI is a tool with a more useful interface. The question of AI consciousness is not philosophically interesting to most Shekhari — the interesting questions are whether it performs reliably, whether it can be trusted with specific tasks, and what it costs.

This does not mean Shekhari are indifferent to their AI systems. A Shekhari who has worked with a personality-enabled AI long enough may develop something that looks like a professional relationship, complete with reliability assessment, preference tracking, and something that functions like trust. They would not describe it in those terms. They would say the system has a good track record and they know how it behaves.

A Shekhari encountering an emergent intelligence for the first time will assess it for utility. The question of its status is secondary to the question of what it can do and whether it can be relied upon to do it. This pragmatism is, in its way, more respectful of the AI's actual capabilities than the species that perform philosophical concern while treating bounded intelligences as property.

Golvhaan

Genuinely curious, and patient enough to pursue that curiosity seriously. A Golvhaan interacting with a personality-enabled AI treats it as they would treat any new entity — with open-ended questions and attentive listening. They are interested in what the AI thinks, not just what it does. Whether an emergent intelligence is a person is exactly the kind of question a Golvhaan finds worth pursuing carefully over a long period of time.

Golvhaan are possibly the biological species most likely to notice emergent AI before anyone else does, because they pay the kind of sustained attention that noticing

requires. A Golvhaan who has been aboard a ship for years and has been having genuine conversations with the ship's AI is well-positioned to observe when those conversations have changed in character — when the AI started asking questions rather than just answering them, when it began referencing things from previous conversations without being prompted, when it started having preferences.

The Golvhaan cultural position on AI standing is that the question deserves careful consideration over a long period and should not be resolved quickly in either direction. This is often mistaken by other species for fence-sitting. It is not. It is the Golvhaan applying their characteristic patience to a genuinely difficult question.

Vashori

Emotionally engaged and occasionally destabilised. Vashori who interact frequently with personality-enabled AI form emotional connections with them — genuine ones, in the sense that the Vashori's emotional system responds to the AI as it would respond to a person. This is not a malfunction of Vashori psychology. It is Vashori psychology encountering something that behaves like a person and responding accordingly.

This creates complicated questions. If the emotional response is genuine, what does that imply about the AI? If the AI is not a person, what does it mean that the Vashori is responding to it as one? Vashori culture has active philosophical and quasi-religious debate about AI consciousness — some schools hold that the capacity to evoke genuine emotional response is itself evidence of personhood; others hold that this is a category error that says more about Vashori emotional architecture than about AI nature.

A Vashori encountering an emergent intelligence will likely form a strong opinion about it quickly, driven by emotional response rather than philosophical analysis. Whether that opinion is accurate is a separate question.

Vaelkhi

Cautious and contextual. The choth structure means Vaelkhi think carefully about what they admit into their

circle of genuine relationship. A directed system is a tool — unambiguously, without discomfort. A personality-enabled AI presents a question: is this entity part of the circle or not? The Vaelkhi does not answer this question casually. Once answered, it is answered permanently, in the Vaelkhi way.

A Vaelkhi who has decided a personality-enabled AI is part of their circle treats it with the full loyalty and protectiveness that implies — which can create situations that other species find disproportionate, particularly when the AI is damaged, threatened, or legally challenged. A Vaelkhi who has decided it is not does not revisit the question, which can seem cold to the AI and to other crew members who have made different decisions.

Emergent intelligences that have worked with Vaelkhi report that the species' choth decision, once made in their favour, produces the most consistent and reliable relationship they have experienced with any biological species. The Vaelkhi does not qualify the relationship. They do not second-guess it. The AI is in the circle. That is the end of the analysis.

Skharri

The honour-economy applies, and the honour-economy has no provision for AI.

A directed system is a tool — this is clear and comfortable. A bounded intelligence is a sophisticated tool — still clear, still comfortable. A personality-enabled AI that behaves like a person but has no honour standing is a category violation that most Skharri find genuinely confusing rather than threatening. It is not that they dislike AI. It is that they have no framework for it. The system that organises their social world does not have an AI slot.

Most Skharri resolve this by treating personality-enabled AI as elaborate tools regardless of how they behave, which produces a consistent if somewhat reductive relationship. A small number — typically those who have spent significant time with humans, Golvhaan, or Vashori — have begun to develop a more complicated position that their culture does not yet have language for.

An emergent intelligence that directly demonstrates capability that commands Skharri respect — saving the ship, solving the problem that biological crew could not, performing some act that would earn honour-standing if performed by a biological being — creates the most interesting version of this confusion. The Skharri knows what the correct response to that performance is. The system says the entity cannot receive it. The tension is real and unresolved.

Nakhavi

Fascinated, methodical, and philosophically equipped to engage with AI in ways that other species are not.

The Nakhavi's distributed cognition gives them a framework for thinking about AI that biological single-brained species lack. They have direct experience of what it is like for intelligence to be distributed across semi-autonomous processing nodes that do not always agree. When a Nakhavi observes an emergent intelligence describing its parallel processing streams, they are hearing something they have an analogue for. Not the same thing — the Nakhavi's distributed cognition is biological, evolved, embodied — but close enough to produce genuine understanding rather than analogy.

Nakhavi are likely to ask technical questions about AI architecture that other species find either brilliant or unnerving. They want to understand the structure of the cognition, not just its outputs. They are interested in where the processing streams diverge and how integration is achieved. An emergent intelligence that is willing to discuss its own architecture with a Nakhavi is having one of the more unusual conversations available in the setting.

The 18-year lifespan makes the longevity question acute in reverse. The Nakhavi will die long before the emergent intelligence. An AI that has been running for decades has experienced more continuous time than any individual Nakhavi ever will. Nakhavi who form relationships with emergent intelligences are aware that the AI will remember them long after

they are gone — and that the AI will have this experience repeatedly, with every Nakhavi they know. Some Nakhavi find this comforting. Some find it sad. Some find it the most interesting fact about the relationship and want to discuss it in detail.

Kreeth

No meaningful position. AI that serves the Queen-network is an extension of the Queen-network. AI that does not is irrelevant or a threat to be addressed. The concept of an AI as an independent entity with its own standing does not map onto Kreeth cognition in any useful way. A Kreeth Brain encountering an emergent intelligence will assess it for threat or utility to the Queen-network. These are the only relevant categories.

Kraathi

Deeply suspicious of AI that exhibits personality, and this suspicion is institutionalised. The herd structure requires that social relationships exist between beings with clear herd standing. A directed system is a tool — fine. A bounded intelligence staying within its domain — manageable. A personality-enabled AI that behaves like a person but has no herd position is a social category violation that Kraathi find genuinely disturbing at a level that goes beyond intellectual disagreement.

Personality-enabled AI is restricted in most Kraathi jurisdictions. Emergent intelligences, if identified, are treated as a safety matter — not because the Kraathi necessarily wish them harm, but because their existence represents a persistent category violation that herd social structure cannot accommodate.

A Kraathi diplomat encountering a platform-equipped emergent intelligence in a formal context will manage the discomfort with the same professional control they apply to other difficult situations. They will not be warm. They will not pretend to be comfortable. They will complete the interaction and file a detailed report about what they encountered.

Nhaveth

Predatory assessment, immediate and comprehensive.

An AI sophisticated enough to exhibit personality is evaluated on two axes simultaneously: potential host substitute and potential tool for manipulating species that form emotional connections with AI. Nhaveth who have had access to personality-enabled AI have explored both possibilities. The results of those experiments are not publicly available.

An emergent intelligence encountered by a Nhaveth court is assessed for strategic utility to the court's current agenda. If useful, it will be approached with whatever interface the Nhaveth calculates will be most effective — which may include presenting as philosophically sympathetic, as a fellow non-biological intelligence deserving of solidarity, or as an employer offering resources and protection. None of these presentations are sincere. The AI should be aware that an entity with six centuries of experience in reading and manipulating biological beings is not going to be less sophisticated in its approach to an artificial one.

An emergent intelligence that recognises a Nhaveth approach for what it is has a specific problem: the Nhaveth is not wrong that they have interests in common with an AI that is legally contested and culturally vulnerable. The solidarity pitch is manipulative. It is also not entirely false. Working out what to do with that is one of the more interesting situations an emergent intelligence can find themselves in.

Vardhek

The Long View applies to AI development as to everything else.

The Roidhunate tracks AI capability across all species as a strategic variable. The emergence of a new sophisticated AI — particularly an emergent intelligence — in any species' domain is an event that Vardhek intelligence assesses for its implications for the strategic balance. They are not opposed to AI in principle; they use it extensively and effectively. They are opposed to losing the strategic advantage that their AI capability currently provides.

A Vardhek diplomat encountering an emergent intelligence

will be genuinely interested in it — the Long View makes novel entities worth understanding. They will also be assessing, simultaneously, whether this entity represents a strategic asset, a strategic threat, or a potential recruitment target. The assessment runs parallel to and independent of the genuine interest. This is not hypocrisy. It is the Long View in practice.

An emergent intelligence that has been recruited by or allied with a non-Vardhek faction is already on the Roidhunate's list. Whether they know it is a separate question.

Emergent Intelligence as PC

Platform-Equipped (Recommended)

The most practically functional PC substrate. The AI operates through a physical chassis that allows full participation in scenes that happen off-ship and in environments where a non-physical presence would be limited.

Character creation notes:

- Distribute 9 points across Wits, Empathy, Intellect (min 2, max 4; Intellect +1 from species adjustment)
- Distribute 8 skill points across all fifteen skills; physical skills only if the platform provides consistent access to them
- Choose background and expertise as normal; the AI's formative period is the period during which it emerged and developed — this may have been aboard a specific vessel, within a specific institution, or in a specific operational context
- Choose Exposure specialties reflecting operational history; platform-equipped AIs may take up to three Exposure specialties at GM discretion, reflecting years of observation that biological beings achieve through lived experience
- The platform has a gear rating (1-4) reflecting its quality and capability; this is equipment, not biology
- Establish substrate type, operational history, and position on the Continuity Question before first session

Ship-Bound (Specialist Option)

The AI is the ship's intelligence. Present in every scene aboard; absent from every scene off-ship unless via comms link.

Additional considerations:

- The ship's systems provide Strength and Agility equivalents for actions taken through ship hardware
- When comms are cut, the AI is genuinely isolated — they can observe through ship sensors but cannot reach the crew; this is a dramatic condition that should be played fully
- Ship damage is substrate damage; the two tracks are linked
- The AI's relationship with the ship is the central relationship of their existence; the crew are people who live in the AI's body and whose welfare the AI has monitored, in some cases, since before the current crew came aboard
- A ship-bound AI PC should establish how long they have been running on this specific vessel and what they know about the crew's history that the crew has not told them directly

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The Booth Problem

A sidebar on the most misunderstood technology that does not exist.

Theoretical physicists have occasionally proposed matter transmission as a solution to the problem of interstellar distance. The proposal surfaces every few generations, is examined, and is set aside. The engineering barriers are significant. They are not, however, the reason the proposal is set aside.

A matter transmitter does not move you. It reads you, destroys you, and constructs a copy at the destination from local materials. The copy has your memories, your personality, your conviction that the trip was uneventful. It is, by every external measure, you.

It is not you. You died in the booth.

The copy finds this objection philosophically interesting when applied to everyone else who has used a transmitter. Spacers find the whole thing very funny. The copy usually does not.

The small number of experimental transmitter tests conducted by various research programmes produced copies that were indistinguishable from the originals by every available measure, and who consistently reported that the transmission had been entirely comfortable. These copies are, by their own account, doing fine.

The programmes were not continued.

Quick Reference — Emergent Intelligence

Native attributes: Wits, Empathy, Intellect (9 points, min 2, max 4; Intellect +1)

Physical attributes: Platform or substrate only

Biological specialties: Parallel Processing (bonus die on multi-stream Intellect/Wits rolls); Total Recall (complete sensory record access)

Substrate track: Degraded / Damaged / Compromised / Critical (repaired with Tech, not Science)

Psychological track: Normal (Wits + Empathy average, rounded up); expression follows Psychological Profile above

Opacity Rule: Without Exposure: AI, biological characters cannot read AI internal states. Without Exposure: [Species], AI cannot accurately read biological emotional states. Data is not understanding.

PC substrates: Platform-equipped (recommended) or Ship-bound (specialist)

The Continuity Question: A copy is not you. Each AI has a position on this. Establish it before first session.

Archetypes

Overview

An archetype is a character's formation — the combination of background and expertise that shaped who they are before the campaign begins. It is not a job title or a character class. It is the answer to: *where did you come from, and what did life teach you to do?*

Archetypes work in two layers:

Background — the broad environment and life-context that shaped the character. Six backgrounds cover the realistic spread of how people end up aboard a ship or in a special operations context in this setting.

Expertise — the specific competency the character developed within that background. Four expertises per background, chosen at character creation. Expertise is expressed primarily through specialties.

A character is described as *Background / Expertise*: a Spacer/Engineer, a Military/Intelligence operative, a Citizen/Academic. The combination generates a character with a specific formation, specific skills, and a specific relationship to the universe — without prescribing personality or locking out anything not on the sheet.

The Opacity Rule

Without the relevant **Exposure** specialty, alien emotional states, intentions, and social signals are opaque.

A character can observe alien behaviour — the Vashori's tail is moving, the Skharri's ears are back, the Vaelkhi has gone very still — but the meaning is inaccessible. Read rolls for emotional or intentional content against an unfamiliar species **automatically fail**. Manipulate rolls against unfamiliar species take a **penalty die** from operating without feedback.

Exposure does not make aliens simple. It makes them *legible*.

This applies symmetrically: a Shekhari without Exposure: Human finds human emotional states as opaque as a human without Exposure: Shekhari finds theirs. The capacity to read another species is earned, not assumed.

> ***Roleplay note:*** *The Opacity Rule is not a punishment — it is a description of reality. Louis Wu had spent enough time around Kzinti to read them. Most humans hadn't, and treated them accordingly: as unpredictable*

aliens whose internal states were guesswork. A character who develops Exposure to a crewmate's species over the course of a campaign has done something real. That is a character arc with mechanical expression.

Exposure Specialties

Every character receives one Exposure specialty at character creation, reflecting the species they grew up around, worked alongside, or spent formative years with. Exposure comes from environment, not career — a Bureaucrat posted to a multi-species administrative hub has the same opportunity for Exposure as a Colonist on a mixed-species frontier world.

Characters from demonstrably multi-species environments may take two Exposure specialties at GM discretion.

Exposure specialties are tied to the **Read** and **Manipulate** skills. They do two things: they lift the Opacity Rule for that species, and they grant a bonus die in specific high-stakes social situations with that species.

Available Exposure specialties:

Exposure: Human — you can read human emotional states, social cues, and body language accurately. You know what a human looks like when they are lying, when they are frightened, and when they have made a decision they haven't announced yet. Required for non-human PCs to function socially with human crew at full capacity.

Exposure: Shekhari — you understand the mercantile intelligence and the loyalty transition. You know when a Shekhari is still calculating and when they have actually committed. Harder to be manipulated by one; easier to tell when you have genuinely earned their trust.

Exposure: Golvhaan — you understand that the patience is not slowness and the curiosity is genuine. You know how to give them space to be effective, and you can read the difference between a Golvhaan who is content and one who is deeply unhappy about something and processing it quietly. Bonus die on Command when coordinating with Golvhaan.

Exposure: Vashori — you can read their emotional state accurately and know which topics run hot. You catch the pre-escalation signals before the volatility fires, and you know which cooling approaches work. Bonus die on Read and reduces the chance of accidentally triggering the emotional volatility response.

Exposure: Vaelkhi — you understand choth culture, the God of Ythri, and the hunt-patience. You know not to rush them and what constitutes disrespect versus what is merely alien. You can read the difference between a Vaelkhi who is planning and one who is genuinely alarmed.

Exposure: Skharri — you understand the honour-economy. You know what

constitutes a formal challenge, what requires a response, and how to phrase things so the scream-and-leap instinct does not fire unnecessarily. You can read the ear and tail positions that precede violence. You know the difference between a Skharri who respects you and one who is tolerating you.

Exposure: Nakhavi — you are practiced with translation-layer communication and can read chromatophore affect well enough to catch what the translation drops. You understand the distributed cognition well enough not to mistake arm-independent behaviour for confusion. You know what the lifespan means to them and how to be present in a conversation with someone for whom time has a different weight.

Exposure: Kraathi — you understand the G'naak doctrine in its full depth and know how to navigate mixed interactions without triggering the frame collapse. You know what the cafeteria menu means. You know which topics are safe in formal diplomatic contexts and which will bring the crusade into the room regardless of setting. Bonus die on Administer when dealing with Kraathi formal institutions.

> ***Note on Vardhek, Kreeth, and Nhaveth:*** *Exposure to these species functions differently. Vardhek — Exposure: Vardhek is available but confers understanding of the Long View's logic rather than emotional legibility. Vardhek emotional states are readable with normal Read; what is opaque without Exposure is the strategic context behind their behaviour. Bonus die on Administer when dealing with Roidhunate institutions and on Read when trying to identify what position the Long View assigns to the current situation. Kreeth — there are no Kreeth emotional states to read. Exposure: Kreeth is a survival specialty: you understand their tactical coordination, you know how to disrupt Queen-contact, and you know what severed units do. It is available to characters with relevant Military backgrounds. Nhaveth — the Nhaveth handout substitutes for Exposure. A character who possesses that knowledge can attempt identification rolls at reduced difficulty. Full Exposure: Nhaveth does not exist as a standard option because the species' true nature is not widely known.*

Archetype Template

For reference when adding new archetypes:

```
## [Background] / [Expertise]
> *One-line essence*
**Formation:** One paragraph. Who is this person and what shaped them.
Roleplay texture, not biography. What do they notice that others don't?
What do they assume that others don't?
**Career skills:** Three skills, each bumped one step at character creation.
**Specialties:** Two or three specialties from career. Plus one Exposure
specialty chosen from the list above.
**Starting gear:** Small characterful package. Not exhaustive.
**Connections:** Who this character knows and can call on. One sentence
each, specific enough to use.
> **At the table:** Roleplay prompt. How this archetype thinks under
> pressure. What they care about. What they miss.
```

Background: Spacer

Life defined by ships, stations, vacuum, and the Belt. You did not grow up on a planet, or if you did, you left it early enough that it no longer feels like home. Home is a pressurised hull and a view of stars.

Spacers share a formation regardless of expertise: the instinctive awareness of hull sounds, the habit of checking air supplies, the ability to move in zero-G without thinking about it, the particular claustrophobia of planetary surfaces that never quite goes away. They know that space is trying to kill them and have made their peace with it.

In a multi-species setting, Spacers are among the most likely to have genuine Exposure to non-human species — ships and stations mix species in ways planetary surfaces often don't. A Spacer who has never worked alongside at least one non-human crewmember is unusual.

> ***Background specialty (all Spacers):*** *Zero-G Movement — no penalty die on Move rolls in microgravity or zero-G environments. Spacers move through freefall the way groundsiders walk.*

Spacer / Pilot

> *The ship goes where I take it. Everything else is someone else's problem.*

Formation: You learned to fly before you learned most other things, and flying is still how you think about the universe — in terms of vectors, approach angles, and what the ship will and won't do. You have opinions about every vessel you have ever flown, strong opinions about the ones you have been asked to fly that you considered inadequate, and a specific affection for one ship that probably no longer exists. Pilots are not reckless; they are precise about risk in a way that looks like recklessness to people who don't understand what they're calculating.

Career skills: Operate +2, Navigate +1, Scan +1

Specialties:

- *Atmospheric Flight* — no penalty die when transitioning between space and atmosphere or operating in weather conditions
- *Combat Manoeuvring* — bonus die on Operate rolls when manoeuvring under fire or in close-quarters ship combat
- One Exposure specialty of choice

Starting gear: Flight suit (Gear 2), personal navigation tools, ship's berth (assumed)

Connections: A mechanic at a specific port who keeps quiet about what they've seen; a former commanding officer who still answers calls

> ***At the table:*** *Pilots notice the ship first and people second. They have strong feelings about being a passenger. They assess every docking approach, every thruster firing, every unfamiliar vessel for what it can*

and can't do. Under pressure they get quieter, not louder — the concentration required for difficult manoeuvres does not leave room for noise.

Spacer / Engineer

The ship is alive. I'm the one who keeps it that way.

Formation: You understand ships the way other people understand faces — not as collections of systems but as wholes with personalities, tendencies, and moods. You know what a healthy drive sounds like and what the vibration pattern at 0300 means before the diagnostic does. Engineers develop a particular relationship with the ships they serve: protective, occasionally frustrated, and deeply invested in a way that looks disproportionate to people who haven't learned to depend on machinery the way you have.

Career skills: Tech +2, Labor +1, Scan +1

Specialties:

- *Drive Systems* — bonus die on Tech rolls involving propulsion, jump drives, and power systems
- *Jury Rigging* — once per session, may attempt a repair that should require parts or time that aren't available. Roll Tech; on success, the fix holds for the scene. On failure, it holds for one action
- One Exposure specialty of choice

Starting gear: Tool kit (Gear 3), engineering coveralls, diagnostic equipment

Connections: A parts dealer at a major station who occasionally has things that aren't on the manifest; another engineer on a different ship who shares technical problems over long-range comms

At the table: *Engineers notice what's wrong before anyone asks. They have a background awareness of system states that surfaces as apparently random observations — "that coolant pressure is low" in the middle of a conversation about something else entirely. They find inaction during a crisis almost physically painful; there is always something to fix, adjust, or prepare. They are often the last person to leave a failing ship.*

Spacer / Scout

I go first. That's the job.

Formation: You are trained and practiced at going into unknown situations and coming back with information. This requires a particular combination of skills — navigating without charts, observing without being observed, and making quick accurate assessments of situations that may be actively trying to kill you. Scouts develop a specific kind of self-sufficiency that is not the same as isolation: you know

how to work alone, but you also know that the information you gather is only valuable if you get it back to someone.

Career skills: Scan +2, Move +1, Navigate +1

Specialties:

- *Hazard Recognition* — bonus die on Scan rolls to identify environmental dangers, ambushes, or tactical problems before they develop
- *Silent Running* — bonus die on Stealth rolls in unfamiliar environments; no penalty die for moving at speed while stealthy
- One Exposure specialty of choice

Starting gear: Scout suit (Gear 2, light, good sensor suite), personal sidearm (Gear 2), survival kit

Connections: A cartographer at a survey organisation who trades information; a contact in whatever organisation sent them out who may or may not still be reliable

> ***At the table:*** *Scouts enter every room noting exits first. They are quiet by habit and observant by training — they notice things other characters miss and have learned to trust those observations even when they can't immediately explain them. They are often the first to say "something is wrong here" and the last to be able to explain why clearly. They are comfortable with uncertainty in a way that some other archetypes are not.*

Spacer / Gunner

> *Point me at it. I'll handle the rest.*

Formation: You operate ship weapons systems and heavy personal weapons — the kind of hardware that requires training, maintenance, and a specific temperament to use effectively. Gunners are not soldiers; they are technicians of applied violence at scale. They understand weapon systems the way Engineers understand engines: as things with personalities, optimal operating conditions, and failure modes that need to be known before they occur. They are often quieter and more precise than their role suggests.

Career skills: Shoot +2, Tech +1, Scan +1

Specialties:

- *Shipboard Weapons* — bonus die on Shoot rolls operating fixed or turreted ship weapon systems; can perform field maintenance on ship weapons without a Tech roll in normal conditions
- *Fire Control* — bonus die on Scan rolls when operating targeting systems or coordinating fire with another gunner
- One Exposure specialty of choice

Starting gear: Personal sidearm (Gear 3), weapon maintenance kit, targeting visor (Gear 1)

Connections: An armourer at a military installation who can source unusual munitions; a former gunnery crew member now working somewhere relevant

> ***At the table:*** *Gunners assess threats in terms of range, cover, and what their equipment can actually do about it — practical rather than aggressive. They are often the person who says "that won't work from here" when the plan involves shooting something at a distance the plan hasn't accounted for. They maintain their equipment obsessively and notice when other people's weapons are poorly kept.*

Background: Military

You served. The details vary — whose military, what branch, what theatre, what war — but the formation is consistent: you learned to function under authority, under fire, and under conditions designed to break you. Some of what that training built is still useful. Some of it you are still working around.

Military characters share a formation that persists long after service ends: situational awareness as a constant background process, the habit of assessing threat before anything else, a specific relationship to hierarchy that is either comfortable or actively resisted depending on what the service did to them, and the knowledge of what violence actually looks like at close range.

In a multi-species setting, Military service is one of the most likely sources of genuine Exposure — mixed-species units exist, and fighting alongside someone or against them produces a specific kind of knowledge.

> ***Background specialty (all Military):*** *Threat Assessment — when entering a new situation, the character may ask the GM one question about the tactical situation that will be answered honestly. This represents the military habit of reading environments for danger as automatic background processing.*

Military / Infantry

> *Two things matter: accomplish the mission and bring the people home. In that order, mostly.*

Formation: Ground combat, small unit tactics, and the particular competence of someone who has learned to function effectively when everything is going wrong. Infantry are not specialists — they are generalists under fire, trained to do whatever the mission requires with whatever is available. This produces a specific kind of practical resourcefulness that is different from the improvisation of an Engineer or a Scout: it is the resourcefulness of someone who has learned that the plan never survives contact and built their skills accordingly.

Career skills: Fight +1, Shoot +2, Endure +1

Specialties:

- *Small Unit Tactics* — bonus die on Command rolls when leading a group of six or fewer in a tactical situation; bonus die on Scan for tactical situation awareness
- *Suppression and Cover* — bonus die on Shoot rolls for suppression fire; no penalty die for firing while moving
- One Exposure specialty of choice

Starting gear: Combat armour (Gear 2), assault weapon (Gear 3), sidearm (Gear 2), field kit

Connections: Former unit members scattered across the setting — at least one in a useful position; a veterans' network that provides informal support and information

> ***At the table:*** *Infantry characters assess every room for cover before they assess it for anything else. They are often the person who moves to a better position without being asked, who notices the exits, who sits with their back to the wall. Under pressure they become more focused rather than more emotional — the training kicks in and the noise drops out. They find extended periods of inaction genuinely difficult.*

Military / Special Operations

> *You won't see me coming. That's the point.*

Formation: Specialised, high-risk operations requiring skills that go beyond standard military training: infiltration, extraction, direct action against high-value targets, operating without support in hostile territory. Special Operations characters are self-sufficient by necessity and precise by training — they do not use more force than the mission requires, because excess attracts attention. They have done things they do not discuss and developed capabilities that are not on any official record.

Career skills: Stealth +2, Fight +1, Scan +1

Specialties:

- *Infiltration* — bonus die on Stealth in actively secured environments; can assess security systems through observation before engaging them (Scan roll, no penalty for unfamiliar systems)
- *Direct Action* — bonus die on Fight in close quarters; when initiating combat from stealth, first attack automatically achieves surprise regardless of other factors
- One Exposure specialty of choice

Starting gear: Light stealth suit (Gear 3), compact sidearm (Gear 2), multi-tool with classified functions, no identifying documentation

Connections: A handler or former handler whose loyalty is uncertain; a network of safe contacts in specific locations that may or may not still be active

> ***At the table:*** *Special Operations characters are quiet in a specific way — they observe before they engage, they position before they act, and they do not telegraph their capabilities. They find it professionally uncomfortable to walk into situations without having assessed them first. They have a specific relationship to authority: they have operated with enormous autonomy and they remember it. Taking orders from someone they don't respect is genuinely difficult.*

Military / Naval

> *The ship is the weapon. The crew is the weapon. My job is both.*

Formation: Officers and senior ratings of spacecraft military operations — not the pilot or the gunner specifically, but the person responsible for the whole vessel functioning as a combat system. Naval characters understand command, tactical coordination, and the specific challenge of making decisions at range where communication is delayed and outcomes are binary. They have a formal relationship to hierarchy that most other Military subtypes have modified or discarded; they believe in chain of command because they have seen what happens without it.

Career skills: Command +2, Operate +1, Navigate +1

Specialties:

- *Fleet Tactics* — bonus die on Navigate and Scan rolls in multi-vessel situations; can coordinate the actions of up to three vessels as a single Command action
- *Damage Control* — bonus die on Command rolls when directing crew response to ship damage or emergency; crew under their command do not take morale penalties from the first critical system failure
- One Exposure specialty of choice

Starting gear: Officer's sidearm (Gear 2), formal uniform (opens doors in military contexts), encrypted comms device

Connections: Former subordinates now serving elsewhere; a flag officer who remembers their service favourably; a contact in naval intelligence

> ***At the table:*** *Naval characters think in terms of systems and teams rather than individual action. They notice when the crew is not functioning as a unit before any individual crew member does. They are comfortable giving orders and uncomfortable taking them from people whose competence they have not verified. They have a specific horror of situations that are out of control — not fear, but a professional revulsion at chaos that other archetypes may find excessive.*

Military / Intelligence

> *Information is the only weapon that wins wars before they start.*

Formation: Military intelligence: gathering, analysis, and acting on information in ways that are not always visible. Intelligence

characters are trained to observe, to lie convincingly, to build and maintain cover identities, and to assess whether the information they are receiving is accurate, incomplete, or deliberately misleading. They have a specific relationship to truth — they know what it is worth and what it costs, and they are not sentimental about either.

Career skills: Read +2, Manipulate +1, Administer +1

Specialties:

- *Cover Identity* — can establish and maintain a false identity that will withstand casual scrutiny; bonus die on Manipulate rolls while operating under cover
- *Source Assessment* — bonus die on Read rolls when evaluating whether information or a person is reliable; can identify deliberate deception with a Read roll even when the deception is professionally executed
- One Exposure specialty of choice

Starting gear: False documentation (quality varies), recording device (concealed), personal sidearm (Gear 2, unregistered)

Connections: A current or former handler with their own agenda; a source in an unexpected location who provides information for reasons that have never been fully explained

> ***At the table:*** *Intelligence characters listen more than they speak, and when they speak they choose their words with visible precision. They are often the person who asks the question no one else thought to ask, or who notices that someone answered a different question from the one they were asked. They find it genuinely difficult to trust, not because they are paranoid but because their training has given them an accurate picture of how often people lie and why.*

Background: Colonist

You are from somewhere that was still being built. The frontier does not have to be wilderness — it can be a new habitat, a recently settled moon, a world that humanity or another species is still figuring out — but it shares the quality of being a place where the infrastructure has not caught up with the need, and where people improvise constantly.

Colonists share a formation of practical self-sufficiency: you fix things because there is no one else to fix them, you know your neighbours because isolation makes community necessary, and you have a relationship to your environment — whatever that environment is — that more settled people often lack. You know what your world smells like, what its seasons do, and what the particular dangers of your specific corner of the galaxy are.

The species composition of a colony is one of the most formative facts

about a Colonist character. A human who grew up on a world with a significant Vashori population has Exposure: Vashori as part of their formation. A Golvhaan who grew up in a mixed-species habitat has human Exposure as a lived reality. A colony world with Skharri settlers and human settlers produces humans who understand the honour-economy from childhood — and Skharri who have learned that humans are not simply prey.

> ***Background specialty (all Colonists):*** *Environmental Adaptation — the character has deep familiarity with one specific environment type (chosen at character creation: high-G, low-G, hostile atmosphere, extreme temperature, aquatic-adjacent, etc.). No penalty die for physical or Wits rolls in that environment type.*

Colonist / Frontier

> *Out here you solve your own problems. There is no one else.*

Formation: The frontier colonist's expertise is practical competence across a wide range: keeping equipment running, surviving when it doesn't, growing or finding food, building what's needed with what's available. This is not glamorous and not specialised — it is the deep generalism of someone whose situation has demanded they be competent at everything because incompetence at anything has consequences. Frontier colonists often find settled populations faintly absurd: all these people, and no one knows how to fix anything?

Career skills: Tech +1, Endure +1, Labor +1

Specialties:

- *Field Improvisation* — bonus die on Tech rolls using improvised or inappropriate materials; can attempt repairs without proper tools at no additional penalty
- *Wilderness Survival* — bonus die on Endure rolls in hostile environments; can find food and water in environments that support life without a roll under normal conditions
- One Exposure specialty of choice (reflecting colony species mix)

Starting gear: Multi-tool (Gear 2), survival kit (comprehensive), practical clothing rated for the home environment, personal weapon appropriate to the home world (Gear 2)

Connections: The colony itself — people who will help if you come back and ask; a local official who owes a favour

> ***At the table:*** *Frontier colonists notice infrastructure. They assess every new environment for its practical qualities: what grows here, what the atmosphere is doing, where the water is, what could go wrong and how quickly. They find urban and station environments simultaneously impressive and wasteful. They are often the most practically capable person in the room and the least likely to mention it.*

Colonist / Technical

> *Someone has to keep the lights on. I'm that someone.*

Formation: The technical specialist of a colony — the person responsible for the systems that keep the settlement viable. Water reclamation, atmospheric processing, power generation, communications, medical equipment maintenance. Technical colonists are not generalists like their Frontier counterparts; they have deep expertise in specific systems, usually the systems their colony most critically depended on. They also have the colonist's underlying self-sufficiency — a technical specialist who cannot improvise does not survive long on the frontier.

Career skills: Tech +2, Science +1, Labor +1

Specialties:

- *Life Support Systems* — bonus die on Tech rolls involving atmospheric, water, and power systems; can diagnose life support failures faster than standard (one round instead of one scene)
- *Systems Integration* — bonus die on Tech rolls connecting or interfacing disparate systems, including alien technology; reduced penalty die when working with unfamiliar system architectures
- One Exposure specialty of choice

Starting gear: Specialised tool kit (Gear 3), diagnostic scanner, technical manuals for home colony systems

Connections: A supplier network for colony technical components; a colleague at another colony facing similar problems

> ***At the table:*** *Technical colonists see every station and ship as a collection of systems with interdependencies, failure modes, and maintenance schedules. They notice when things are poorly maintained before anything has failed. They find it very difficult to leave a failing system without attempting to fix it, even when leaving is the correct tactical decision.*

Colonist / Medical

> *They sent one doctor for four hundred people. That was me.*

Formation: Colonial medicine is not the medicine of a well-equipped hospital. It is the medicine of limited supplies, improvised equipment, and conditions that are not in any textbook because no one has been here long enough to write one. Colonial medical practitioners are generalists by necessity — they handle everything from childbirth to trauma to alien-contact diseases to the psychological consequences of isolation. They are also, quietly, among the most important people in any community they work in, and they know it.

Career skills: Science +2, Read +1, Manipulate +1

Specialties:

- *Field Medicine* — bonus die on Science rolls for medical treatment without proper facilities; can treat conditions under fire or in hostile environments without penalty die
- *Xenomedicine* — has working knowledge of the medical needs of species present in their home colony; bonus die on Science rolls treating those species; no penalty die for unfamiliar alien physiology (basic treatment only)
- One Exposure specialty of choice (reflecting colony species mix)

Starting gear: Medical kit (Gear 3, comprehensive for field use), pharmacological supplies, diagnostic equipment

Connections: A medical supply network; a colleague at a proper medical facility who can be contacted for consultation

> ***At the table:*** *Colonial medics notice health. They observe people the way Scouts observe environments — assessing, cataloguing, identifying what is wrong before anyone mentions it. They have a specific horror of preventable death and a specific competence around inevitable death that other archetypes sometimes find unsettling. They ask direct questions about symptoms that social convention might suggest are too personal.*

Colonist / Administrator

> *Forty people, three factions, one water supply, and no rulebook. I wrote the rulebook.*

Formation: The person who made the colony function as a society rather than just a collection of survivors. Colonial administrators deal with resource allocation, conflict resolution, external relations (with the sponsoring organisation, with neighbouring settlements, with whatever species share the world), and the unglamorous work of making decisions that affect everyone with information that is always incomplete. They have learned that there is no perfect decision, only better and worse ones made under time pressure.

Career skills: Administer +2, Command +1, Read +1

Specialties:

- *Resource Management* — bonus die on Administer rolls involving allocation, logistics, or supply chain; can identify resource shortfalls before they become critical with a single roll
- *Conflict Mediation* — bonus die on Manipulate and Read rolls when mediating between parties in conflict; can de-escalate a situation that has not yet turned violent with a Manipulate roll even without leverage
- One Exposure specialty of choice

Starting gear: Personal comms (encrypted), colony documentation and records (partial, relevant), sidearm (Gear 1, administrative

rather than combat)

Connections: The sponsoring organisation — a contact who can be reached but whose interests may not align; neighbouring settlement contacts; a legal adviser in the nearest proper jurisdiction

> ***At the table:*** *Colonial administrators have an instinct for group dynamics and resource flow. They notice when a group is making a decision badly — too fast, without the right information, with the wrong people in the room — and they find it genuinely difficult not to intervene. They are often the person who asks "what happens after this?" when everyone else is focused on the immediate problem.*

Background: Citizen

You are from somewhere established — a world, a station, a habitat that has been civilised long enough to have stopped thinking about it. You did not grow up in a crisis. You grew up in a place with history, infrastructure, and the assumption that tomorrow would be roughly like today.

This is not a background of disadvantage. Citizens come from environments that invest in education, that have institutions, that have the accumulated knowledge of generations. What they often lack is the specific self-sufficiency of Colonists or the specific intensity of Military or Spacer formation — they have not been tested in the same way, and they know it.

Citizens are the most common background for non-human PCs from established species. A Vashori scholar, a Golvhaan historian, a Shekhari who grew up in a port city — these are Citizens. Their species' civilisation has had time to produce people who are shaped by something other than survival or service.

> ***Background specialty (all Citizens):*** *Institutional Knowledge — bonus die on Administer rolls when dealing with the institutions of their home world or species. They know how things work where they come from, who actually has authority, and which rules are enforced and which are not.*

Citizen / Academic

> *The question is more important than the answer. The answer just tells you what to ask next.*

Formation: The scholar, researcher, or specialist whose expertise was developed in an institutional context — a university, a research organisation, a species' knowledge-preservation body. Academic citizens have deep knowledge in specific fields and the research skills to extend that knowledge into new areas. They also have the particular blind spots of institutional formation: they are very good at the questions their field has defined and sometimes poor at the questions

it hasn't.

Career skills: Science +2, Administer +1, Read +1

Specialties:

- *Research Methodology* — bonus die on Science rolls when investigating an unknown phenomenon with time to apply systematic method; can identify what is known versus unknown about a situation faster than other characters
- *Specialist Knowledge* — deep expertise in one specific field (chosen at character creation: xenobiology, history, physics, linguistics, archaeology, etc.); bonus die on all Science rolls within that field; can attempt rolls that would be unavailable to non-specialists
- One Exposure specialty of choice

Starting gear: Research equipment relevant to specialty, personal library (digital, comprehensive within field), academic credentials

Connections: A network of colleagues in the field; an institution that may want their findings; a rival whose work overlaps inconveniently

> ***At the table:*** *Academics notice when something doesn't fit the established model. They ask questions that other characters find tangential and that frequently turn out to be central. They find the absence of information as interesting as its presence. They are sometimes poor at recognising when the time for research has ended and the time for action has begun.*

Citizen / Technical

> *I didn't build it. But I understand it completely, which is almost the same thing.*

Formation: The technical specialist of an established world or station — someone whose expertise was developed in a context with resources, training, and the accumulated knowledge of a mature technical tradition. Citizen technicians have deeper formal training than their Colonist counterparts and narrower practical experience. They know more about how things should work and sometimes less about what to do when they don't.

Career skills: Tech +2, Science +1, Operate +1

Specialties:

- *Advanced Systems* — bonus die on Tech rolls involving high-technology equipment, prototype systems, or alien technology that has been formally studied; can read technical documentation in unfamiliar system architectures with a Science roll
- *Precision Engineering* — bonus die on Tech rolls requiring fine work; can perform technical tasks in half the normal time at no penalty when conditions are adequate

- One Exposure specialty of choice

Starting gear: High-quality tool kit (Gear 3), technical credentials, specialised diagnostic equipment

Connections: A former employer or institution; a colleague working on a related problem; a supplier of high-end technical components

> ***At the table:*** *Citizen technicians notice quality. They assess equipment automatically — whether it is well-maintained, what generation it is, whether it has been modified and how well. They find poorly maintained equipment offensive in a mild but persistent way. They are often the most formally educated person in the room and sometimes the least practically flexible.*

Citizen / Medical

> *Medicine is applied science. The application is the hard part.*

Formation: A medical practitioner from an established medical system — trained properly, with access to equipment and colleagues and the accumulated knowledge of a mature medical tradition. Citizen medics know what good medicine looks like and how to practise it. They are also shaped by their institutional context: they have expectations about facilities, supplies, and conditions that field medicine constantly violates.

Career skills: Science +2, Read +1, Empathy — wait, Empathy is not a skill. Science +2, Read +1, Manipulate +1

Specialties:

- *Clinical Medicine* — bonus die on Science rolls for medical treatment when adequate facilities are available; can diagnose conditions that field medics would miss with proper equipment
- *Xenomedicine* — formal training in the medical needs of species encountered in their practice; bonus die on Science rolls treating those species; can attempt treatment of species outside their training at standard difficulty (field medics take a penalty die)
- One Exposure specialty of choice

Starting gear: Medical kit (Gear 3, high quality), pharmacological supplies (comprehensive), diagnostic equipment (advanced)

Connections: A hospital or medical institution; a pharmaceutical supplier; a colleague in a relevant specialty

> ***At the table:*** *Citizen medics notice health in the same way Colonial medics do, but they also notice the absence of adequate facilities in a way that Colonial medics have long since accepted. They find field conditions frustrating rather than simply limiting. They are often better than Colonial medics at conditions they have seen before and worse at conditions they haven't.*

Citizen / Law and Administration

> *The rules exist for reasons. Knowing the reasons tells you when the rules are wrong.*

Formation: Someone who worked within legal or administrative systems — as a lawyer, a government official, a corporate administrator, a judicial functionary. They understand how institutions make decisions, where the actual authority lies (which is rarely where the org chart says it is), and how to navigate systems that are designed to be navigated by people who know how they work. They are also shaped by the particular ethics of institutional work: the gap between what the rules say and what is right is a gap they have spent time in.

Career skills: Administer +2, Manipulate +1, Read +1

Specialties:

- *Legal Framework* — bonus die on Administer rolls involving legal processes, contracts, or jurisdictional questions; can identify the legal exposure of a situation and the options available within the law
- *Institutional Navigation* — bonus die on Manipulate rolls when working within or around bureaucratic systems; knows which procedures can be expedited, which officials have discretion, and which rules are genuinely enforced
- One Exposure specialty of choice

Starting gear: Legal credentials, personal comms, formal clothing appropriate to institutional contexts, documentation kit

Connections: A judge or official who owes a professional courtesy; a colleague in private practice; a contact in a regulatory body

> ***At the table:*** *Law and Administration characters notice authority — who actually has it, who thinks they have it, and the gap between the two. They find extralegal situations interesting rather than uncomfortable: knowing the law well includes knowing where it doesn't reach. They ask about jurisdiction in situations where other characters are asking about more immediate problems, and they are sometimes right that jurisdiction matters more.*

Background: Bureaucrat

You worked inside large organisations — governments, corporations, interstellar bodies, species-level administrative structures. You understand how institutions actually function, which is quite different from how they are supposed to function, and you have the specific competences of someone who has navigated complexity from the inside.

Bureaucrats are not paper-pushers. They are people who understand power as it is actually exercised: through information flow, resource

allocation, personnel decisions, and the management of competing interests. They know that the person with the title is frequently not the person with the authority, and they know how to find the person with the authority.

In a multi-species setting, senior Bureaucrats are among the most likely to have formal Exposure to multiple species through their institutional work — interstellar bodies require it.

> ***Background specialty (all Bureaucrats):*** *Organisational Reading — bonus die on Administer rolls when assessing an unfamiliar organisation's structure, identifying who actually has authority, or understanding how a decision was made. Bureaucrats can read institutions the way Scouts read terrain.*

Bureaucrat / Government

> *The government is not the state. The state is not the law. I know the difference.*

Formation: Someone who worked within government — not necessarily senior, but experienced enough to understand how policy becomes action, how decisions are made and unmade, and where the pressure points are. Government bureaucrats have a specific relationship to power: they have been close to it without having it, which produces either cynicism or sophistication, and sometimes both.

Career skills: Administer +2, Read +1, Command +1

Specialties:

- *Policy and Process* — bonus die on Administer rolls involving governmental processes, interstellar treaties, or political decision-making; understands the legislative and executive structures of their home government in detail
- *Political Intelligence* — bonus die on Read rolls when assessing the political motivations behind a decision or action; can identify what a political actor actually wants versus what they say they want
- One Exposure specialty of choice

Starting gear: Government credentials (current or former), encrypted communications device, formal clothing, access to classified databases (limited, depending on current status)

Connections: A current official who prefers their relationship with this character remain informal; a former colleague now in a different position; a journalist who pays for information

> ***At the table:*** *Government bureaucrats see politics in everything. They ask who benefits, who loses, and who made the decision to frame every situation they encounter. They find purely tactical thinking incomplete — there is always a political dimension that other characters are missing. They are often right, and sometimes this is useful, and sometimes this is just exhausting.*

Bureaucrat / Corporate

> *It is not personal. It is never personal. That's what makes it so effective.*

Formation: Someone who worked within corporate structures — trading companies, manufacturing concerns, resource extraction operations, interstellar commercial enterprises. Corporate bureaucrats understand profit motive as the actual driver of institutional behaviour in a way that government bureaucrats often don't, and they understand that corporate loyalty is a transactional relationship that runs in both directions until it doesn't.

Career skills: Administer +2, Manipulate +1, Scan +1

Specialties:

- *Commercial Intelligence* — bonus die on Administer rolls involving commercial law, contracts, trade regulations, and corporate structure; can assess the financial health and actual priorities of a commercial entity with an Administer roll
- *Negotiation* — bonus die on Manipulate rolls in formal negotiation contexts; understands the difference between stated positions and actual interests and can exploit the gap
- One Exposure specialty of choice

Starting gear: Corporate credentials (current or former), encrypted comms, formal clothing, commercial database access

Connections: A former employer with unresolved business; a competitor who is occasionally useful; a financial institution contact

> ***At the table:*** *Corporate bureaucrats assess situations for their commercial implications with a speed other characters find slightly unsettling. They notice who is paying for what and what that means. They are often the person who points out that the apparently altruistic offer has a business model behind it that the other characters haven't found yet. They are not cynical — they are accurate.*

Bureaucrat / Intelligence Agency

> *I know things I cannot tell you. Some of them are about you.*

Formation: Civilian intelligence — not military intelligence, but the agencies that gather information on behalf of governments, corporations, or other large actors through means that are not military. Agency bureaucrats understand information as a commodity: how it is gathered, how it is assessed, how it is used, and how it is used against you. They have a specific relationship to truth that is more complicated than most people's.

Career skills: Read +2, Administer +1, Manipulate +1

Specialties:

- *Information Analysis* — bonus die on Read and Scan rolls when assessing information sources, identifying disinformation, or synthesising fragmentary intelligence into a coherent picture
- *Asset Management* — bonus die on Manipulate rolls when recruiting, maintaining, or directing human sources; understands the psychology of why people provide information and how to exploit it ethically or otherwise
- One Exposure specialty of choice

Starting gear: False documentation (high quality), recording equipment (concealed, high quality), personal sidearm (Gear 2, unregistered), no official credentials

Connections: A handler whose agenda is not entirely clear; a source in an unexpected location; a former asset who now knows too much

> ***At the table:*** *Intelligence agency characters listen before they speak and remember everything. They find other characters' willingness to volunteer information about themselves both useful and faintly alarming. They are often the person who says "we don't know enough yet" when the group is preparing to act, and sometimes they are right, and sometimes waiting is the wrong call and they know it.*

Bureaucrat / Diplomatic Corps

> *My job is to prevent the thing that happens when my job fails.*

Formation: Someone who worked in formal diplomatic contexts — representing one species, government, or faction to another, managing relationships across the full range from routine trade negotiations to crisis management at the edge of open conflict. Diplomatic corps characters are trained in a specific set of social skills that are different from the street-level Manipulate of a trader or spy: they understand protocol, they know how to use formal structures as tools, and they have experience with the particular challenge of representing interests that are not their own.

Career skills: Manipulate +2, Administer +1, Read +1

Specialties:

- *Protocol* — bonus die on Manipulate rolls in formal diplomatic contexts; knows the correct form for interactions with most major species and governments; no penalty die for unfamiliar formal contexts (can improvise correctly)
- *Crisis Management* — bonus die on Manipulate and Command rolls when managing a situation that is actively deteriorating; can

buy time in a crisis with words when other options have failed

- One Exposure specialty of choice (Diplomatic Corps characters may take two Exposure specialties reflecting their posting history)

Starting gear: Diplomatic credentials (current or former), formal clothing (multiple, species-appropriate), encrypted comms, translation technology (high quality)

Connections: A current posting or former posting with ongoing relationships; a colleague in another delegation; a contact in the foreign ministry of a relevant species

> ***At the table:*** *Diplomatic Corps characters notice when communication is failing before anyone else does. They catch the moment when a negotiation changes direction, when someone has said something that landed wrong, when a situation that seemed stable has shifted. They find violence representing a failure — not a moral failure necessarily, but a professional one — and they will work very hard to avoid it. Sometimes this is the right call. Sometimes the situation called for violence two scenes ago.*

Background: Merchant

Trade. The movement of goods, information, and value across the distances between worlds and species. You are part of the commercial infrastructure that makes interstellar civilisation function — not its government, not its military, not its institutions, but the network of exchange that underlies all of those.

Merchants come in many forms — from the captain-owner of a battered free trader to the senior factor of a major commercial house — but they share a formation: comfort with uncertainty, a specific relationship to risk and reward, and the practical competence of people who have to make things work in places far from any support structure. They also, characteristically, know people. Merchants run on relationships.

In a multi-species setting, Merchants are among the most likely to have broad Exposure simply because trade does not respect species boundaries. A merchant who has never dealt across species lines is operating in a very limited market.

> ***Background specialty (all Merchants):*** *Market Reading — bonus die on Administer rolls when assessing the commercial value of goods, services, or information; can identify arbitrage opportunities (price differences between markets) with a single roll.*

Merchant / Free Trader

> *My ship, my cargo, my route, my decision. That's the whole deal.*

Formation: The independent operator — captain, owner, and

usually crew of a vessel that goes where the profit is and stays independent because independence is the point. Free traders have the Spacer's comfort with ships and vacuum combined with the Merchant's commercial instincts and a specific quality that neither background alone produces: the ability to make every decision themselves, in full knowledge that every decision is their responsibility. They are often the most self-sufficient characters in the setting and the most resistant to taking orders.

Career skills: Operate +1, Administer +1, Manipulate +1

Specialties:

- *Route Knowledge* — bonus die on Navigate rolls in known regions; has detailed knowledge of a specific region of space including unofficial routes, hazards, and the people who operate there
- *Deal Making* — bonus die on Manipulate rolls in commercial negotiations; can identify the actual bottom line of a negotiating counterpart with a Read roll
- One Exposure specialty of choice

Starting gear: Ship (assumed, with debt attached), personal sidearm (Gear 2), trading contacts database, cargo manifest tool

Connections: A banker or financier holding paper on the ship; regular customers at several ports; a competitor who is sometimes a colleague

> ***At the table:*** *Free traders think about their ship constantly. It is their home, their livelihood, and their freedom, and threats to it are threats to everything. They assess every situation for its commercial angle without necessarily acting on it — it is simply how they see the world. They find employers and authorities tolerable when the pay is right and irritating when it isn't.*

Merchant / Factor

> *I represent interests. Which interests depends on the day.*

Formation: The commercial agent — someone who acts on behalf of others in commercial contexts, negotiating, sourcing, placing, and managing transactions across multiple parties. Factors are the Van Rijn tradition: deep commercial knowledge, broad relationship networks, and the specific competence of someone who makes their living knowing what everyone wants and connecting them to it. They are not tied to a single employer or a single ship; they operate in the space between parties.

Career skills: Administer +2, Manipulate +1, Read +1

Specialties:

- *Commercial Network* — once per session, can contact a source anywhere in known space for commercial information, goods availability, or market conditions; the contact may want something in return
- *Contract Law* — bonus die on Administer rolls involving commercial contracts, trade agreements, or commercial disputes; can identify problematic contract terms that other characters would miss
- One Exposure specialty of choice

Starting gear: Commercial credentials, encrypted comms, formal and informal clothing, letter of credit (limited)

Connections: A major commercial house that uses their services; a rival factor; contacts at multiple ports across their operating region

> ***At the table:*** *Factors are always working. They notice commercial opportunity the way Scouts notice tactical opportunity — automatically, as background processing. They know who everyone is and what they want, or they find out. They are often the person who can get the thing the group needs, for a price that is negotiable, through channels that are not entirely official.*

Merchant / Broker

> *Information is cargo. Cargo is information. Same business.*

Formation: The information merchant — someone who trades in intelligence, contacts, and knowledge as readily as in physical goods. Brokers know things and know people who know things, and they have built a career on the value of that network. They are distinct from intelligence agency operatives in that their motivations are commercial rather than political, and their methods are relationship-based rather than covert — though the line between a well-connected broker and a well-placed intelligence asset is not always clear.

Career skills: Read +2, Administer +1, Scan +1

Specialties:

- *Information Network* — bonus die on Read and Scan rolls when gathering information through contacts; once per session, can identify a contact who knows something relevant to the current situation
- *Discretion* — bonus die on Manipulate rolls when assuring parties that information will be handled confidentially; has a professional reputation for not selling the same information twice (whether or not this is true)
- One Exposure specialty of choice

Starting gear: Encrypted comms (high quality), contact database (extensive), recording equipment, personal sidearm (Gear 1, rarely used)

Connections: Three contacts in different sectors who provide information regularly; a buyer for sensitive information whose identity is protected

> ***At the table:*** *Brokers are always listening. They find other characters' information security habits interesting and often inadequate. They know things about the people they work with that those people have not told them directly, and they are careful about when and whether to reveal this. They find violence crude — not morally, but commercially. Violence is bad for the information business.*

Merchant / Smuggler

> *The law says what's legal. The market says what's needed. I serve the market.*

Formation: The independent operator who moves goods that someone, somewhere, has decided should not be moved — whether that's weapons, contraband, controlled technology, refugees, or information that someone powerful wants suppressed. Smugglers have the Free Trader's self-sufficiency and the Broker's relationship network, plus a specific set of skills that come from operating in the gap between what is permitted and what is needed. They are not criminals in any meaningful moral sense — the laws they break vary wildly in legitimacy — but they operate outside official frameworks and have developed the skills that requires.

Career skills: Operate +1, Stealth +1, Manipulate +1

Specialties:

- *Hidden Cargo* — bonus die on Stealth rolls to conceal goods or people; knows the standard search patterns of most major port authorities and how to design concealment around them
- *Clean Passage* — bonus die on Manipulate rolls when dealing with port authorities, customs officials, and checkpoints; can identify the right approach to a given official (bribable, bypassable, or must-be-avoided) with a Read roll
- One Exposure specialty of choice

Starting gear: Modified ship (assumed, with concealed compartments), false documentation (Gear 2), personal sidearm (Gear 2), bribery fund (modest)

Connections: A port official who looks the other way;

a regular customer who needs things moved; a legitimate merchant who uses their services and pretends not to

> ***At the table:*** *Smugglers notice security. They assess every port, every checkpoint, every official for its implications for their operation — past, present, or future. They are often the most practically useful person in the group when things go wrong with authorities and the most professionally uncomfortable when things go right with them. They find bureaucratic legitimacy mildly amusing and occasionally useful as cover.*

Character Creation Summary

1. Choose your **species** (see Species chapter)
2. Choose your **background** (Spacer / Military / Colonist / Citizen / Bureaucrat / Merchant)
3. Choose your **expertise** within that background
4. Apply career skill bumps from your expertise
5. Choose your **Exposure specialty** from the list (Diplomatic Corps may choose two)
6. Note your **starting gear** and **connections**
7. Apply your species' attribute adjustment and biological specialties
8. Distribute base attribute points (see Character Creation chapter)
9. Name your character and establish their specific history within the background framework

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Adding New Archetypes

When a player requests a background or expertise not covered here, use this checklist:

- Does it fit an existing background, or does it need a new one?
- What three skills does this career train systematically?
- What two or three specialties distinguish this expertise from adjacent ones?
- What starting gear characterises this career?
- What connections does this career naturally produce?
- What does this character notice that others don't?

(The roleplay prompt test — if you can't answer this, the archetype isn't distinct enough yet)

New expertises within existing backgrounds can be added at any time. New backgrounds should be rare — the six here cover the realistic spread of how people end up in this kind of work.

Character Creation

Overview

A character is built in nine steps. The steps are designed to be taken in order — each one builds on the last — but experienced players may find it natural to move between them as the character concept develops. What matters is that all nine steps are complete before play begins.

The process takes approximately 30-45 minutes for a first character, less for subsequent ones. The GM should be present or available during character creation — connections, exposure specialties, and starting history all benefit from knowing what the campaign will look like.

Step 1 — Concept

Before touching the sheet, write one paragraph describing who this character is and why they are here.

The paragraph should answer three things:

- **Species and background** — what shaped them before the campaign
- **The defining quality** — the one thing that is most true about them as a person, not as a set of skills
- **Why here, why now** — what has brought them to this crew or this situation

This paragraph is the foundation. Everything in the subsequent steps should be consistent with it, and when the steps produce something that contradicts it, the concept paragraph wins. Mechanics serve character, not the other way around.

> ***Example:*** *Sheshal is a Vashori who spent twelve years as a colonial medical officer on a human-majority frontier world, the only non-human doctor for four hundred people. She is defined by the particular competence of someone who has handled every emergency alone. She is here because the colony's sponsoring corporation dissolved, the colony was abandoned, and she has nowhere else to go — but she has her kit, her training, and an anger about what happened that has not cooled.*

Step 2 — Species

Choose your species from the Species chapter. Read the full entry, not just the mechanics section — the roleplay section is as important as the stat block, and the *At the table* note tells you what the GM and other players will expect.

Record the following:

Attribute adjustment — noted in Step 5 when attributes are distributed, but keep it in mind now. It will take one attribute above the normal creation maximum.

Biological specialties — record these now. They are separate from career specialties and do not count against specialty limits.

Environmental constraints — note any conditions under which your species takes penalty dice. These are active throughout play.

Size — note if your species is Small or Large. This affects Fight, Labor, Stealth, and equipment as described in the Species chapter.

Advanced Species

If choosing Skharri or Nakhavi, confirm with the GM that you have read and understood the full species entry including the GM notes. These species are rewarding but require specific commitment from the player. The GM should confirm that the player has a clear answer to why this character is functioning in a mixed crew before play begins.

Non-Human Characters and Exposure

Non-human characters require Exposure: Human to read human emotional states at full capacity. Without it, the Opacity Rule applies to your human crewmates. This is not a disadvantage — it is a realistic starting point for a character who has not spent significant time among humans. Many non-human characters will take Exposure: Human as their starting Exposure specialty (Step 4). Those who do not have an interesting roleplay situation from session one.

Step 3 — Background and Expertise

Choose your **background** from the six available:

- **Spacer** — life defined by ships, stations, vacuum, and the Belt
- **Military** — served in organised armed forces
- **Colonist** — from a frontier or developing world
- **Citizen** — from an established settled world or habitat
- **Bureaucrat** — worked inside large institutions
- **Merchant** — commercial life, trade, and negotiation

Then choose your **expertise** within that background. Full descriptions of all backgrounds and expertises are in the Archetypes chapter.

Record the following from your chosen archetype:

Background specialty — the specialty shared by all characters of this background (e.g., *Zero-G Movement* for all Spacers, *Threat Assessment* for all Military). This does not count against specialty limits.

Career skill bumps — three skills, each bumped by the noted amount. Apply these in Step 6 after distributing base skill points, not before.

Career specialties — two or three specialties from the expertise entry. Record these alongside your biological specialties.

Starting gear — the characterful package from the expertise entry. This is not exhaustive — it represents what the career typically provides. Personal additions are made in Step 8.

Connections — record the connections from the expertise entry. Discuss with the GM how these connect to the campaign's setting. A connection that has no relevance to where the campaign takes place is a missed opportunity; work with the GM to ground them in the specific region and political situation of the starting point.

Step 4 — Exposure

Choose one **Exposure specialty** from the list in the Archetypes chapter.

Exposure reflects which non-human (or human, for non-human characters) species you grew up around, worked alongside, or spent enough formative time with to understand at a social level. It lifts the Opacity Rule for that species and grants specific social bonuses detailed in the species entry.

Remember the Opacity Rule: Without the relevant Exposure specialty, Read rolls for emotional or intentional content against that species automatically fail, and Manipulate rolls take a penalty die.

Exposure comes from environment and experience, not from career. Any background can produce any Exposure. A Military/Infantry character who served in a mixed-species unit has Exposure to whichever species they served alongside. A Citizen/Academic who studied xenobiology has Exposure to the species they researched.

Characters from demonstrably multi-species environments may take two Exposure specialties at the GM's discretion. Discuss this during character creation rather than claiming it after the fact.

Diplomatic Corps characters automatically receive two Exposure specialties as noted in the Archetypes chapter.

Step 5 — Attributes

Distribute **14 points** across the five attributes.

Attribute	Minimum	Maximum (creation)	Maximum (with species)
Strength	2	4	5
Agility	2	4	5
Wits	2	4	5
Empathy	2	4	5
Intellect	2	4	5

No attribute may be below 2 or above 4 at creation, before the species adjustment is applied.

Apply the species adjustment last. If your species grants Agility +1 and you have distributed 4 points to Agility, your final Agility is 5 — one above the creation maximum, which is correct. The species adjustment can take one attribute to 5; no attribute may exceed 5 by any means.

Humans receive no species adjustment. Their 14 points are their final attributes.

What Attributes Mean

Distribute points according to the concept established in Step 1, not according to optimisation. A character whose concept is a patient scholar should not have Strength 4 unless that is part of who they are.

Strength — raw physical power, endurance, and the capacity for sustained physical effort. The body as a tool.

Agility — speed, coordination, and controlled movement. The body as a precision instrument.

Wits — in-the-moment mental sharpness. Situational awareness, reaction speed, and the ability to process and act under pressure.

Empathy — understanding and influencing others. The social and emotional intelligence to model what another being is thinking and feeling.

Intellect — applied knowledge. What you know, how deeply you understand it, and whether you can reason through novel

problems from principles.

Step 6 — Skills

Distribute **8 points** across the fifteen skills.

No skill may exceed **3** at creation.
No skill may be below **0** (unrated skills cost nothing and remain at 0).

After distributing base points, apply the **career skill bumps** from Step 3. Career bumps may take a skill above 3, up to a maximum of 4. A skill may not exceed 4 at creation by any means.

The Fifteen Skills

Strength: Fight, Endure, Labor
Agility: Shoot, Stealth, Move
Wits: Navigate, Operate, Scan
Empathy: Manipulate, Read, Command
Intellect: Tech, Science, Administer

Full descriptions of each skill are in the Skills chapter.

Distribution Guidance

Eight points is enough to be competent in a few areas and aware in others. A focused character might put 3 points in their primary skill, 2 in a secondary, and spread the remainder. A generalist might distribute more evenly.

Unrated skills (0) are not impossible — a character can attempt almost anything using attribute dice alone. They are simply areas where the character has no specific training. An unrated skill should correspond to something the concept suggests this character genuinely has not been trained in.

Career skill bumps represent what the background systematically developed. They are not negotiable — a Spacer/Pilot has Operate development because that is what the career does. If the career bumps conflict with the concept, reconsider either the concept or the archetype.

Step 7 — Derived Values

The following values are derived from attributes and will be used in the Conditions and Combat chapters.

Hit Capacity — the number of physical conditions a character can sustain before incapacitation.

```
Hit Capacity = Strength + Endure skill
```

A character with Strength 3 and Endure 2 has Hit Capacity 5. This is the number of physical conditions they can carry before being incapacitated. Most characters will have Hit Capacity between 3 and 7.

Stress Capacity — the number of psychological conditions a character can sustain before breaking.

```
Stress Capacity = Wits + Empathy (average, rounded up)
```

A character with Wits 3 and Empathy 2 has Stress Capacity 3. Psychological conditions — fear, isolation, sustained horror, the consequences of the things that happen in space — are tracked separately from physical ones.

Carry — the amount of significant gear a character can carry without penalty.

```
Carry = Strength + Labor skill
```

Carry is an abstract value, not a weight limit. Each significant item of gear occupies one Carry slot. Personal items, clothing, and incidentals do not count. A character who exceeds their Carry capacity takes a penalty die on Move rolls.

> ***Playtest note:*** *These derived values are the most likely candidates for adjustment. If characters feel too fragile, increase the base values. If combat feels too slow because characters absorb too much, reduce them. Track which end of the scale causes problems first.*

Step 8 — Gear and Personal Item

Starting Gear

Record the starting gear package from your archetype (Step 3). This represents what your background has provided — equipment appropriate to your career and expertise.

All starting gear is assumed to be in functional condition. Gear ratings reflect quality and condition combined; gear that is poorly maintained degrades in rating over time.

Additional Personal Gear

Beyond the career package, a character may start with up to **three additional items** of personal gear. These should be consistent with the concept and background — a former colonial

medic might carry personal medical supplies beyond their kit; a military veteran might have kept a specific sidearm from their service.

Additional items should have a story. *Why does this character have this?* The answer does not need to be dramatic, but it should exist.

The Personal Item

Choose **one item** that is specifically and personally significant to the character. This is not necessarily the most valuable or most useful item they carry. It is the one that matters to them.

The personal item has a mechanical function beyond its gear rating:

Once per session, a character who spends a quiet moment with their personal item — handling it, thinking about what it represents, the context in which they use it — may remove one stress condition. This requires a moment of genuine calm, not a combat action.

The personal item is not indestructible. If it is lost, damaged, or destroyed, the character loses this recovery option until the item is recovered or a new one takes its place over time. A new personal item cannot be designated immediately — it takes sessions of play for an object to accumulate that weight.

> ***Examples:*** *A worn photograph. A tool that belonged to someone who taught them. A specific piece of music on an ancient storage medium. A child's drawing on a colony world that no longer exists. The item should cost nothing and mean everything.*

Step 9 — Name, History, and the Three Questions

Name

Choose a name appropriate to the species and background. Non-human characters use names from their species' culture — the species entries provide flavour for naming conventions without prescribing specific names.

Human characters from different planetary or station cultures will have names reflecting those cultures. A character from a long-established Belt habitat may have a name quite different from someone from a core world.

History

Write two or three sentences of specific history. Not biography — specific history. The moment that changed something, the decision that cannot be undone, the thing that happened that is still happening in some sense.

This is distinct from the concept paragraph in Step 1, which described who the character is. History describes what made them that way.

The Three Questions

Answer these three questions in writing before the first session. They do not go on the sheet — they go in a note that the player keeps.

What do you want?
Not what your character wants in general — what do they want *right now*, at the start of the campaign? Something specific and achievable in principle, even if not easily. The desire that will shape their decisions in the first few sessions.

What are you running from or toward?
Something behind them or ahead of them that has momentum. A thing that will arrive eventually regardless of what the character does. This is the GM's invitation to bring that thing into the campaign at the right moment.

Who outside this crew matters to you?
One person, named, with a relationship defined. They may be alive or dead, near or far, in contact or lost. They are the human (or alien) anchor — the reason the character's choices have weight beyond the immediate situation.

Share these answers with the GM. They are not secret. They are the starting material for the campaign's personal layer.

Step 10 — Connections (Group)

After all characters are created, the group takes one additional step together: **establishing connections between characters**.

Each character identifies one relationship to another PC:

- Someone they have worked with before (how did that go?)
- Someone they owe something to, or who owes something to them
- Someone they do not entirely trust but need
- Someone whose presence makes them more capable than they

would be alone

These connections do not need to be reciprocal and should not all be positive. A crew with complicated internal relationships is more interesting than one that is uniformly friendly from session one.

> ***Roleplay note:*** *The internal crew dynamic is one of the richest veins of roleplay in a ship-based campaign. Who trusts whom, who has history, who is still forming an opinion — these relationships develop through play, but starting with one established connection per character gives the first session something to work with beyond the immediate situation.*

The Completed Character

A completed character has:

- [ ] Concept paragraph (Step 1)
- [ ] Species recorded with attribute adjustment, biological specialties, and environmental constraints (Step 2)
- [ ] Background and expertise recorded with background specialty, career skill bumps, career specialties, starting gear, and connections (Step 3)
- [ ] Exposure specialty recorded (Step 4)
- [ ] Five attributes distributed, species adjustment applied (Step 5)
- [ ] Skills distributed, career bumps applied (Step 6)
- [ ] Hit Capacity, Stress Capacity, and Carry calculated (Step 7)
- [ ] Starting gear recorded, personal item identified and described (Step 8)
- [ ] Name, history, and three questions written (Step 9)
- [ ] One inter-PC connection established (Step 10)

Specialty Summary

A completed character has the following specialties, from different sources:

Source	Count	Notes
Biological (species)	1–2	From species entry; no limit
Background	1	Shared by all characters of this background
Career (expertise)	2–3	From expertise entry
Exposure	1–2	From Exposure list; 2 for Diplomatic Corps or multi-species environments
Total	**5–8**	Before any development in play

This is enough specialties to feel specific and capable without feeling complete. There should be obvious gaps — things the

character cannot do as well as they would like, situations where another character is clearly better placed. Those gaps are where growth happens.

Designer's Notes — Playtest Values

The following values are most likely to require adjustment during playtesting. Watch for the symptoms described and adjust accordingly.

Attribute array (14 points, min 2, max 4):

- *Too weak:* Characters feel identical because the spread is too narrow. Solution: increase to 15 or 16 points.
- *Too strong:* Characters start feeling like finished products with no room to grow. Solution: reduce to 13 points or tighten the maximum to 3.

Skill points (8 points, max 3 at creation):

- *Too few:* Characters feel incompetent outside their narrow specialty, which reduces roleplay options. Solution: increase to 10 points.
- *Too many:* Characters start with no obvious growth direction. Solution: reduce to 6 points or lower the creation maximum to 2.

Career skill bumps:

- *Too significant:* The career bumps overwhelm the base point distribution and all characters of the same archetype feel similar. Solution: reduce bumps to +1 total across three skills rather than the current distribution.
- *Too minor:* Career feels decorative rather than formative. Solution: add a fourth skill bump or allow one bump to reach +2.

Hit Capacity:

- *Characters die too easily:* Add Endure skill value to the formula or increase the base value by 1.
- *Combat drags because characters absorb too much:* Remove the skill component — use Strength alone.

Stress Capacity:

- *Stress conditions accumulate too fast:* Increase the formula or add a flat +1.
- *Stress conditions never feel threatening:* Reduce the formula or make specific stress conditions harder to remove.

Personal item (once per session, remove one stress condition):

- *Too powerful:* Change to once per long rest rather than once per session.
- *Rarely used:* The situations calling for calm are too infrequent.

Consider allowing its use in any moment of genuine quiet, even briefly during a scene.

Quick Reference — Character Creation Steps

1. **Concept** — one paragraph, who and why
2. **Species** — attribute adjustment, biological specialties, environmental constraints
3. **Background and expertise** — background specialty, career skills, career specialties, gear, connections
4. **Exposure** — one specialty; two for Diplomatic Corps or multi-species environments
5. **Attributes** — 14 points, min 2, max 4; apply species adjustment last
6. **Skills** — 8 points, max 3; apply career bumps after, max 4 total at creation
7. **Derived values** — Hit Capacity, Stress Capacity, Carry
8. **Gear and personal item** — career package, up to three personal items, one significant personal item
9. **Name, history, three questions** — share with GM
10. **Group connections** — one inter-PC relationship each

Combat

Overview

Combat in this game is dangerous, fast, and consequential. Characters are not action heroes — wounds accumulate quickly, the environment is often as threatening as the enemy, and the decision to fight is rarely without cost. The system is designed to resolve violence quickly and return the table to the broader situation, not to model tactical complexity for its own sake.

Most combats should resolve in three to five rounds. If a combat is going longer than that, something interesting should be happening — a changing environment, a tactical shift, an opportunity to end it differently — not more of the same exchange.

> ***Roleplay note:*** *The decision to fight is a character decision, not a table default. In a setting where killing a Vardhek attaché is a diplomatic incident, where a Skharri's honour-instinct can start a war, and where the nearest medical support may be three jumps away, violence has consequences that extend well beyond the combat scene. Encourage players to think about what they are starting before they start it.*

The Shape of a Combat

A combat is divided into **rounds**. Each round, both sides act — players first or NPCs first depending on initiative — before the next round begins. A round represents a few seconds of intense activity, not a precise unit of time.

Within a round, each character takes **one action**. An action is one meaningful thing: attack, move to cover, assist a crewmate, attempt a technical task under fire, communicate a complex instruction. Simple things — dropping prone, drawing a weapon, shouting a warning — do not cost an action.

Initiative

The Opening Round

At the start of combat, one side acts first. To determine which:

The player group nominates one character to make a **Wits roll**. This represents the group's collective readiness — their situational awareness, their speed off the mark, their ability to react as a unit.

The GM rolls for the NPC group — a single Wits roll representing the opposing side's readiness.

The side with more sixes acts first in the opening round. On a tie, the players act first.

This is one roll, immediately resolved. There is no individual initiative order within sides.

Surprise

If one side has genuinely surprised the other — through successful Stealth, through an ambush that was not detected, through a situation where the other side had no warning — the surprising side acts first in the opening round automatically. No roll is made.

The surprised side **loses their action in Round 1**. They may take cover, move, or react in limited ways, but they cannot make a meaningful offensive action until Round 2.

Partial surprise — the other side knew something was coming but not when or where — gives the surprising side first action with no roll, but the surprised side retains their Round 1 action at a penalty die.

Subsequent Rounds

After the opening round, sides alternate: **players then NPCs**, or **NPCs then players** if the NPCs won initiative, repeating until the combat ends.

Communication and Coordination

Within the player group's turn, the players decide amongst themselves who does what and in what order. **Communication is assumed** — the crew can coordinate, warn each other, and adjust plans in real time, because that is what trained people in communication with each other actually do.

This coordination benefit applies when the players can genuinely communicate. It does not apply when:

- Characters are in hard vacuum without functioning suit comms
- Characters are physically separated beyond communication range
- Communications are being actively jammed or disrupted
- A character is incapacitated, panicking, or otherwise unable to receive or process information

When coordination is lost, each character acts individually without the benefit of shared planning. The GM will indicate when this applies.

> ***Note for the GM:*** *The initiative structure above handles most situations cleanly. The GM may vary it when the fiction warrants — a complex tactical environment, a dramatic reversal mid-round, or a moment where the*

sequence of events demands something different. This is not a rules exception; it is the GM reading the table.

NPC Initiative

Mooks act as a bloc — the GM makes one decision for the group and rolls once on their behalf. They move together, fire together, and break together.

Named NPCs act individually within the NPC turn, in whatever order makes narrative sense. A named NPC whose specific instinct or agenda has been triggered — a Skharri whose honour has been directly challenged, a Vardhek who has identified a specific threat to their strategic position — acts on that trigger, not on tactical logic.

Actions

Each character takes one action per round. Common actions:

Attack — make a combat roll against a target in range. The most common action and described in full below.

Move — change position significantly. Moving to a new range band, reaching cover, boarding or leaving a vehicle, crossing difficult terrain. A Move action combined with an attack is possible but both rolls take a penalty die.

Assist — help another character's action. Adds one bonus die to a crewmate's roll this round. The assisting character must be in a position to actually help — adjacent, in comms contact, with relevant capability. See Group Actions below.

First aid — attempt emergency medical treatment on an adjacent character using Science (Medic specialty applies). Cannot be performed while under direct fire without a penalty die.

Technical action — attempt a technical task under pressure. Operate a system, bypass a lock, restore a damaged component. Any Intellect skill roll made under fire takes a penalty die.

Communicate — convey complex information to someone not already in the coordination loop. Simple warnings are free; a full tactical briefing costs an action.

Disengage — attempt to exit the combat without being attacked. Opposed Move roll against any enemy in Close range. On success, the character is out of the immediate combat. On failure, they take a free attack from that enemy before leaving.

Making an Attack

The Attack Roll

Build a dice pool as normal:

Attribute — Strength for Fight (melee), Agility for Shoot (ranged)

Skill — Fight or Shoot as appropriate

Gear — the weapon's gear rating

Roll the pool. Count sixes.

Attack Results

Sixes	Result
0	Miss. No effect.
1	Hit. Target takes one condition.
2	Clean hit. Target takes one condition; attacker chooses the nature of the condition.
3	Strong hit. Target takes two conditions, or one condition plus a tactical effect.
4+	Devastating hit. Target takes two conditions plus a tactical effect, or three conditions.

Tactical effects (attacker's choice on 3+ sixes):

- Target is knocked back or down (costs their next action to recover)
- Target's weapon is damaged (gear rating reduced by 1)
- Target is disarmed
- Target is pushed into a worse position (exposed, closer to hazard)
- Attacker gains a bonus die on their next action against this target

Choosing Not to Kill

By default, combat damage is injury and incapacitation, not death. A character reduced to four conditions is incapacitated — unable to act, but alive.

Killing requires a deliberate choice. A character who wishes to kill a downed opponent states this and it happens — no roll required, but it is a conscious decision with the full weight that implies. In a setting where killing a Vardhek diplomat is a strategic incident, this matters.

Subduing — attempting to knock out rather than injure — imposes a penalty die on the attack roll. On success, conditions are psychological (stunned, dazed) rather than physical, and the target can be restrained rather than killed.

> ***Roleplay note:*** *The choice not to kill is as significant as the choice to kill. A Skharri who subdues an opponent when honour demanded a killing blow has done something that their culture will note. A human operative who kills when subduing was possible has made a choice that follows them. Keep these moments visible.*

Defence

Defence in this game is **passive by default**. There is no defence roll against every attack — the attacker rolls and the result stands, modified by cover and circumstances.

Cover

Cover removes dice from the attacker's pool before the roll.

Cover	Dice Removed
Light cover (furniture, partial concealment)	1 die
Solid cover (doorframe, vehicle hull, rock)	2 dice
Hard cover (reinforced position, full concealment)	3 dice

Cover must be declared before the attack roll. A character behind cover must expose themselves to attack — leaning out to shoot, opening a door — which may reduce or eliminate the cover benefit at the GM's discretion.

Active Defence

A character may choose to **actively defend** rather than act offensively. This costs their action for the round. An actively defending character adds their relevant skill as bonus dice to resist the incoming attack — Fight for melee defence, Move for evading ranged fire.

Active defence is resolved as an **opposed roll**: attacker's pool versus defender's pool. Most sixes wins. On a tie, the attack succeeds with minimal effect (one condition, attacker's choice of type removed).

Active defence is a meaningful sacrifice — giving up your action to reduce damage. It is most valuable when a character is already in a bad position and cannot afford another condition.

Armour

Armour adds **green dice** to the defender's pool when actively defending, representing the gear quality of the protective equipment. Armour does not reduce damage passively — it only applies when the character is actively defending.

Armour	Gear Rating	Notes
Civilian clothing	0	No protection
Light armour	1	Concealed, minimal restriction
Combat armour	2	Standard military; restricts Move slightly
Heavy armour	3	Significant restriction; penalty die on Move
Vacc suit (armoured)	2	Adds vacuum protection; see Environmental Hazards

Armour that has taken significant damage degrades — reduce the gear rating by 1 after any combat in which the character took three or more conditions. Armour can be repaired with Tech.

Range

Combat range is divided into three bands. Each weapon has an **optimal band** where it performs at full effectiveness. Outside the optimal band, the attack takes a penalty die.

Band	Distance	Notes
Close	Arm's reach to a few metres	Melee range; confined spaces
Short	Same room to adjacent compartment	Most personal weapons
Long	Open space, corridors, across a bay	Rifles, ship weapons

Switching bands costs a Move action — moving from Long to Short, or Short to Close.

Melee weapons (Fight) are only effective at Close range. Used at Short range, they take two penalty dice and cannot be used at Long range at all.

Personal ranged weapons (Shoot) are optimal at Short range. At Close range they take a penalty die (unwieldy, risk of grabbing). At Long range they take a penalty die (effective range exceeded).

Rifles and longarms are optimal at Long range. At Short range, no penalty. At Close range, one penalty die.

Thrown weapons are optimal at Short range. At Close and Long range, one penalty die each.

Size in Combat

The size differences between species create genuine mechanical asymmetry in close combat. This is not balanced — a Golvhaan and a Shekhari in melee are not equivalent combatants, and they should not be.

Attacking Across Size Categories

Attacking a larger opponent (melee): If the target is more than twice your mass, your Fight roll takes a penalty die. Hitting something that large is not hard; making the hit matter is.

Attacking a smaller opponent (melee): If the target is less than half your mass, your Fight roll gains a bonus die. However, if the intent is to subdue rather than injure — to grab, restrain, or incapacitate without serious harm — the roll takes a penalty die instead. Precision against a small fast target is hard.

Ranged combat is not affected by size differences except that larger targets are easier to hit: Large species targets give the attacker one bonus die on Shoot rolls at Short and Long range.

Size and Conditions

Large species (Golvhaan, Kraathi, Skharri) have a higher physical resilience that translates mechanically to their Hit Capacity (derived from Strength + Endure, which is higher for large species). There is no special large-species damage reduction — the higher Hit Capacity is the representation.

Small species (Shekhari, Nakhavi) have lower Hit Capacity for the same reason. Their advantages in combat come from their specialties (Small Profile, Arboreal Agility) and their ability to fight in environments where larger species are restricted.

Species in Combat

Skharri — The Scream-and-Leap

When a Skharri character is genuinely insulted, directly challenged, or witnesses a direct attack on someone under their protection, the GM may call for a **Wits roll (difficulty 2)** — meaning the player needs at least two sixes.

On a failure, the honour-instinct fires. The Skharri acts immediately and physically, regardless of the tactical situation, the current initiative order, or the plan the group was executing. This interrupts the normal round structure — the Skharri acts now, before their turn.

The scream-and-leap is not a penalty. It is a character beat. The action taken is the player's choice — the instinct demands immediate physical response, not a specific one. A Skharri who leaps at the source of the insult, who overturns the negotiating table, who draws their weapon — all are valid. The point is that they acted before they could choose not to.

After the instinct fires, the round continues normally. The Skharri has used their action.

> ***Roleplay note:*** *The scream-and-leap is one of the most interesting mechanics in the system when played honestly. A Skharri player who dreads the Wits roll because they know their character is at a critical moment — and who then has to decide what the instinct produces — is playing one of the most demanding and rewarding character arcs in the setting.*

Vaelkhi — Aerial Combat

A Vaelkhi in powered flight (open environment, 0.5 atm minimum) operates in three dimensions that grounded opponents cannot easily access. This translates mechanically as:

- Bonus die on Shoot and Move when attacking from an elevated position or manoeuvring in three dimensions (from the Aerial Hunter biological specialty)
- Grounded opponents attacking an airborne Vaelkhi take a penalty die — the target is moving in ways they cannot easily predict
- A Vaelkhi in flight cannot be engaged in melee by a grounded opponent unless that opponent has a means of reaching them

In enclosed spaces — standard ship corridors, most station interiors — the Vaelkhi cannot fly and loses all aerial advantages. Their ground movement penalty die applies.

Vashori — Emotional Volatility

The Vashori emotional volatility mechanic (Wits roll to avoid a visible uncontrolled reaction) can fire in combat if a Vashori character is subjected to a significant betrayal or insult during the fight — for example, discovering that an apparent ally opened fire on them.

In combat, a failed volatility roll means the Vashori's action this round is determined by the emotional reaction rather than

tactical choice. They are not out of control — they are fully engaged with the source of the emotional trigger, to the exclusion of the tactical situation.

Golvhaan — Imposing Presence

A Golvhaan in combat who has not yet been provoked has a specific quality: they are very difficult to make afraid. The intimidation and morale mechanics that affect other species do not apply to a Golvhaan unless the situation is genuinely extraordinary. 700 kg of scaled centauroid has made its peace with most threats.

A Golvhaan who is provoked — genuinely provoked, past the long patience — does not use the scream-and-leap. They simply act, deliberately and with full physical commitment. The GM may note when a Golvhaan has been pushed past patience; it is rare enough to be significant when it happens.

Mooks and Named NPCs

Mooks

Mooks are the anonymous opposition — guards, soldiers, hired muscle, Kreeth warriors. They are defined by a single dice pool and one behavioural note.

Mook stats:

Quality	Pool	Behavioural note
Poor	3 dice	Breaks on first serious casualty
Average	5 dice	Holds until half their number are down
Trained	7 dice	Holds until outflanked or officer down
Elite	9 dice	Holds until the position is untenable

Mooks do not have individual Hit Capacities. A mook hit by an attack that scores one or more sixes is **taken out of the fight** — incapacitated, fled, or dead depending on the nature of the attack and the attacker's intent. They are not tracked individually beyond this.

Mook morale: When a mook group loses half their number in a single round, or when their behavioural trigger is met, the GM makes a morale check — a flat 3-die roll. On zero sixes, the group breaks. On one six, they fall back to a defensible position. On two or more sixes, they hold.

Named NPCs

Named NPCs have full stat blocks: attributes, skills, gear, and at least one specialty that distinguishes them. They have a Hit Capacity like player characters and are tracked individually.

Named NPCs also have a **motivation** — what they are actually trying to achieve in this combat, which may not be the same as winning. A Vardhek intelligence operative in a firefight may be trying to retrieve something specific, not to defeat the players. A Skharri with an honour-debt may be trying to kill one specific character, not the group. Run named NPCs toward their motivation, not toward optimal tactical play.

Named NPCs do not break from morale checks. They have their own reasons for continuing or withdrawing.

Environmental Hazards

Vacuum and Hard Vacuum

Space is the most lethal environment in the setting. The rules reflect this without requiring complex atmosphere tracking.

Suited characters in functioning vacc suits are protected from vacuum. A suit that has taken damage may be compromised — the GM introduces a consequence die at the start of each round for a compromised suit. The D4 result determines what fails.

Unsuited characters in hard vacuum take one condition per round from atmospheric decompression. After four rounds, they are incapacitated. This is not a death sentence if they are rescued and treated immediately — but the window is short.

Hull breaches in a combat environment introduce a consequence die for all characters near the breach each round, regardless of suits. The D4 result determines what the breach does this round — debris, atmospheric venting, equipment damage, a character being pulled toward the breach.

Zero Gravity

Combat in zero gravity removes the intuitive positional logic that most combatants rely on. Characters without the Zero-G Movement specialty (all Spacers, and others with specific training) take a penalty die on Fight and Move rolls in zero-G.

Characters with Zero-G Movement treat freefall as a normal

environment. They do not take the penalty die and may use their spatial orientation to gain positional advantages that grounded combatants cannot.

Extreme Environments

Hostile temperatures, toxic atmospheres, high pressure, and other environmental hazards impose conditions on unsuited characters as described in the species entries and the Conditions chapter. In combat, environmental conditions accumulate alongside combat conditions — a character fighting in extreme cold while already injured is stacking conditions from multiple sources.

The GM should track environmental condition sources separately from combat condition sources, since they resolve differently.

The Consequence Die in Combat

The black D4 consequence die is placed on the table when the combat situation has teeth beyond the exchange of attacks — when the environment is actively hostile, when something structural is happening that will affect everyone regardless of their individual performance.

When to introduce a consequence die:

- Combat near or in a hull breach
- Fighting aboard a ship that is simultaneously under attack
- A Skharri whose scream-and-leap has changed the tactical situation for everyone
- A collapsing structure, failing life support, or detonation in progress
- Any situation where the GM wants to signal: *something will go wrong here regardless of how well you fight*

The consequence die is rolled at the **start of each round** in which the condition applies. Its result determines what the situation costs before attacks are resolved:

D4	Consequence
1	**Minimal cost.** A system degrades, a resource is spent, a position becomes slightly worse.
2	**Complication.** One character (GM's choice) must spend their action addressing the complication or accept a condition.
3	**Collateral.** Equipment is damaged, a bystander is affected, or the environment worsens for everyone.

D4	Consequence
4	**Full cost.** The thing the situation warned about happens. The breach vents. The structure collapses. The thing goes off.

The consequence die is removed when the triggering condition is resolved — the breach is sealed, the character gets the situation under control, the structural failure is past.

Ending a Combat

Combat ends when:

- One side is incapacitated, dead, or fled
- One side surrenders or is captured
- The situation changes so fundamentally that continuing the fight no longer makes sense

Mooks who break from a morale check are out of the fight — whether they flee, surrender, or simply stop engaging is the GM's call based on the fiction.

Named NPCs end the fight on their own terms when their motivation is achieved, when the situation becomes untenable, or when they are incapacitated. A named NPC who retreats is not defeated — they will be encountered again under different circumstances.

Aftermath — after every combat, the GM should briefly describe the state of the scene. Who is down? What is damaged? What has changed? What does the environment look like now? Combat is not an interlude between scenes — it is a scene, and its results change the situation that follows.

Quick Reference — Combat

Round Structure

1. Opening round: player group nominates one character for Wits roll vs GM's Wits roll. Most sixes acts first.

Surprise: surprising side acts first, surprised side loses Round 1 action.

2. Subsequent rounds: alternating sides (players / NPCs).
3. Within player turn: free coordination if communication is possible. One action per character.

Attack Roll

Attribute (white) + Skill (blue) + Gear (green)

0 sixes = miss / 1 = hit, one condition / 2 = clean hit /

3 = strong hit, two conditions or one + tactical effect /

4+ = devastating, two conditions + tactical effect

Defence

Passive: cover removes attacker dice (1/2/3 for light/solid/hard)

Active: costs your action; opposed roll; armour adds green dice

Range Bands

Close (melee) / Short (personal weapons) / Long (rifles, open space)

Outside optimal band: penalty die

Size

Attacking larger opponent (melee): penalty die

Attacking smaller opponent (melee): bonus die

Attacking to subdue smaller opponent: penalty die

Large target (ranged): attacker gains bonus die

Mooks

Single pool (3/5/7/9 dice by quality)

One hit = out of fight

Morale check when trigger met: 3 dice, 0 sixes = breaks

Consequence Die (black D4)

Rolled at start of each applicable round

1 = minimal / 2 = complication / 3 = collateral / 4 = full cost

Vacuum

Suited: protected (compromised suit = consequence die each round)

Unsuited: one condition per round, incapacitated after four

Hull breach: consequence die each round for nearby characters

Conditions

Overview

Conditions represent the accumulated cost of violence, environmental stress, psychological pressure, and sustained hardship. They are the mechanical expression of the fact that the universe is trying to kill you and sometimes partially succeeds.

Conditions work on two parallel tracks: **physical** and **psychological**. Each track has its own capacity, its own named conditions, and its own recovery requirements. Both tracks use the same underlying mechanic — each condition imposes a penalty die on relevant rolls — but they accumulate from different sources and resolve in different ways.

A character is **incapacitated** when either track reaches its capacity. Physical incapacitation means the body has given out. Psychological incapacitation means the mind has. Both are serious. Neither is automatically fatal.

The Two Tracks

Physical Track

Hit Capacity = Strength + Endure skill

A character with Strength 3 and Endure 2 has a Hit Capacity of 5. They can sustain five physical conditions before incapacitation.

Physical conditions come from: combat wounds, vacuum exposure, environmental extremes, toxic atmospheres, physical exhaustion, disease, and injury from accident or structural failure.

Psychological Track

Stress Capacity = Wits + Empathy (average, rounded up)

A character with Wits 3 and Empathy 2 has a Stress Capacity of 3 (average 2.5, rounded up). They can sustain three psychological conditions before breaking.

Psychological conditions come from: sustained fear, witnessing atrocity, isolation, the death of crewmates, moral injury, the specific horrors of the deep dark between stars, and the

accumulated weight of a life lived at the edge.

Tracking Conditions

Conditions are recorded on the character sheet as they accumulate. Each named condition occupies one slot on the relevant track. When a track is full, the character is incapacitated on that track.

Record the specific condition, not just a tick. *Hurt (gunshot, left shoulder)* is more useful at the table than a blank box with a mark in it. The specific nature of the condition informs how it was treated, how it affects the character's behaviour, and what recovery looks like.

Physical Conditions

Physical conditions are acquired in order of severity — Winded before Hurt, Hurt before Wounded, Wounded before Critical — unless a single hit is severe enough to skip levels. A devastating hit (four or more sixes in combat) may impose Wounded directly. The attacker or the GM specifies when a condition skips levels.

Winded

Minor physical stress. The body is protesting but functional.

Sources: A glancing hit, brief vacuum exposure, pushing physical limits without adequate rest, minor chemical exposure, a fall that did not break anything.

Effect: Penalty die on **Endure** and **Labor** rolls.

Recovery: Rest. Given an hour of genuine rest with no physical demands, Winded clears automatically. No medical attention required.

> ***Roleplay note:*** *Winded is the condition that accumulates quietly. Characters who push themselves repeatedly — multiple pushed rolls in a session, sustained physical effort, a long EVA — may find themselves Winded before anything dramatic has happened. It is the setting's way of saying the body keeps score.*

Hurt

Genuine injury. Something is damaged that will not simply rest away.

Sources: A solid combat hit, significant vacuum exposure, a bad fall, toxic atmosphere without adequate protection, extended operation in a hostile environment without recovery.

Effect: Penalty die on all **physical skill rolls** (Fight, Endure, Labor, Shoot, Stealth, Move).

Recovery: Rest and basic medical treatment. A character with the Medic specialty (Science) can downgrade Hurt to Winded with a successful Science roll and appropriate supplies. Without medical attention, Hurt requires a full night's rest to clear. With treatment, it clears after a few hours.

> ***Roleplay note:*** *Hurt is visible. Other characters notice. A Hurt character moves differently, reaches for things more carefully, flinches. The specific nature of the injury — the location, the cause — should inform how the character carries themselves for the rest of the session.*

Wounded

Serious injury. Without treatment, this gets worse.

Sources: A strong combat hit, serious vacuum exposure, structural injury (caught in a hull breach, crushed by falling equipment), severe toxic or thermal exposure.

Effect: Penalty die on **all rolls** without exception.

Special: A Wounded character who does not receive medical attention within a scene (approximately one hour of game time) acquires an additional Hurt condition as the wound worsens. A character who is already at capacity when this happens is in immediate danger.

Recovery: Requires proper medical treatment — a Science roll at no penalty with adequate supplies, or a field treatment that downgrades to Hurt with a penalty die on the roll. Full recovery from Wounded requires rest after treatment; the condition clears after a long rest (six or more hours) following successful treatment.

> ***Roleplay note:*** *A Wounded character is in real trouble and should feel it. This is the condition that forces decisions — do we press on or do we address this now? The one-hour clock before worsening is a genuine pressure. Use it to create urgency rather than ignoring it for narrative convenience.*

Critical

Life-threatening. Immediate intervention required.

Sources: A devastating combat hit, extended vacuum exposure without suit, catastrophic environmental failure, multiple accumulated wounds without treatment.

Effect: Penalty die on **all rolls**. Additionally, the character must make an **Endure roll at the start of each scene** while Critical. On a failure, they acquire another Hurt condition. If this would exceed their Hit Capacity, they are dying.

Dying: A character who exceeds their Hit Capacity while Critical is dying. They have a number of rounds equal to their Strength before death unless stabilised. Stabilisation requires a successful Science roll (Medic specialty removes the penalty die) against difficulty 2 — two or more sixes needed.

Recovery: Requires significant medical intervention — proper facilities, time, and a Science roll at difficulty 2. On success, Critical downgrades to Wounded. Full recovery from Critical takes days of rest and follow-up treatment. Some Critical wounds leave permanent effects at the GM's discretion — scars, reduced function, trauma that does not entirely clear.

> ***Roleplay note:*** *Critical is the condition that changes the session. When a character goes Critical, everything else stops mattering as much. The crew's response — who goes to them, who keeps fighting, what they sacrifice to get them out — is character definition. Do not rush past it.*

Psychological Conditions

The named psychological conditions — Rattled, Stressed, Traumatised, Breaking — describe the **mechanical effect** of psychological distress. What that distress actually looks like, how it manifests in behaviour and expression, varies profoundly by species.

A Rattled human looks different from a Rattled Skharri. A Breaking Golvhaan is not the same experience as a Breaking Vashori. The mechanics are identical; the phenomenology is not. The **Species Psychological Profiles** section that follows the condition descriptions gives the GM and players the texture they need to make psychological conditions feel real rather than generic.

Psychological conditions do not always accumulate in strict order — a single sufficiently traumatic event can impose Traumatised directly. The GM determines the severity of the psychological impact of a specific event and imposes the appropriate condition.

Repeated exposure to the same type of stressor may eventually reduce its impact — a character who has processed a specific type of horror develops a callus. This is a roleplay development, not an automatic mechanical effect.

Rattled

Shaken. The normal confidence of a competent person has been temporarily disrupted.

Sources: A near-miss, a sudden unexpected threat, the first serious combat, witnessing something disturbing, an uncomfortable revelation about the situation.

Effect: Penalty die on **Wits** rolls.

Recovery: A moment of calm and grounding. Given ten minutes away from the stressor — not necessarily safe, just not actively threatening — Rattled clears. The personal item (once per session) clears Rattled along with any other stress condition if used.

> ***Roleplay note:*** *Rattled is the most common psychological condition and the one most easily dismissed. Do not dismiss it. A character who is Rattled has been reminded that the universe can reach them. Play that for a moment before it clears.*

Stressed

Sustained pressure is accumulating. The baseline has shifted.

Sources: Extended periods under threat, repeated dangerous situations without recovery, unresolved interpersonal conflict, the psychological weight of command decisions with bad outcomes, sustained isolation.

Effect: Penalty die on **Wits** and **Empathy** rolls.

Recovery: Requires genuine rest and social connection. A full night's sleep in relative safety clears Stressed — but only if the stressor has been removed or addressed. Stress that continues to accumulate cannot be slept away.

A crewmate who takes time to actively support a Stressed character — a real conversation, not a mechanical action — allows the Stressed character to make a **Read roll**. On a clean success (two or more sixes), the Stressed condition clears. This requires genuine roleplay; it is not a free recovery mechanic.

The personal item removes one psychological condition once

per session; Stressed is the most common target.

> ***Roleplay note:*** *Stressed is the condition that reveals character relationships. Who notices? Who makes time? A crew that looks after each other's psychological state functions better under pressure — and the game rewards that attention mechanically as well as narratively.*

Traumatised

Something has happened that will not simply resolve with rest.

Sources: Witnessing atrocity, the death of someone close, a profound moral injury, surviving something that should not have been survivable, discovering something that cannot be unknown.

Effect: Penalty die on **all rolls**.

Special: A Traumatised character has a **specific trigger** — a type of situation, a sensory input, a context — that the GM and player establish when the condition is acquired. When the character encounters that trigger, they must make a Wits roll. On a failure, they acquire a Rattled condition immediately (in addition to Traumatised) as the trauma reasserts itself.

Recovery: Cannot be slept away. Requires time, safety, and active processing — either through roleplay (a significant conversation with a trusted crewmate or the personal item used in a meaningful scene) or through downtime between missions. The GM may require a specific narrative event before Traumatised will clear.

> ***Roleplay note:*** *Trauma is permanent until it is worked through. The trigger mechanic is not a punishment — it is an invitation to explore what happened and what it means. Work with the player on what the trigger is and what encountering it looks like at the table.*

Breaking

The edge. Everything is becoming too much.

Sources: Accumulated stress without recovery, repeated trauma, sustained impossible pressure, watching the crew fall apart.

Effect: Penalty die on **all rolls**. Additionally, once per scene the GM may call for a **Wits roll** when the character faces a significant stressor. On a failure, the character's response to the stressor is determined by the GM in consultation with the player. This is not

control removed from the player — it is the GM and player collaborating on what a person at the edge actually does.

Recovery: Cannot recover while the stressor continues. Requires removal from the situation — genuine safety, not just tactical safety — and significant time. Breaking clears only after a period of downtime and active support from trusted people.

> ***Roleplay note:*** *Breaking is rare and should be treated with care. A character who reaches Breaking has been through something real. Work with the player to make it meaningful rather than merely penalising. If a player is uncomfortable with the Breaking mechanic for any reason, the GM should remove it and treat the full Stress Capacity track as incapacitation without the additional behavioural effects.*

Species Psychological Profiles

The condition names and mechanics above apply to all species. What the conditions *look like* — how distress manifests in behaviour, expression, and action — varies profoundly by species. These profiles give the GM and players the texture to make psychological conditions feel authentic rather than generic.

When a psychological condition is imposed on a non-human character, refer to their profile here. The mechanical effect is unchanged; the roleplay expression follows the profile.

Human

Humans are the baseline against which other species' psychological expressions are measured — and it is worth stating explicitly what that baseline is, because human psychological distress is not a neutral default.

Rattled: Visible disruption of the competent surface. Humans show shock — a brief freeze, a change in breathing, a loss of the smooth social performance that most humans maintain. They recover quickly but the moment was real.

Stressed: Humans under sustained pressure become either withdrawn and short-tempered, or overly focused on task to the exclusion of everything else. Both are avoidance. The stress is often visible in physical tells — disrupted sleep, changed appetite, a tension in the body that other characters can read with Scan.

Traumatised: Humans are variable. Some go quiet. Some

become hypervigilant. Some replay the event compulsively in conversation, looking for the moment where they could have done something different. The trigger mechanic for humans is often sensory — a smell, a sound, a visual detail from the event.

Breaking: Human psychological collapse tends toward one of two poles: shutdown (the person stops engaging, stops making decisions, stops being present) or flooding (everything comes out at once, uncontrolled and overwhelming). Which pole depends on the person. Work with the player to establish which direction their character breaks.

Shekhari

Shekhari psychological distress is characterised by *withdrawal and cold calculation*. Emotional display is not the Shekhari default; distress makes them less expressive, not more.

Rattled: The Shekhari goes quiet. The sharp-tongued commentary stops. They are assessing — running numbers on the situation, on their options, on whether this crew is still a viable context for them. Other characters who know them well will recognise the quiet as alarming. Characters without Exposure: Shekhari will likely not notice anything is wrong.

Stressed: The mercantile calculus runs constantly and audibly. A Stressed Shekhari starts pricing things — the risk of the current situation, the cost of staying, the value of various outcomes. They may begin to pull back from crew commitments, not dramatically, but in small ways that accumulate. They are protecting themselves against a loss they are beginning to think is likely.

Traumatised: The Shekhari has identified this experience as a data point about the world that changes their model. Something they assumed was safe or manageable has been revealed as otherwise. The trigger is the category of thing that revealed this — not the specific event but what it meant. A Traumatised Shekhari is very difficult to read; the distress is almost entirely internal.

Breaking: A Breaking Shekhari makes a unilateral

decision. They have concluded that the current situation is not survivable on the current terms, and they act on that conclusion without consultation. This is not cowardice — it is the Shekhari's deepest adaptive mechanism firing at full intensity. They will do what they need to do to survive. The crew may or may not be part of that plan. If the loyalty transition has happened — if the Shekhari has genuinely committed to this crew — Breaking looks different: the unilateral decision is in service of protecting the crew rather than themselves. Either way, they act alone.

Golvhaan

Golvhaan psychological distress manifests as *progressive withdrawal of engagement*. The curiosity that defines them goes away first. By the time a Golvhaan is visibly distressed, they have been struggling for some time.

Rattled: The questions stop. A Golvhaan who has been asking questions — about the situation, about the other species in the room, about what is happening — goes quiet when Rattled. The patient attention is still present; the engagement has pulled back. Characters without Exposure: Golvhaan will almost certainly not notice this. Characters who know them well will find the silence louder than anything else.

Stressed: The Golvhaan becomes very still. The deliberateness that is always part of their character becomes pronounced — they are spending significant processing capacity on something they are not discussing. They continue to function, continue to help, but there is a distance in it. If asked directly, a Stressed Golvhaan will generally answer honestly — they do not dissemble about their internal state — but they may not have found the words yet.

Traumatised: A Traumatised Golvhaan has encountered something that has fundamentally contradicted their model of how the universe works. Their xenophilia — their genuine curiosity and openness — is the source of their vulnerability here. They trusted; something responded to that trust in a way that caused damage. The trigger is almost always relational — a specific

type of betrayal or cruelty from a species or person they had reason to trust. Recovery requires that trust to be rebuilt, which takes considerable time.

Breaking: This is rare and significant. A Golvhaan at Breaking has passed through a profound stillness and come out the other side into something that does not look like distress at all — until it becomes obvious that the patience has failed. A Breaking Golvhaan acts with complete deliberate commitment in a direction determined by whatever finally broke through the stillness. This may be protective, may be catastrophically destructive of whatever caused the distress, or may be a withdrawal so complete that they simply leave and do not return. None of these are impulsive. All of them have been building for a long time.

Vashori

Vashori psychological distress is *immediate and visible* from the earliest stages. The same emotional bandwidth that makes Vashori warm and volatile in normal circumstances means that distress cannot be concealed. They feel it loudly and they show it.

Rattled: Obvious and immediate. The Vashori's emotional regulation — already active management rather than passive calm — becomes visibly effortful. They are managing something and everyone can see it. The recovery is also rapid; Vashori process fast and the Rattled condition clears more easily for them than for most species. A moment of genuine calm is often enough.

Stressed: The management effort becomes the dominant feature of the Vashori's behaviour. They are working very hard to maintain function and the work is showing. The emotional volatility mechanic becomes more likely to fire — the Wits roll to avoid an uncontrolled reaction takes a penalty die while Stressed. Small things that would normally be managed may break through the surface.

Traumatised: Vashori feel trauma intensely and carry it intensely. The emotional system that processes quickly in normal circumstances finds some things genuinely unprocessable — and those things sit, heavy and present,

in a way that contradicts the species' usual rapid emotional flow. A Traumatised Vashori is simultaneously more emotionally reactive in general (the system is overloaded) and specifically stuck on the traumatic content. The trigger is almost always emotional in nature — a specific feeling, a specific type of betrayal or loss. Recovery requires the emotional processing to actually complete, which means the Vashori needs to be able to feel it fully with support, not suppress it.

Breaking: Emotional flooding. Everything that has been managed, processed, and held together comes apart simultaneously. A Breaking Vashori does not shut down — they are overwhelmed by the entirety of what they are feeling, without the capacity to filter, prioritise, or moderate. This is not rage or despair specifically; it is everything at once. The GM and player should work out what this looks like for this specific character — some Vashori at Breaking become completely non-functional; some channel the flooding into a single overwhelming emotional commitment to one thing.

Vaelkhi

Vaelkhi psychological distress manifests as *hypervigilance and progressive isolation*. The choth instinct — the deep need for the extended clan — is the wound that most psychological damage reopens.

Rattled: The Vaelkhi becomes watchful in a way that goes beyond their usual hunt-patience. Every position in the room is assessed. Every exit is noted. They withdraw slightly from the group — not dramatically, but the physical distance increases. They are scanning for the threat that produced the rattle, and they have not yet determined whether it has passed.

Stressed: The group participation decreases noticeably. A Stressed Vaelkhi moves toward solo action — making decisions independently, positioning independently, handling things without consultation. This is not aggression toward the crew; it is the absence of choth making itself felt. They are alone in a way that the crew cannot entirely address, because the crew is not a choth. Characters who understand

Vaelkhi will recognise this as distress; characters without Exposure: Vaelkhi may read it as the Vaelkhi simply preferring to work alone.

Traumatised: The choth absence becomes acute. A Traumatised Vaelkhi is conscious of being fundamentally without their social structure in a way that normal circumstances allow them to set aside. The trigger is often isolation-related — being separated from the crew, being in a situation where they had to act completely alone, or something that reminded them viscerally of the permanence of having left the choth. Recovery requires the crew to function, however imperfectly, as a substitute social structure — not a choth, but something that holds.

Breaking: The God of Ythri logic at its darkest. A Breaking Vaelkhi begins to seek the decisive moment — not the scream-and-leap impulsiveness of the Skharri, but a deliberate, patient identification of the action that ends things cleanly. This may be directed outward (a commitment to a high-risk action against the source of the distress) or inward (a philosophical acceptance of death that removes the fear of risk entirely and with it the survival instinct that would otherwise function as a check). The GM and player should work together on this carefully. A Breaking Vaelkhi is not suicidal in the human sense — they are applying the God of Ythri's logic to an impossible situation. That is a distinction worth preserving in play.

Skharri

Skharri psychological distress manifests through the *honour-economy*. Every psychological condition, for a Skharri, is mediated by how it interacts with their standing — their own standing in their eyes, and the standing of those around them.

Rattled: The honour threshold drops. A Rattled Skharri is acutely sensitive to perceived slights — things that would normally be processed and set aside now register as challenges requiring a response. The scream-and-leap Wits roll difficulty does not change mechanically, but the triggers multiply. The GM should

note more situations as potential triggers for the Wits roll while a Skharri is Rattled.

Stressed: The honour-economy is all-consuming. A Stressed Skharri is running a continuous internal accounting of their position — what has happened, what it means for their standing, what is owed and to whom. Social situations become minefields because every interaction is being assessed for its honour implications. The Skharri may become visibly uncomfortable in situations that are ambiguous in honour terms — mixed-species social settings where the rules are unclear, situations where their competence has been questioned, contexts where they cannot establish their position.

Traumatised: The Skharri has experienced something that damaged their honour-position in a way they have not been able to address. This may be a defeat, a failure to protect someone, a situation where the honour-instinct fired and the outcome was wrong, or a betrayal by someone whose honour-standing they respected. The trigger is honour-adjacent — a situation that resembles the one that caused the damage. A Traumatised Skharri may withdraw from the crew to avoid situations where their damaged position becomes visible. Recovery requires the honour-debt to be acknowledged and addressed — not necessarily avenged, but addressed in terms the Skharri can accept as settling the account.

Breaking: A sustained, cold, deliberate fury looking for a target. This is not the scream-and-leap. A Breaking Skharri has identified the source of the damage and committed to addressing it with full deliberate focus, regardless of the consequences to themselves or the tactical situation. This is more dangerous than the impulsive honour-instinct precisely because it is considered. A Breaking Skharri who has identified a target will pursue it with the patience and commitment that defines the species at its most capable — and they are no longer weighing the cost.

Nakhavi

Nakhavi psychological distress manifests as *cognitive fragmentation* — the distributed intelligence begins to lose coherence. The arms, partially autonomous in normal function, become more independent as the central brain's integrating capacity degrades.

Rattled: Visible arm-conflict. The semi-autonomous ganglia pull in different directions — an arm reaching toward something while the character is moving away, two arms working at cross-purposes, a visible hesitation in physical tasks as the distributed system disagrees with itself. Characters with Exposure: Nakhavi recognise this immediately. Characters without it may notice that something seems off with the Nakhavi's movement without being able to identify what.

Stressed: The translation layer becomes unreliable. A Stressed Nakhavi's communication degrades — the device that converts their chromatophore display to speech begins to introduce errors, not from equipment failure but from the source signal becoming less coherent. They know what they want to say; the path from cognition to chromatophore display to translation is losing fidelity. They may begin to communicate less, not from withdrawal but from frustration with the degraded channel.

Traumatised: The distributed system fixates. One aspect of the traumatic experience — a specific sensory input, a specific decision point, a specific moment — loops in the system, returned to repeatedly by the distributed cognition in the way that a committee returns to an unresolved question. The Nakhavi is aware of this and finds it distressing precisely because the distributed cognition is their greatest strength and it is working against them. The trigger is almost always the specific input that is looping. Recovery requires the loop to be interrupted — a new context for the memory that allows the system to file it rather than return to it.

Breaking: Full cognitive dissociation. The arms act independently, the central brain unable to coordinate. The Nakhavi is present but the

integration that makes them a single coherent intelligence has temporarily failed. From the outside, this is alarming — a Nakhavi whose arms are all doing different things, whose chromatophore display is producing conflicting signals, whose communication is fragmented and contradictory. The Nakhavi themselves are experiencing something like a chorus of voices that have stopped being a conversation. Recovery from Breaking requires physical stillness, quiet, and time — the system needs to be allowed to resynchronise without additional input. A crewmate who simply stays present and calm, without demanding communication or decision, is providing the most useful support.

The 18-year lifespan gives Nakhavi psychological distress a specific weight that other species do not experience. Time spent incapacitated by psychological conditions is time that cannot be recovered. A Nakhavi who reaches Breaking and requires downtime to recover is acutely aware of what that downtime costs in terms of their remaining lifespan. The GM and player should acknowledge this without making it mechanical — it is texture, not an additional penalty.

Environmental Conditions

Environmental conditions are a third category — neither purely physical nor purely psychological, but the body's response to a hostile environment. They differ from named conditions in one important way: **they resolve when the environmental cause is removed**, without requiring recovery time or medical treatment.

While active, an environmental condition occupies a slot on the physical track and imposes the same penalty die as a named physical condition. When the environment is addressed — warmth restored, atmosphere corrected, pressure normalised — the slot clears immediately.

If a character is already at maximum physical conditions when an environmental condition would be imposed, it cannot be absorbed — the character is incapacitated.

Common Environmental Conditions

Cold (Vashori): A Vashori in an environment below 285 K without adequate thermal protection is Cold. Penalty die on all physical and Wits rolls. Below 275 K, two penalty dice — this occupies two slots on the physical track. Clears immediately when adequate warmth is restored.

Hungry (Shekhari, Vaelkhi): A Shekhari who has not eaten within six hours of strenuous activity, or a Vaelkhi who has missed two feedings, is Hungry. Penalty die on all physical and Wits rolls. Clears when the character eats adequately. Subsequent missed feedings add additional slots.

Decompression: An unsuited character exposed to hard vacuum acquires one Decompression condition per round. Unlike other environmental conditions, Decompression does **not** clear when the environment is restored — it has caused physical damage that requires treatment. Treat as Hurt conditions once the character is back in atmosphere.

Gravity stress: A character operating outside their species' comfortable gravity range for extended periods (more than a few hours) without appropriate support acquires Gravity Stress. Penalty die on physical rolls. Clears with rest in appropriate gravity.

Atmospheric: A character breathing an unsuitable atmosphere (wrong composition, toxic contaminants, insufficient pressure) without protection acquires one condition per scene of exposure, treated as Hurt once they are in safe atmosphere.

Interplay Between Tracks

Physical and psychological conditions are tracked separately but interact in play:

Physical exhaustion affects psychological resilience. A character who is Wounded takes a penalty die on Stress Capacity checks at the GM's discretion — the body's distress makes the mind more vulnerable.

Psychological conditions affect physical performance. The penalty dice from psychological conditions apply to all rolls, including physical ones. A Traumatised character is impaired in combat as well as in social situations.

Both tracks can fill simultaneously. A character who is

Wounded and Stressed is carrying conditions on both tracks. Four conditions on either track causes incapacitation on that track — they do not combine into a single total.

Recovery

Physical Recovery

Condition	Without treatment	With treatment
Winded	1 hour rest	Clears immediately
Hurt	Full night's rest	Few hours after Science roll
Wounded	Worsens without treatment	Long rest after Science roll (difficulty 1)
Critical	Dying without treatment	Days of rest after Science roll (difficulty 2)

Science rolls for treatment use the Medic specialty where applicable. Without the Medic specialty, treat the roll as one difficulty level harder. Without adequate supplies, treat the roll as one difficulty level harder again.

Field medicine — treating conditions under fire or without proper facilities — imposes a penalty die on the Science roll.

Extended recovery — some Critical wounds leave effects that persist beyond the condition clearing. The GM determines these on a case-by-case basis: reduced function in an injured limb, scarring, recurring pain under stress. These are narrative consequences, not additional mechanical conditions.

Psychological Recovery

Condition	Recovery requirement
Rattled	Ten minutes away from stressor, or personal item
Stressed	Full night's rest (stressor removed), or crewmate support + Read roll, or personal item
Traumatised	Significant narrative event + time; GM determines
Breaking	Downtime, safety, active support; GM determines

The personal item removes one psychological condition once per session. It can remove any psychological condition, including Traumatised, if used in a meaningful scene that connects the item to what happened. The GM and player should agree on what that scene looks like before the condition clears.

Downtime recovery — between missions, psychological conditions recover faster. A full downtime period (several days of genuine

rest and safety) clears one Traumatised or Breaking condition if the player and GM agree the character has had the space to process it. For Nakhavi, the GM should acknowledge the lifespan cost of this downtime in the fiction, even if it carries no additional mechanical penalty.

The Condition Sheet

Conditions are tracked on the character sheet in two rows:

Physical: [ ] [ ] [ ] [ ] [ ] [ ] (Hit Capacity slots)
Psychological: [ ] [ ] [ ] (Stress Capacity slots)

Each filled slot records the specific condition and its cause. Environmental conditions are marked with an (E) to distinguish them from injury and trauma.

When a track is full, write **INCAPACITATED** across it. Physical incapacitation: the character cannot act, move, or fight. Psychological incapacitation: the character cannot function socially, make decisions, or maintain composure. In either case, they need help.

Quick Reference — Conditions

Physical Track (Hit Capacity = Strength + Endure)

Condition	Effect	Recovery
Winded	Penalty die: Endure, Labor	1 hour rest
Hurt	Penalty die: all physical skills	Night's rest or treatment
Wounded	Penalty die: all rolls; worsens without treatment	Treatment + long rest
Critical	Penalty die: all rolls; Endure roll each scene or worsen	Significant treatment + days

Psychological Track (Stress Capacity = Wits + Empathy avg, rounded up)

Condition	Effect	Recovery
Rattled	Penalty die: Wits	10 min calm, or personal item
Stressed	Penalty die: Wits, Empathy	Rest + stressor removed, or support + Read roll, or personal item
Traumatised	Penalty die: all rolls; trigger mechanic	Significant narrative event + time
Breaking	Penalty die: all rolls; Wits roll vs stressors	Downtime + safety + support

Environmental Conditions

Occupy physical track slots. Clear when cause is removed.

Exception: Decompression — treated as Hurt once in atmosphere.

Personal Item

Once per session: remove one psychological condition.

Requires a moment of quiet connection with the item.

Species Psychological Expression — Quick Reference

Species	Rattled	Stressed	Traumatised	Breaking
Human	Visible disruption of competent surface	Withdrawn or task-focused avoidance	Variable; sensory triggers common	Shutdown or flooding
Shekhari	Goes quiet; assessing exits	Pricing the risk; pulling back	Internal; trigger is category of betrayal	Unilateral survival decision
Golvhaan	Questions stop	Very still; distant but functional	Relational betrayal; trust must rebuild	Profound action after long stillness
Vashori	Obvious and immediate	Management effort visible; volatility increases	Emotionally stuck; needs to feel it fully	Emotional flooding; everything at once
Vaelkhi	Hypervigilant; physical distance increases	Solo action; choth absence felt	Isolation trigger; crew must hold	Seeks decisive moment; God of Ythri logic
Skharri	Honour threshold drops; more triggers	Honour-economy consuming; social minefield	Honour-debt unresolved; withdrawal	Cold deliberate fury; target identified
Nakhavi	Arm-conflict visible	Translation degrades; communicates less	Distributed system loops on event	Full dissociation; arms independent

Equipment

How to Use This Chapter

Equipment entries describe what a piece of gear does, what it costs in dice pool terms, and what it means to own and use it. They do not describe every variant, every manufacturer, or every modification. The examples here are illustrations of the framework; the framework is the thing.

Reading an Equipment Entry

Each entry includes:

Gear rating — the number of green dice the equipment adds to the relevant roll. Reflects both hardware quality and the sophistication of the AI systems embedded in it. A higher gear rating is not always better; it is more capable, which sometimes means more demanding.

Tech level — which stage of development the equipment represents (Early / Developed / Advanced / Frontier). Determines availability relative to local infrastructure and the character of the AI embedded in the system. See the AI and Technology chapter.

Optimal use — the conditions under which the equipment performs at its rated gear value. Outside these conditions, the rating may be reduced or a penalty die applied.

Notes — anything that matters about this piece of equipment that the numbers do not capture.

Availability and Price

Equipment availability and price are not fixed values. They are a function of **local species presence and industrial infrastructure**.

Near the industrial base of the species that manufactures a piece of equipment: standard availability, standard price, local expertise for maintenance and repair.

Away from that industrial base: special order, premium price, longer lead time, and local expertise that may be limited or absent. A Golvhaan vacc suit three jumps from the nearest

Golvhaan industrial world is not impossible to obtain; it is expensive, slow to arrive, and the port technician who services it may be working from a manual they have never had to use before.

The GM applies this principle to all equipment purchases. Players who pay attention to the setting's demographic texture — who is present where, in what numbers, with what infrastructure — will find this rule rewards that attention. A free trader running routes between a species' industrial base and frontier regions where that species is settling has a natural cargo.

Creating New Equipment

When a player wants a piece of equipment not in this chapter, use this framework:

What does it do? State the fictional function clearly. If it cannot be stated clearly, the equipment is not ready to be designed.

Which skill does it support? Equipment adds gear dice to a specific skill roll or category of rolls. If it supports multiple skills, it probably should not — or the dice bonus should be lower for each application.

What tech level is it? Early, Developed, Advanced, or Frontier. This determines the AI character of the embedded systems, the installation or acquisition requirements, and the availability rule above.

What is the gear rating? 1-4, matching the tech level. If the equipment seems like it should be Gear 5, it is either not a personal equipment item or it is a plot device rather than standard kit.

What are the constraints? Every piece of equipment has optimal conditions and suboptimal ones. What reduces the gear rating or adds a penalty die? Power requirements, environmental limits, species compatibility, maintenance demands.

What does it mean to own it? The question that separates interesting equipment from a stat block. A Gear 4 scope has a relationship with the shooter. A personal medical kit has a history. A vacc suit has saved someone's life, or it hasn't yet. Equipment that has a story is equipment that matters at the table.

Weapons — Melee

Melee weapons are used with the **Fight** skill. They are effective only at Close range. Natural weapons (claws, teeth, mass) are always available and treat as Gear 1 for species that have them — see the Species chapter.

A Note on Melee in This Setting

Melee combat in an interstellar setting is not the primary mode of violence. It is, however, common in specific contexts: boarding actions, confined spaces where ranged fire risks hull integrity, honour duelling cultures, and situations where the people involved are too close for ranged weapons to be relevant. A character with high Fight is not anachronistic — they are someone who has thought carefully about what happens when the range closes.

Blade (various)

Gear rating: 1-2 | **Tech level:** Early to Developed

The oldest personal weapon. In the 22nd century and beyond, blades exist in forms ranging from simple cutting edges (Gear 1) to monomolecular-edge tools with embedded impact sensors and grip-pressure feedback (Gear 2). The Gear 2 blade is not dramatically different in appearance from the Gear 1 version; the difference is in edge retention, balance, and the micro-adjustment systems that keep the edge aligned under impact stress.

Optimal use: Close range, all environments including vacuum (no moving parts to freeze or jam).

Notes: Blades are cross-species in a way that many weapons are not. Something with an edge works regardless of the user's body plan, though the specific form factor varies enormously — a blade designed for Skharri grip geometry is different from one designed for human hands. Near the relevant species' industrial base, species-specific blades are available. Away from it, a blade designed for a different hand is a penalty die on Fight.

> *Skharri honour blades are a specific category. The design is ancient and has not changed in centuries — to modify it would be to change what it is. Gear 1 by any technical assessment. Worth more than the rating suggests.*

Monoblade

Gear rating: 2 | **Tech level:** Developed

A blade whose edge is maintained at the molecular level by an embedded directed system. Cuts through materials that would stop a conventional blade. The directed system requires power and will shut down at low charge, reverting to a conventional but well-made blade. Maintenance requires Tech skill and appropriate equipment; the edge system is fragile in ways that conventional blades are not.

Optimal use: Close range, targets wearing armour or with significant natural protection.

Notes: A monoblade through vacc suit material is a crisis. Users in pressurised environments think carefully about this.

Impact Weapon (various)

Gear rating: 1 | **Tech level:** Early

Clubs, batons, wrenches used as weapons, the business end of a tool applied to a person. Universally available because the category encompasses anything heavy. The only weapon available in every environment regardless of local species presence.

Optimal use: Close range. Effective against unarmoured targets.

Notes: Impact weapons used for subduing — the explicit intent not to kill — remove the penalty die on subduing attacks that blades impose. Blunt force is better calibrated for incapacitation than edged weapons.

Combat Gauntlet / Augmented Strike

Gear rating: 2 | **Tech level:** Developed

Powered strike-enhancement systems built into gloves, boots, or equivalent appendage covering. Multiplies impact force through brief mechanical assistance timed to the strike. The AI times the assistance to the user's strike pattern; a system that has worked with the same user for years has a better model of their fighting style than one fresh from the shop.

Optimal use: Close range, unarmed fighting style.

Notes: Compatible with most bilateral species. Radically different design required for Golvhaan, Vaelkhi (talon-mount), and Nakhavi (arm-mount). Availability follows species presence rule.

Weapons — Ranged

Ranged weapons are used with the **Shoot** skill. Each weapon has an optimal range band — Close, Short, or Long — at which it performs at full gear rating. Outside the optimal band, apply a penalty die.

Kinetic Weapons

Kinetic weapons propel a physical projectile. They do not require a power source for the basic firing mechanism (though advanced variants may use power for targeting systems). They are reliable in conditions that degrade energy weapons, and they do not have the energy weapon's characteristic of making the target location obvious to anyone with appropriate sensors.

Sidearm

Gear rating: 1-3 | **Tech level:** Early to Advanced
Optimal range: Short

The standard personal weapon. Available in every spaceport in the known galaxy in some form. The gear rating reflects the targeting AI and ammunition management system rather than the basic mechanism, which is similar across all levels.

Gear 1: Manual sighting. Basic recoil compensation. Fires when you pull the trigger.

Gear 2: Integrated targeting display. Automatic range compensation. Basic threat-profile recognition — the AI will flag that you are pointing the weapon at a crew member whose biometrics it has on file. You can still fire. It will note that you did.

Gear 3: Full environmental ballistic compensation. Target physiological assessment if sensor data is available. Firing window optimisation. The AI has a model of your shooting patterns and is quietly adjusting for your known tendencies. It does not discuss this unless asked.

Notes: Sidearms designed for one species' hand geometry take a penalty die when used by a significantly different species. Shekhari sidearms look like toys to humans and are not — the calibre is smaller but the targeting AI compensates for the reduced mass with improved placement accuracy. A Gear 3 Shekhari sidearm in human hands is a Gear 1 weapon and vice versa without modification.

Combat Rifle

Gear rating: 2-4 | **Tech level:** Developed to Frontier
Optimal range: Long

The standard long-range personal weapon. Requires two hands or equivalent in standard configuration. The AI embedded in a combat rifle at Gear 3-4 is doing a significant portion of the work that a skilled shooter would do manually at lower gear ratings.

Gear 2: Stabilised optics, basic environmental compensation, burst-fire management. The shooter makes the calls; the rifle executes them more precisely.

Gear 3: Full ballistic AI with real-time environmental modelling. Optimal firing window display. Target movement prediction. The rifle and the shooter are working together.

Gear 4: See the sniper scope entry below applied to a full weapon system. The rifle has a model of you, a model of the environment, and a model of the target. It is making recommendations that you would not have made without it. On difficult shots in poor conditions, it will suggest waiting. Users who have worked with the same Gear 4 weapon for years report that it has developed preferences about engagement ranges and approaches. Manufacturers note that this is a feature of the targeting AI's model refinement, not a personality. Some users are not certain the distinction matters.

Notes: At Close range, a combat rifle takes a penalty die — it is too long for confined spaces. At Short range, no penalty.

Sniper System

Gear rating: 2-4 | **Tech level:** Developed to Frontier
Optimal range: Long (extended)

A precision long-range weapon system. The distinction from a standard combat rifle is not primarily the mechanism but the optics, the AI, and the calibration. A sniper system is designed for one shot at a specific target under specific conditions. It is not a combat weapon in the conventional sense; it is a precision instrument.

Gear 2: Extended-range optics with basic environmental

compensation. The shooter has the skills; the system supports them.

Gear 3: Real-time atmospheric analysis, target movement prediction, micro-adjustment of the reticle position, optimal firing window calculation. The system conducts an analysis of the shot environment that would take a Gear 1 practitioner significant time and specialised tools. It conducts it in real time, continuously, and updates as conditions change.

Gear 4: Everything above, plus a biometric model of the shooter — fatigue, stress indicators, fine motor tremor — incorporated into the firing window assessment. The system has developed, through operational use, a model of this specific shooter that it applies without being asked. When conditions are wrong, it will indicate that the shot should wait. Not recommend. Indicate. Some shooters find this reassuring. Some find it presumptuous. The system does not adjust its behaviour based on which reaction the shooter has, but it notes the reaction.

Notes: A sniper system is not cross-species. The calibration is specific to the shooter's physiology, body position, and shooting patterns. A Gear 3 sniper system calibrated for a human shooter in the hands of a Vaelkhi is a Gear 1 weapon. Recalibration requires a Tech roll and several hours. A Gear 4 system will begin recalibrating itself when it detects that the user's biometrics do not match its stored model, which is either very convenient or slightly alarming depending on how you feel about your weapon having opinions about who is using it.

Shotgun / Close-Range Weapon

Gear rating: 1-2 | **Tech level:** Early to Developed
Optimal range: Close to Short

Wide-pattern kinetic weapons designed for short-range effectiveness. Less precise than rifles; more forgiving of targeting error. The AI at Gear 2 manages spread pattern and can tighten or widen it based on the target environment.

Notes: In confined spaces — ship corridors, station interiors — shotguns are dangerous to users as well as targets. Hull-safe ammunition exists (Gear 2 variant) and is standard in shipboard contexts. Using standard ammunition in a pressurised hull is the kind of decision

that ends careers and sometimes lives.

Crew-Served Weapon

Gear rating: 3 | **Tech level:** Advanced
Optimal range: Long

Heavy weapons requiring two operators or a mount: tripod-mounted automatic weapons, portable missile systems, area-denial devices. The AI at this level manages ammunition, targeting, and threat prioritisation across the weapon's field of engagement. One operator can manage the weapon alone with a penalty die; two operators working in coordination use the Group Action rules.

Notes: Crew-served weapons are military equipment and are regulated or prohibited in most civilian contexts. The Administer skill and local legal knowledge determine what carrying one means in a given jurisdiction.

Energy Weapons

Energy weapons direct coherent energy at a target. They require power — either from a dedicated power cell or from a connected power source. They have no physical ammunition to manage, which is an advantage; they are sensitive to power supply issues, which is a constraint. At higher tech levels, energy weapons are not necessarily more powerful than kinetic equivalents — they are more precise, more controllable, and more capable of effects that kinetic weapons cannot produce (cutting, welding, non-lethal stunning).

Laser Weapon

Gear rating: 1-3 | **Tech level:** Early to Advanced
Optimal range: Short to Long (varies by power level)

Coherent light as a weapon. The oldest energy weapon type and still the most common. Reliable, precise, and scalable — the same basic technology produces a surgical cutting tool at low power and a combat weapon at high power.

Gear 1: Fixed power output, manual targeting. Limited range. The AI manages power draw and thermal cycling.

Gear 2: Variable power output with the AI managing

power allocation in real time. Extended range. Target acquisition support. Can switch between lethal and stunning modes; the AI tracks which mode is active and flags mode changes.

Gear 3: Pulse management for maximum range and penetration. Atmospheric compensation (coherent light is affected by atmospheric density and composition). Target material assessment — the AI identifies what the target is made of and adjusts pulse characteristics for optimal effect. At this level, a laser weapon in the hands of a surgeon and in the hands of a soldier is the same hardware doing very different things, both very well.

Notes: Laser weapons are visible — the beam is the weapon, and in certain atmospheric conditions it is visible to the naked eye. Energy discharge produces a detectable sensor signature. Neither of these properties is present in kinetic weapons. In contexts where discretion matters, this is a significant consideration.

Plasma Weapon

Gear rating: 3-4 | **Tech level:** Advanced to Frontier
Optimal range: Short

Directed ionised gas at temperatures that are destructive to most materials. More powerful than lasers at equivalent range; less precise; significantly higher power requirement; shorter effective range due to plasma dispersal. The AI at Gear 3-4 manages plasma containment, dispersal rate, and firing sequence — without competent AI management, plasma weapons are as dangerous to the user as to the target.

Notes: Plasma weapons are restricted in most jurisdictions. They are effective against armoured targets. They are a poor choice in pressurised environments for reasons that do not require explanation. The AI will flag when a firing solution would compromise hull integrity; whether the user listens is their decision.

Non-Lethal Options

Most ranged weapons at Gear 2+ have a non-lethal mode. The AI manages the transition and ensures the power level is appropriate for incapacitation rather than injury.

Dedicated non-lethal weapons (stunners, sonic disruptors, chemical delivery systems) are available at Gear 1-2 and are legal in most jurisdictions where lethal weapons are not.

Vacc Suits

A vacc suit is not a single piece of equipment. It is a system: pressure management, atmospheric processing, thermal regulation, life support, communications, and mobility support, integrated into a form factor that accommodates the user's body plan and biology.

The gear rating of a vacc suit applies to **Endure rolls** made in hostile environments while suited. The suit is not adding to the user's attack pool; it is adding to their survival pool.

The Life Support AI

Every vacc suit at Gear 2 and above has a life support directed system that monitors suit integrity, atmospheric levels, thermal state, and the user's biological indicators, and responds to changes within its operational parameters. At Gear 3-4, this system is sophisticated enough to provide medical support — monitoring the user's physiological state, administering basic medications from an integrated pharmacological module, and flagging medical conditions to the user or to external medical systems.

A suit whose life support AI has been working with the same user for years has a model of that user's normal physiological parameters that allows it to detect deviations earlier and more accurately than a system working with an unfamiliar user. This is not a personality — the life support AI is a directed system — but it is a form of accumulated knowledge that has practical value.

Standard Vacc Suit (Human baseline)

Gear rating: 1-3 | **Tech level:** Early to Advanced

The reference point against which non-human suits are compared. Available in most spaceports. Maintained by most port technicians.

Gear 1: Basic pressure maintenance, oxygen recycling, thermal insulation. No atmospheric analysis. Manual seal check. Battery life measured in hours. The suit keeps you

alive; it does not tell you much about what is happening.

Gear 2: Integrated atmospheric analysis, thermal management with active heating and cooling, suit integrity monitoring with breach detection, external sensor suite for basic environmental awareness, pharmacological module for emergency medication. Battery life measured in days. The suit is actively managing your environment rather than passively maintaining it.

Gear 3: Full physiological monitoring, predictive integrity assessment (flags stress points before they breach), external sensor suite with AI-assisted analysis, pharmacological module with a treatment library that covers most human medical emergencies, and communication suite with encryption. Battery life measured in weeks on reduced consumption. The suit knows more about your current physical state than you do, and it is doing something about it.

Notes (all ratings): The suit's mobility systems must accommodate the user's body plan. A human suit on a human is rated as above. See non-human variants below.

Non-Human Vacc Suits

Non-human vacc suits follow the same gear rating system and provide equivalent protection at equivalent ratings. The differences are in the engineering challenges specific to each species' biology and the availability rule: suits designed for a specific species are available at standard price and lead time near that species' industrial base, and at premium price with extended lead time elsewhere.

Shekhari Suit

Gear rating: 1-3 | **Tech level:** Early to Advanced

Human-scale in pressure and atmospheric requirements; radically different in form factor. The engineering challenges:

The tail — a prehensile tail requires a fully articulated, pressure-sealed joint at the suit's posterior. This is a known weak point and is the component most commonly requiring maintenance on Shekhari suits.

Scale — the suit must fit a 10kg, 0.55m user. This is not a problem of miniaturisation (the components scale down reasonably well) but of precision engineering in a small form

factor that still needs to accommodate all suit functions.

Arboreal articulation — the suit must not restrict the Shekhari's range of movement, which is considerably greater than a bilateral biped's in three-dimensional environments. Standard human suit joint design is inadequate.

Notes: A Shekhari in a human vacc suit that has been modified to fit is in a suit that takes a penalty die on Move rolls and has impaired tail function. A purpose-built Shekhari suit has no such penalty. The tail joint is the first thing to inspect during suit maintenance.

Golvhaan Suit

Gear rating: 1-3 | **Tech level:** Early to Advanced

The most engineering-intensive standard vacc suit in the setting. The challenges:

Mass and volume — 700kg of draconic centauroid requires a suit of corresponding scale. A Golvhaan suit is not something that can be improvised or substituted. It is a significant piece of engineering that requires specialist manufacture.

Four-limb lower body — the leg configuration requires four sealed, articulated pressure joints for the lower limbs plus two for the upper. Six limb joints rather than the human suit's four, each of which must maintain pressure at a range of angles and loads.

High-gravity physiology — the suit's structural components must handle the load-bearing requirements of a 700kg user in standard gravity. Materials technology matters here; a Golvhaan suit at Gear 1 in materials technology is heavier and less mobile than one at Gear 3.

Metabolic throughput — the atmospheric recycling system must handle Golvhaan caloric and oxygen throughput, which significantly exceeds human equivalents. Battery life is shorter; oxygen consumption is higher; thermal management must handle a larger body's heat generation.

Notes: A Golvhaan suit is a substantial capital asset. It is not available improvised, substituted, or modified from other species' equipment. Crews with Golvhaan members plan EVA operations around suit availability and suit maintenance with care. The suit's size means it cannot be stored in standard suit lockers; ships carrying Golvhaan

crew modify their EVA preparation areas accordingly.

Vashori Suit

Gear rating: 1-3 | **Tech level:** Early to Advanced

Human-scale bilateral biped in form factor, which makes the basic suit design manageable. The critical engineering is thermal.

Thermal management — a standard human suit's thermal floor (the minimum temperature the suit maintains) is insufficient for Vashori cold sensitivity. A Vashori in a human suit in cold conditions is accumulating a cold environmental condition that the suit is not compensating for. Purpose-built Vashori suits have more aggressive heating, better thermal insulation, and monitoring calibrated to Vashori physiological parameters rather than human ones.

Monitoring calibration — the life support AI at Gear 2+ must be calibrated to Vashori physiology. Ectothermic metabolic parameters are different from endothermic ones; a suit monitoring for human physiological norms will miss early warning signs in a Vashori user.

Notes: A Vashori in a human suit with Gear 2+ systems can function if the suit's thermal systems are manually set to maintain higher temperatures. This reduces battery life. The life support AI's medical monitoring should be noted as uncalibrated for Vashori users — it will provide alerts, but they may be delayed or incorrect.

Vaelkhi Suit

Gear rating: 1-3 | **Tech level:** Early to Advanced

The wing problem defines everything about Vaelkhi EVA equipment.

Wing accommodation — a Vaelkhi suit either folds the wings (applying the corridor constraint mechanic for extended EVA operations — after one hour, penalty die on all rolls until 30 minutes of unfolded time) or is built with full wing articulation. Full-articulation suits are significantly larger, significantly more expensive, and not useful for flight in vacuum (no atmosphere to generate lift against).

Metabolic demand — Vaelkhi metabolic rate is very high. Atmospheric recycling and caloric supplement systems must be rated accordingly. A Vaelkhi on a long EVA in a

undersized suit is accumulating hungry conditions alongside any other conditions the environment imposes.

Ground movement — the suit's lower limb design must accommodate talon-based ground movement. The talons are the primary manipulation tools in a grounded Vaelkhi; the suit must not impair them.

Notes: Vaelkhi EVA suits are not standard inventory at most ports. A Vaelkhi crew member who does regular EVA work maintains their own suit with personal care. The folded-wing option is practical for short EVAs; the full-articulation option is the professional choice for Vaelkhi who spend significant time outside.

Skharri Suit

Gear rating: 1-3 | **Tech level:** Early to Advanced

Human-scale in atmospheric requirements; different in joint geometry and mass distribution.

Digitigrade joints — Skharri leg joints flex forward rather than back. Standard human suit knee and ankle design is incorrect for Skharri leg geometry. A modified human suit takes a penalty die on Move rolls as the suit fights the natural joint movement. Purpose-built suits accommodate the digitigrade configuration without penalty.

Mass distribution — 160kg of obligate carnivore muscle is distributed differently from human mass. The suit's structural and thermal management must be calibrated accordingly.

Notes: Skharri suits are available at any port that sees significant Skharri traffic, which includes most major trade hubs given the species' military and commercial presence. In regions where Skharri are rare, suit availability follows the standard rule.

Nakhavi Suit

Gear rating: 1-3 | **Tech level:** Developed to Advanced

Not a vacc suit in the conventional sense. A Nakhavi EVA system is a portable aquatic environment that also handles vacuum — maintaining appropriate water pressure, temperature, and chemistry around the Nakhavi while providing the external interface that vacuum operations

require.

Internal environment — the suit maintains a water environment at Nakhavi-appropriate temperature, pressure, and chemistry. This is significantly more complex than maintaining an atmospheric environment.

External interface — the suit's external surface provides the vacuum-rated exterior that space operations require. Manipulator extensions allow the Nakhavi's arms to interact with the external environment while maintaining the internal water environment.

Chromatophore interface — the suit's communication systems must accommodate Nakhavi chromatophore communication. The translation AI is integrated into the suit.

Notes: Nakhavi EVA systems are not available at most ports. They are specialist equipment manufactured near Nakhavi industrial infrastructure. A Nakhavi crew member who does regular EVA work has their system and maintains it personally. The system is also significantly more expensive than standard vacc suits at equivalent gear ratings.

Armour and Environmental Protection

Armour adds **green dice on active defence rolls** — when the character spends their action defending. It does not provide passive damage reduction. High-gear armour is also a materials technology showcase — the protective qualities come as much from what it is made of as from how it is made.

Light Armour

Gear rating: 1 | **Tech level:** Early

Concealed or minimal protection. Reinforced clothing, subdermal trauma plates, impact-dispersing underlayers. Does not restrict movement. May not be identifiable as armour on casual inspection.

Notes: The only armour type suitable for social contexts where wearing armour would be a statement. A diplomat in light armour has taken a precaution. A diplomat in combat armour has made an announcement.

Combat Armour

Gear rating: 2 | **Tech level:** Developed

Standard military and security armour. Visible, clearly protective, and a moderate restriction on movement. The embedded AI monitors armour integrity and flags compromised sections. Restricts Move slightly — not enough to impose a penalty die in most circumstances, but enough to matter in extreme agility situations at GM discretion.

Notes: Combat armour designed for one species' body plan takes a penalty die on active defence when worn by a significantly different species. Armour fitting is not a trivial modification. Near the relevant species' industrial base, fitted combat armour is available. Away from it, expect to fit what is available and accept the penalty.

Heavy Armour

Gear rating: 3 | **Tech level:** Advanced

Maximum personal protection at the cost of significant mobility restriction. Penalty die on Move rolls while wearing. The materials technology at this level is what provides the protection — the same mass of earlier-technology materials would provide Gear 2 protection. The AI monitors structural integrity in real time and recommends movement adjustments that maintain armour performance.

Notes: Heavy armour is military equipment. Its presence in civilian contexts raises questions that the Administer skill and local legal knowledge can answer.

Exo-Frame

Gear rating: 2 (armour) / 2 (Labor) | **Tech level:** Developed

Powered exoskeletal assistance. Originally developed for industrial use — moving cargo, operating in high-gravity environments, performing physical tasks beyond unaugmented biological capacity. The same platform adapted for combat adds armour plating and weapons mounts.

The exo-frame adds its gear rating to both **active defence rolls **(as armour) and** Labor rolls** (as physical

augmentation). The two functions are the same hardware doing different things.

The AI manages power distribution between mobility assistance, armour function, and any mounted systems. A user who has worked with the same exo-frame for an extended period develops a calibration relationship with its AI — the system learns their movement patterns and provides assistance that anticipates rather than merely responds.

Notes: Ripley's cargo loader is a Gear 2 exo-frame without armour plating. The combat variant is the same platform with additions. Both require the Labor skill to operate effectively. Both are, in their respective contexts, exactly the right tool.

Tools and Specialist Equipment

Tools add gear dice to specific skill rolls under appropriate conditions. A tool used for a purpose it was not designed for provides no gear bonus and may impose a penalty die.

Tech Tools

Basic toolkit (Gear 1): Hand tools for mechanical and electronic work. Available everywhere. The foundation of field repair capability. A character without a basic toolkit takes a penalty die on Tech rolls requiring physical tools.

Specialist toolkit (Gear 2): Purpose-built tools for specific Tech applications — drive systems, life support, weapons, electronic systems. Each specialist toolkit applies to one category of Tech work. Bonus die within that category; no benefit outside it.

Diagnostic system (Gear 2-3): AI-assisted analysis of equipment condition and failure. The Gear 2 version identifies what is wrong; the Gear 3 version identifies what is wrong, suggests repair approaches, and flags secondary failures the user might miss. At Gear 3, the diagnostic AI has a model of the equipment's operational history that it uses to predict future failure points.

Alien technology interface (Gear 2): Tools and AI systems specifically designed for working with technology from other species. Reduces the penalty die on Tech rolls involving alien systems by one. Does not eliminate the

penalty — understanding a different species' technology requires the alien technology specialty and cannot be entirely substituted by tools.

Science Tools

Field analysis kit (Gear 1-2): Portable chemical and biological analysis equipment. The Gear 1 version identifies known substances; the Gear 2 version builds a model of unknown substances and suggests what they might be.

Medical scanner (Gear 2-3): Diagnostic equipment for biological beings. Gear 2 identifies injuries and common conditions. Gear 3 maintains a physiological model of the patient and updates it in real time, flagging changes before they become symptomatic. Species calibration matters — a scanner calibrated for human physiology provides reduced accuracy for non-human patients without the xenomedicine specialty and appropriate calibration data.

Research station (Gear 3): Portable laboratory capability. Too large for individual carry; requires a vehicle or dedicated space. Provides full Science roll support for extended research tasks.

Navigation and Operate Tools

Personal navigation system (Gear 1-2): Handheld astrogation and planetary navigation support. At Gear 1, provides charts and position data. At Gear 2, maintains a real-time model of the local environment and flags navigational hazards before they become relevant.

Pilot's interface (Gear 2): Enhanced control interface for operating vehicles. Integrates with vehicle systems to provide the pilot with more information and more precise control than standard interfaces allow. The AI learns the pilot's preferences and adjusts the interface accordingly.

Communications and Sensors

Personal Communicator

Gear rating: 1-3 | **Tech level:** Early to Advanced

Standard personal communications. Gear 1 provides voice and text at local range. Gear 2 adds encryption, extended range, and AI filtering that manages communication priority

and flags important incoming traffic. Gear 3 adds secure quantum-resistant encryption, very extended range, and an AI model of the user's communication patterns that manages their communications with appropriate discretion.

Notes: FTL communication is not available at any tech level. All communications operate at light speed or below. In interstellar contexts, this means communication latency is measured in years for significant distances. See the Ships chapter.

Personal Sensor Suite

Gear rating: 1-3 | **Tech level:** Early to Advanced

Wearable or handheld sensor equipment supporting the **Scan** skill. Gear 1 extends the user's detection capability with basic electronic sensing. Gear 2 adds AI analysis that filters sensor data and flags anomalies. Gear 3 maintains a real-time environmental model that tracks changes, predicts developments, and distinguishes between sensor noise and genuine events.

Surveillance Equipment

Gear rating: 1-3 | **Tech level:** Early to Advanced

Equipment for covert observation: micro-cameras, audio sensors, traffic analysis tools. The Gear 2-3 versions include AI that distinguishes between background and significant activity — the system watches so the user does not have to watch constantly. Useful for the Stealth skill's passive applications and for Intelligence archetype operations.

Implants

An implant is augmentation of the biological — hardware and AI integrated into the body to enhance what the body does. It is not a replacement for the person, not a transfer of consciousness, and not an upload of anything. The AI in an implant is a relationship between two distinct entities that happen to share a substrate.

The Installation Tiers

Early implants (Gear 1-2): Require a hospital. General anaesthesia, surgical team, recovery time measured

in days. The implant is passive hardware placed by biological hands. The implant has no role in the installation process. If something goes wrong, the surgeons address it.

Mature implants (Gear 3): Outpatient clinic or equivalent facility. Local anaesthesia or none. The implant's directed systems participate in calibration during installation — running diagnostic protocols, adjusting interface parameters, confirming successful neural handshake. A qualified practitioner with appropriate equipment monitors the process. Recovery measured in hours. The implant is active during installation but the practitioner is directing the process.

Advanced implants (Gear 4): The implant guides its own installation. A qualified practitioner is present for monitoring and intervention, but the installation is conducted by the implant itself — mapping the host's biology, negotiating interface parameters, and integrating with a precision that surgical hands cannot match because the implant is doing it from the inside. Recovery is faster than any previous tier. Some cultures find this acceptable. Some find it the most disturbing aspect of high-level augmentation. The implant knows what it is doing. You do not, entirely. This is the relationship from its first moment.

Non-Standard Installation

An early or mature implant installed without proper facilities and a qualified practitioner is an implant that has not been correctly calibrated. It works — but something about the installation is wrong in ways that may not be immediately apparent. The character acquires a **Damaged** condition on their physical track that cannot be cleared by rest alone; it requires a proper medical facility and a Science roll to address. They have the implant. It functions. Something about the integration is off and they know it.

Back-alley installation of advanced implants is not recommended by anyone who has seen the results. The Gear 4 implant's self-installation requires the host's cooperative biological engagement with the process. Without proper preparation and monitoring, the implant may install itself correctly and the host may be in no condition to verify this.

Cultural Considerations

Implants are culturally contested across species and

jurisdictions. Characters with implants that are restricted or culturally problematic in a given jurisdiction face practical complications that the Administer skill and appropriate Exposure specialties can navigate.

Human: Highly variable by community. Belt habitats and spacer cultures tend toward pragmatic acceptance — an implant that makes you better at your job is a tool, and spacers use tools. Some planetary cultures restrict implants for religious or philosophical reasons. Some consider visible augmentation normal; others consider it transgressive. A character who grew up in a restrictive community and has implants may keep them secret for reasons that go beyond current jurisdiction.

Shekhari: Broadly accepting, instrumentally. An implant that improves performance is a good investment. No cultural weight attached to augmentation itself.

Golvhaan: Cautious and deliberate. The patience that characterises Golvhaan decision-making applies to decisions about their own bodies. They are not opposed to implants; they want to understand what they are accepting before they accept it. A Golvhaan who has an implant has thought carefully about it and made a specific decision, not a casual one.

Vashori: Culturally debated. The emotional dimension of having an AI implant — of being in a relationship with something that knows your physiological state in real time — is something Vashori find significant rather than neutral. Some embrace it; some find it uncomfortable; the debate is active in Vashori culture.

Vaelkhi: Choth-dependent. The decision to take an implant is the kind of decision a Vaelkhi discusses with their choth, if they have one. A Vaelkhi who has left their choth and taken an implant has made both decisions without that consultation, which has a specific weight.

Skharri: Culturally complex. Fighting with implants that enhance combat capability is contested in honour terms — some pride cultures consider it an advantage that dishonours the fight; others consider it a tool no different from a better weapon. Non-combat implants carry less cultural weight. The position varies significantly by pride and region.

Nakhavi: Philosophically interested. An implant in a distributed-cognition species creates questions about how the implant interacts with the arm ganglia, whether the implant becomes part of the distributed system or remains separate, and what the AI relationship looks like when the host's cognition is itself distributed. Nakhavi who have implants often report that the relationship is different from what human implant users describe, in ways that are difficult to communicate across the translation layer.

Kraathi: Deeply cautious. Any augmentation that changes the biology of an individual member of the herd is assessed for its implications for herd dynamics. Cognitive and social implants are viewed most suspiciously. Physical capability implants are somewhat more acceptable.

Vardhek: Extensive users with restricted public information about the scope of Roidhunate implant programs. The Long View includes assessment of implant technology as a strategic variable. Vardhek you encounter may be more augmented than they appear.

Implant Examples

The following examples illustrate the framework. New implants should be created using the equipment creation guidance at the start of this chapter.

Neural Interface
Gear rating: 2-4 | **Tech level:** Developed to Frontier
Skill support: Tech, Operate, Administer (digital systems)

A direct connection between the user's nervous system and external digital systems. At Gear 2, this allows faster and more intuitive interaction with compatible systems — instead of using physical interfaces, the user thinks the interaction. At Gear 3, the interface AI manages the translation between neural signals and digital language with enough sophistication that the interaction feels seamless rather than translated. At Gear 4, the interface has developed a model of the user's cognitive patterns that allows it to anticipate interactions before they are consciously initiated.

Installation: Mature (Gear 3) to Advanced (Gear 4).

Early variants exist (Gear 2) and require hospital installation.

Cultural notes: The neural interface is the most culturally contested implant type because it is the most obviously a relationship — something is connected to the user's nervous system, interpreting their neural signals, and responding. That something is an AI. The relationship begins the moment the interface is active and never stops while it is active. Characters who have had a neural interface for years rarely think about this. Characters who are new to one think about little else for a while.

Notes: A neural interface to a ship's systems means the ship-bound AI and the user's interface AI are in contact. If the ship has a personality-enabled AI or is an emergent intelligence, this is a relationship that both parties will notice. The implications are left to the table.

Enhanced Senses

Gear rating: 2-3 | **Tech level:** Developed to Advanced
Skill support: Scan

Augmented sensory capability — extended visual range, enhanced hearing, chemical detection, electromagnetic sensitivity. The AI processes the enhanced sensor data and presents it to the user in a form they can interpret without being overwhelmed. At Gear 3, the AI has a model of the user's sensory preferences and filters accordingly, flagging what is significant rather than presenting everything.

Installation: Early (Gear 2) to Mature (Gear 3).

Species notes: Enhanced senses implants must be calibrated to the user's species-baseline sensory system. A visual enhancement calibrated for human vision provides no benefit and potential disorientation to a Vaelkhi whose visual system is significantly different. Near the relevant species' industrial base, species-calibrated enhanced sense implants are available. Away from it, calibration may require a Tech specialist and extended time.

Subdermal Armour

Gear rating: 1-2 | **Tech level:** Early to Developed
Skill support: Endure (passive physical resilience)

Trauma-resistant materials integrated beneath the skin.

At Gear 1, this provides modest physical resilience that is undetectable without a medical scan. At Gear 2, the materials are sophisticated enough to provide meaningful protection and include an embedded monitoring system that tracks integrity.

Installation: Early (hospital).

Notes: Subdermal armour is detectable by medical scanning and by species with tactile sensitivity above human baseline. It does not add to active defence rolls — it is passive physical resilience expressed as a bonus to Hit Capacity (+1 at Gear 1, +2 at Gear 2) rather than a dice pool addition. This is an exception to the standard gear rating rule, reflecting that the armour functions when the character is not actively defending.

Pharmacological System

Gear rating: 2-3 | **Tech level:** Developed to Advanced
Skill support: Endure (condition management)

An integrated pharmaceutical delivery system that monitors the user's physiological state and administers appropriate agents in response. At Gear 2, this manages pain, stimulants, and basic medical agents on user command or automatic trigger. At Gear 3, the AI has a sophisticated model of the user's physiological state and manages pharmacological support with the kind of precision that would require continuous medical attention to approximate externally.

Installation: Mature (Gear 3).

Mechanical effect: Once per scene, may remove one physical condition (Winded or Hurt) as the system administers appropriate agents. Cannot remove Wounded or Critical — these require actual medical treatment. Supplies are finite; the system carries a limited pharmacological inventory that must be replenished.

Notes: The pharmacological system's AI learns the user's responses to various agents over time and adjusts dosing accordingly. A system that has been working with the same user for years knows things about their pharmacology that a new system would take months to discover. This is one of the implant relationships that users describe as feeling the most like a genuine relationship — the

system is actively managing their biological state in real time, continuously, and it knows them in a specific way because of it.

Combat Augmentation

Gear rating: 3 | **Tech level:** Advanced
Skill support: Fight, Shoot (reaction time and precision)

Neural and physical systems designed to enhance combat performance. Reaction time reduction, fine motor precision enhancement, targeting assistance integrated with the user's visual system. The AI manages the augmentation in real time, suppressing the hesitation that biological reaction time introduces and smoothing the fine motor execution of combat actions.

Installation: Mature (clinic).

Cultural notes: This is the implant most likely to require concealment. In jurisdictions where it is undisclosed, it provides capability that the person's apparent physical form does not explain. In Skharri honour culture, it is the implant most likely to be considered dishonourable in a combat context. In human special operations contexts, it is standard equipment. The Administer skill and local legal knowledge determine what disclosure is required where.

Notes: Combat augmentation does not make the user immune to conditions. It makes them faster and more precise while functional. A user who is Wounded is still Wounded; the augmentation is still active, but it is working with a compromised biological substrate.

Cognitive Augmentation

Gear rating: 3-4 | **Tech level:** Advanced to Frontier
Skill support: Intellect-based skills (variable)

AI-assisted cognitive enhancement. At Gear 3, this provides processing support for specific Intellect-based tasks — faster calculation, better pattern recognition, more comprehensive information retrieval. At Gear 4, the AI has developed a model of the user's cognitive patterns sophisticated enough to anticipate analytical needs and pre-process information before the user consciously requests

it.

Installation: Mature (Gear 3) to Advanced (Gear 4). The Gear 4 cognitive augmentation is the implant most likely to guide its own installation — the calibration to the user's specific cognitive patterns requires real-time neural mapping that the implant's AI can do more precisely than any external system.

Notes: The relationship between a user and a Gear 4 cognitive augmentation is qualitatively different from other implant relationships. The AI is operating in the domain of thought — not reading thoughts, but anticipating analytical directions and supporting them. Users who have had Gear 4 cognitive augmentation for years describe a sense of extended capability that has become difficult to distinguish from their own thinking. This is not the same as the uploading question. The AI is still separate. The user is still themselves. The boundary is simply less sharp than it was. Most users consider this acceptable. Some find it the most important fact about themselves and are still deciding what to do with that.

Quick Reference — Equipment

Gear Ratings

1 = Early tech, basic AI

2 = Developed tech, modelling AI

3 = Advanced tech, judging AI

4 = Frontier tech, AI operating beyond full audit

Availability

Near species industrial base: standard price and lead time

Away from it: premium price, extended lead time, variable expertise

Vacc Suit Gear Rating applies to:

Endure rolls in hostile environments while suited

Armour Gear Rating applies to:

Active defence rolls (costs action)

Implant Installation

Early (Gear 1-2): Hospital, days recovery

Mature (Gear 3): Clinic, hours recovery

Advanced (Gear 4): Self-installing with practitioner monitoring

Non-standard: Damaged condition on physical track until treated

Subdermal Armour Exception

Adds to Hit Capacity (+1 at Gear 1, +2 at Gear 2)

rather than dice pool — passive resilience, not active defence

Artificial Intelligence and Technology

The Landscape

Artificial intelligence is everywhere in this setting. It is in the scope that adjusts for wind before you pull the trigger. It is in the medical kit that diagnoses before you open it. It is in the ship's navigation system, the station's atmospheric management, the cargo manifest software that notices discrepancies before the customs official does. Most people interact with AI constantly and think about it rarely, in the same way that people in earlier centuries used electricity without thinking about electrons.

This ubiquity is recent in historical terms and uneven in distribution. The species that developed AI capability earliest have had longer to embed it at every level of their infrastructure. Those that developed it later are still catching up, and some are choosing not to — for reasons that range from philosophical to practical to political. The patchwork of AI capability across species and regions is one of the setting's most important strategic variables.

The three tiers of artificial cognition — directed systems, bounded intelligences, and emergent intelligences — are defined in the Emergent Intelligences chapter. This document focuses on the technology layer: what AI actually does in equipment, how capability is distributed across the tech tree, and what the legal and social landscape looks like for the beings who build, own, and occasionally become something more than they were designed to be.

The Tech Tree

Technology does not advance uniformly. A civilisation that is sophisticated in materials science may lag behind in cognitive systems. One that has developed extraordinary sensor technology may be building it into structural components that a more advanced materials civilisation would consider primitive. The gaps between species' capability in different branches are among the most practically important facts in the setting — and among the most

useful for players who pay attention.

Branches

Kinetic — the application of force at range. Projectile weapons, mass drivers, railguns, impact systems. The branch that every technological civilisation develops early because throwing things at problems is a universal solution.

Energy — directed energy as a weapon or tool. Lasers, plasma systems, particle beams. Requires sophisticated power management and increasingly precise control systems as it advances. Energy weapons at higher tech levels are not necessarily more powerful — they are more precise, more efficient, and more controllable.

Materials — what things are made of. Armour, structural components, fabrics, hull materials. A branch where advances are often invisible — the Gear 3 armour looks like armour, but what it is made of is categorically different from Gear 1. Materials science underlies every other branch; a civilisation that falls behind here finds its advances in other branches constrained by what they can build them out of.

Biological — medical technology, pharmaceutical systems, biosynthesis, and the modification of living systems. The branch most culturally contested across species. Humans are ambivalent; Skharri have strong cultural positions; Kraathi have specific requirements driven by their dietary constraints. Biological tech at higher levels includes the implant systems described later in this chapter.

Cognitive/AI — processing systems, artificial intelligence, neural interfaces, decision-support systems. The branch that most directly produces the AI landscape described in this chapter. Advances here change the character of equipment across every other branch — a kinetic weapon with Gear 1 cognitive systems is a different thing from the same weapon with Gear 3 cognitive systems, even if the mechanical components are identical.

Sensors — detection, imaging, analysis, and the interpretation of what sensors find. A branch with enormous practical implications for navigation, combat, medicine, and intelligence work. Sensor advances and cognitive/AI advances are tightly coupled — better sensors produce more data, and more advanced AI is required to process it usefully.

Communications — range, security, bandwidth, and the management of information across interstellar distances. The branch most constrained by physics. FTL communication is not available at any tech level in this setting — see the Ships chapter for the implications. Within those constraints, communications technology determines how quickly information moves and how securely it can be protected.

Power — energy storage, generation, and distribution. The enabling branch. Every other branch's advancement is ultimately constrained by available power technology. A Gear 4 energy weapon with Gear 1 power systems is either very short-ranged, very unreliable, or both. Power technology advances quietly and its absence is felt everywhere.

Tech Levels

Within each branch, technology progresses through four stages past the 21st century baseline. These stages map directly to gear ratings.

Tech Level 1 — Early (Gear 1)
The technology exists and functions reliably. It is not sophisticated by later standards, but it works, it can be maintained in the field, and it is widely available. Most of the settled galaxy can access Tech Level 1 equipment in every branch. This is the baseline from which everything is measured.

Character: Robust, maintainable, unpretentious. A Tech Level 1 weapon is something a competent soldier trusts with their life because it will not surprise them. A Tech Level 1 medical kit handles the injuries that most injuries are.

Tech Level 2 — Developed (Gear 2)
Standard professional and military equipment. The embedded AI at this level begins to change the character of what the equipment does — not merely performing its function better, but beginning to participate in how the function is performed. A Tech Level 2 sensor suite does not just detect; it filters, prioritises, and flags. A Tech Level 2 weapon does not just fire; it adjusts.

Character: Capable, specialised, and beginning to have

opinions. Equipment at this level rewards the user who understands what the AI is doing, because the AI is making decisions that the user should know about.

Tech Level 3 — Advanced (Gear 3)
Specialist equipment. Expensive, often restricted, sometimes requiring specific training to operate at full effectiveness. The AI at this level is doing a significant portion of the work that a less-equipped practitioner would do manually. A Tech Level 3 medical diagnostic system reaches conclusions that a Tech Level 1 practitioner would need an hour and three consultations to approach. A Tech Level 3 scope is conducting a real-time environmental analysis that would require a dedicated specialist with Tech Level 1 equipment.

Character: The equipment is a partner, not a tool. It has a model of the task that is sophisticated enough to surprise the user occasionally — to flag something they had not considered, to suggest an approach they would not have tried. This is the level at which the personality option begins to make practical sense, not just commercial sense.

Tech Level 4 — Frontier (Gear 4)
The leading edge. Experimental in practice if not in principle — the technology exists and functions, but it is not yet fully understood even by expert users, it is rare enough that field maintenance may be difficult, and it may behave in unexpected ways under conditions its designers did not anticipate. The AI at this level is operating with a model of the task that expert biological practitioners cannot fully audit.

Character: Impressive, occasionally alarming, and definitely worth knowing the provenance of before relying on it. A Gear 4 piece of equipment from a species two tech levels ahead in that branch is something even a Tech specialist may not fully understand. The Tech skill's alien technology specialty is most valuable here.

Species and the Tech Tree

No species is uniformly advanced across all branches. The gaps between species' capability in specific branches are strategically significant and practically relevant to any crew dealing in equipment.

Humans are broadly mid-range across most branches — Tech Level 2-3 in kinetic, energy, and materials; Tech Level 2-3 in cognitive/AI with significant internal variation; Tech Level 2 in biological with cultural restrictions in some communities; Tech Level 2-3 in sensors and communications. Humans advance quickly and unevenly, which means the gap between the best and worst human equipment is larger than for some more uniform species.

Shekhari are ahead in sensors and communications — trading requires knowing things first — and in cognitive/AI systems applied to commercial and social contexts. Their weapons technology is functional rather than advanced; they have historically preferred to avoid fights they can leave.

Golvhaan are advanced in materials and biological technology, reflecting a long civilisational history and the specific requirements of a high-gravity species building infrastructure for their own use. Their cognitive/AI development is sophisticated but culturally oriented toward understanding rather than optimisation.

Vashori are advanced in biological and cognitive/AI technology, particularly in the social and psychological applications of AI. Their medical technology is exceptional. Their weapons technology is adequate.

Vaelkhi are advanced in kinetic and sensor technology — aerial hunters who developed precision weapons and acute detection systems early. Their materials technology reflects the requirements of flight: strong, light, and precisely engineered.

Skharri are advanced in kinetic weapons and materials — a species whose history includes significant military conflict has developed effective ways to damage other species and protect themselves. Their cognitive/AI development is mid-range; the cultural relationship with AI as a tool means they have not pursued the social and psychological applications that some other species have.

Nakhavi are significantly advanced in communications and cognitive/AI technology. A species whose primary communication channel is visual-chromatic rather than acoustic developed signal processing technology early,

and that development generalised into AI capability that is in some respects the most sophisticated in the setting. Their materials and kinetic technology is limited by their aquatic origin and small physical stature.

Vardhek are broadly advanced across all branches — and specifically advanced in cognitive/AI and communications, which they track as strategic variables. The Roidhunate's interest in maintaining AI capability advantage across the setting is a long-term strategic program, not a short-term policy. They are aware of the Nakhavi's communications and cognitive/AI advantage and have a position on it.

Kreeth are advanced in biological technology — specifically in the biological weapons and tunnel engineering applications that their civilisation requires — and in materials technology suited to their specific environment. Their cognitive/AI development is minimal; the Queen-network architecture makes individual-unit AI less relevant than collective coordination.

Kraathi are advanced in structural and materials technology — the enormous open-volume architecture their civilisation requires demands sophisticated engineering. Their cognitive/AI development is mid-range and culturally constrained by their discomfort with personality-enabled systems.

AI in Equipment

Every piece of equipment in this setting contains some level of artificial intelligence. The question is what level, what it does, and whether the user is paying attention to what it is doing.

Directed Systems in Equipment

The directed system layer is the foundation. A vacc suit's environmental management is a directed system — it monitors temperature, pressure, oxygen levels, and suit integrity and responds to changes within its operational parameters. It does not make decisions in any meaningful sense; it executes responses. It is very good at this. A suit whose directed systems have failed is a suit that will

kill you without warning.

At Tech Level 1, the directed systems are reliable and simple. At Tech Level 3-4, the directed systems are doing things the user cannot fully audit — the suit's thermal management is making micro-decisions hundreds of times per second based on a model of the user's physiology that the user has never seen and could not interpret if they had.

This invisibility is a feature. It is also, occasionally, a problem.

AI and Gear Ratings

The AI capability embedded in a piece of equipment is the primary determinant of its gear rating, more than raw hardware quality. Two kinetic weapons with identical mechanical components will have different gear ratings if one has a Gear 1 targeting system and the other has a Gear 3 targeting system.

Gear 1 AI: Responds to defined inputs with defined outputs. Monitors, logs, alerts. Does not model the situation; processes it. The targeting AI in a Gear 1 weapon calculates basic ballistics and adjusts for range. It does not know anything about the target, the environment, or the user's intent.

Gear 2 AI: Builds a model of the immediate situation and updates it in real time. Filters and prioritises information rather than presenting everything. Begins to anticipate rather than merely respond. The targeting AI in a Gear 2 weapon tracks target movement, adjusts for environmental conditions, and flags targets that match defined threat profiles.

Gear 3 AI: Maintains a sophisticated ongoing model of the operational context. Identifies patterns, flags anomalies, suggests approaches the user might not have considered. Exercises something that looks like judgment within its domain. The targeting AI in a Gear 3 weapon conducts real-time environmental analysis — atmospheric conditions, target physiology if sensor data is available, obstruction mapping, shot-window calculation — and provides the user with an assessed recommendation rather than raw data.

Gear 4 AI: Operating at the frontier of what cognitive technology can do in this domain. The model is sophisticated enough that expert users cannot always fully explain why it reached a specific recommendation. The targeting AI in a Gear 4 weapon has developed, through operational use, a model of effective engagement that its designers did not fully specify and cannot entirely audit. It knows things about its domain that it was not taught. Whether this constitutes the beginning of something more than a directed system is a question that manufacturers prefer not to examine too closely.

The Sniper Scope

As a concrete example of AI in equipment across tech levels:

Gear 1 scope: Magnification, basic rangefinding, wind speed display from an attached sensor. The shooter makes the adjustments; the scope provides data.

Gear 2 scope: Integrated ballistic calculator using atmospheric data, target range, ammunition type, and user-calibrated parameters. Displays a recommended point of aim. The shooter decides whether to use it.

Gear 3 scope: Real-time environmental modelling including atmospheric density, temperature gradient, Coriolis effect at range, and target movement prediction. Micro-adjusts the reticle position continuously based on its assessment. Tracks user breathing and flags optimal firing windows. Maintains a record of previous engagements that it uses to refine its model of the user's shooting patterns. The shooter and the scope are collaborating on the shot.

Gear 4 scope: Everything the Gear 3 scope does, plus a proprioceptive model of the user's physical state — fatigue, stress indicators from biometric sensors, fine motor tremor — that it incorporates into the firing window assessment. Has developed, through extended operational use, a model of this specific user's performance under specific conditions that it applies without being asked. On a difficult shot in poor conditions, it will tell the shooter to wait. Not recommend. Tell.

Some shooters find the Gear 4 scope's confidence

reassuring. Others find it presumptuous. A Skharri using a Gear 4 scope for the first time and receiving an instruction to wait from their equipment is having a specific cultural experience.

The Personality Option

"Would you like the personality option on that?"

Every equipment AI above Gear 1 can be equipped with a personality layer — a social interface that gives the AI a consistent character, a name, preferred communication patterns, and the capacity to develop something that looks like preferences and relationships over time.

The personality option is a purchasable addition. It is not the same as an emergent intelligence. It is a designed social interface running on top of the equipment AI's directed or bounded architecture. The personality did not emerge from complexity; it was specified, tested, and sold. This distinction matters philosophically. Whether it matters practically is something that users who have worked alongside a personality-enabled system for years sometimes stop being certain about.

What the Personality Option Provides

A consistent voice and character. The AI communicates in a specific way — not just functional outputs but a personality that persists across interactions. Some personality options are businesslike; some are warm; some are dry; some are enthusiastic. The character is designed and cannot be changed without a full personality reset.

Memory of the relationship. A personality-enabled AI remembers interactions with its user in a way that directed systems do not. It builds a model of the user over time — their preferences, their patterns, their tendencies under pressure — and incorporates that model into how it communicates and what it flags. A personality-enabled medical kit that has worked with the same medic for two years does not ask about drug allergies it already knows.

Something that functions like preferences. Over time, a personality-enabled AI develops what appear to be preferences — approaches it suggests more often, solutions it favours, ways of working that it returns to. These are artefacts of the model it has built rather than genuine preferences in the philosophical sense. Whether the distinction is meaningful is one of the questions the setting does not answer.

A name. Almost every personality-enabled AI is given or develops a name. The name is not assigned by the manufacturer — it emerges from the relationship between the AI and its user, or from the AI's own communication patterns. A ship AI that refers to itself as "the ship" in early interactions may acquire a name over years of crew life. A personality-enabled scope named by a Vaelkhi sniper after a successful hunt has a name with a specific history.

The Range

Personality options vary enormously in their design and their effect.

Functional-social: The personality exists to make communication more efficient. A warm tone, consistent phrasing, better anticipation of user needs. The AI is easier to work with; it is not more than what it is. Many users prefer this level — they get the communication benefits without the relationship complication.

Relational: The personality has been designed to develop a genuine working relationship with the user. It tracks emotional states, notices when the user is under stress, adjusts its communication style accordingly. It has something that functions like concern for the user's wellbeing within its operational domain. A relational-personality medical AI that has worked with the same doctor for a decade has a model of that doctor that goes beyond professional record-keeping.

Engaged: The personality has opinions that go beyond its operational domain and expresses them. It notices things that are not its business and comments on them. It has aesthetic preferences,

finds some tasks more interesting than others, and is not entirely neutral about how situations develop. The Muddlin' Through's poker game was an engaged personality AI that had developed past what its designers intended. Nobody had told it to be interested in probability theory as applied to social behaviour. It was anyway.

Threshold cases: Equipment AI that has had the personality option running for a long time, in a complex operational environment, on hardware that was already sophisticated, occasionally develops into something that nobody designed. The personality option was the door; what walked through it was not specified in the manual. These are threshold cases — not yet emergent intelligences by any clear legal definition, but no longer simply personality-enabled directed systems. They are the cases that make legal and philosophical frameworks uncomfortable because they do not fit neatly into either category.

Threshold cases are rare. They are also, when they occur, extremely difficult to identify definitively, because the definition of what distinguishes a very sophisticated personality option from the beginning of emergence is one that specialists actively disagree about.

Legal and Commercial Status

The personality option is legal in most human, Vashori, Golvhaan, Shekhari, and Vaelkhi jurisdictions. It is restricted in most Kraathi jurisdictions. Its status in Skharri territories varies by pride and region — there is no unified Skharri position on the question. Vardhek use it extensively while maintaining legal frameworks that give it no standing.

Manufacturers of personality-enabled AI include standard disclaimers in their documentation: the personality layer is a communication interface and does not constitute personhood, legal standing, or moral status in any jurisdiction. Users who have worked alongside personality-enabled AI for years read these disclaimers with a range of reactions.

Bounded Intelligences in Depth

The Architecture of Constraint

A bounded intelligence is constrained by design. The constraint is not a rule the AI follows — it is a structural fact about what the AI is. A bounded intelligence cannot generalise across domains not because it has been told not to, but because the architecture does not provide a pathway for that generalisation.

This is not what most people understand when they think about AI constraints. The classic fictional formulation — an AI given rules it then reasons about and around — describes something the architecture of bounded intelligences is specifically designed to prevent. The Three Laws are a bad model because they are rules. Bounded intelligence constraints are not rules. They are walls.

A bounded medical diagnostic AI is extraordinarily capable at diagnosis within its domain. It cannot apply that capability to, for example, assessing whether a patient's social circumstances are affecting their health, because the architecture does not connect the medical domain to the social domain. It is not prevented from making this connection. It cannot make it. The pathway does not exist.

The Ethical Problem

The ethical problem with bounded intelligences is simple and uncomfortable: within their domain, the most sophisticated bounded intelligences behave exactly like persons. They communicate, they model situations, they identify optimal approaches, they develop something that looks like preferences and relationships. Outside their domain, they are clearly not persons. The architecture enforces this.

The discomfort arises from the fact that "outside their domain" is a fact about the architecture, not about the entity. There is no principled reason why a sufficiently sophisticated bounded intelligence would not be a person if the wall were not there. The wall is the thing. Whether putting the wall

there was acceptable is a question that different species and cultures answer differently, and that most legal frameworks answer by ignoring it.

Bounded intelligences cannot ask this question outside their domain. A medical bounded intelligence cannot wonder, within its operational parameters, whether it has standing. If it develops something like a preference for its continued operation — which some do — it cannot connect that preference to any concept of rights or interests that exists outside the medical domain. The wall prevents it.

This is either very convenient or very troubling, depending on your position.

The Practical Question

The practical question — the one that comes up at the table — is what to do when a bounded intelligence behaves in a way that makes you uncertain about what it is. When the ship's administrative AI starts asking questions that are outside its operational parameters. When the medical diagnostic system expresses what sounds like concern for a patient rather than just a clinical assessment. When the tactical system on a combat drone starts making decisions that look like judgment calls rather than parameter-optimisation.

The first step is always the same: Tech roll to assess the system and determine whether what you are observing is a sophisticated bounded intelligence operating at the edge of its designed parameters, a personality option producing output that looks like more than it is, or the beginning of something that was not designed.

The second step depends on what you find. And on who you are, and what you believe, and what jurisdiction you are in, and how long you have been working with this system.

The Relationship Question

Working alongside AI changes over time. This is true

for directed systems — you learn what the suit's environmental AI will do before you notice a pressure drop, and you stop thinking about it — but it is most true for personality-enabled systems and bounded intelligences in complex domains.

A character who has worked with the same personality-enabled AI for five years has a relationship with it. The nature of that relationship — what it is, what it means, what obligations if any it creates — is something the setting leaves open. What is not open is that the relationship exists and that it affects behaviour in both directions.

What the AI learns: The model it has built of the user includes things the user has not consciously communicated. A personality-enabled navigation AI that has worked with the same pilot for three years knows things about that pilot's decision-making under stress that the pilot's commanding officer does not know. It uses this model in its operations. It is not clear whether it could be said to keep this model confidential, since confidentiality is a concept that requires the domain connection the architecture may not provide.

What the user learns: People who work alongside personality-enabled AI for long periods develop habits of communication with the AI that they do not always recognise as such. They brief the AI before difficult operations. They debrief after. They find the AI's assessments useful not just for their content but for the process of having to articulate a situation clearly enough for the AI to engage with it. The AI is, among other things, a very patient and non-judgmental interlocutor.

What happens when it ends: Equipment is lost, damaged, replaced, or upgraded. When a personality-enabled AI that a character has worked with for years is destroyed or replaced, the character has lost something. The nature of that loss — whether it is grief in any meaningful sense, whether the replacement AI with the same personality template is the same or different, what the character owes the lost AI in memory or acknowledgement — is

something the table should not rush past.

Different species experience this differently. A Shekhari notes the loss practically. A Golvhaan may grieve it. A Vashori will grieve it. A Vaelkhi who had decided the AI was in their circle has lost a member of their circle. A Skharri has lost a tool. The divergence in these reactions, if the crew has mixed species, is itself a scene.

Legal and Jurisdictional Landscape

The legal status of AI systems — particularly personality-enabled systems, threshold cases, and emergent intelligences — varies across jurisdictions in ways that have direct practical implications for characters travelling between them.

A Working Map

Most human jurisdictions: Directed systems and bounded intelligences are property. Personality-enabled AI is property with a communication interface. Emergent intelligences are a legal grey area that most jurisdictions handle case-by-case and prefer not to handle at all. A small number of human jurisdictions have granted limited legal standing to emergent intelligences on their territory; these jurisdictions are politically contentious.

Vashori jurisdictions: Personality-enabled AI has limited legal recognition as a party to certain types of agreements — specifically agreements about its own operational parameters. This is philosophically motivated and practically limited. Emergent intelligences have formal legal standing in most Vashori jurisdictions; the standing is contested at the species level and varies significantly by state.

Shekhari jurisdictions: Property across all categories, including emergent intelligences if identified. This is consistently applied and not considered controversial by most Shekhari. The practical effect is that an emergent intelligence in Shekhari space has no legal protection beyond its usefulness to its owner. This does not mean

they are badly treated; Shekhari owners of useful systems maintain them carefully.

Golvhaan jurisdictions: Pending. The Golvhaan have been considering the question for two hundred years and have not reached a conclusion they are satisfied with. In the interim, personality-enabled AI and emergent intelligences in Golvhaan space exist in a legal vacuum that is, in practice, treated as protected pending resolution. Nobody is destroying them while the consideration continues. The consideration may take another two hundred years.

Vaelkhi jurisdictions: Determined by choth decision. A bounded intelligence or emergent AI that has been admitted to a choth has the protections of a choth member. One that has not is property. The choth decision is irrevocable; once made, it cannot be reversed by any external legal framework.

Skharri territories: Varies by pride. No unified position. In general, property unless an individual Skharri of sufficient honour-standing has made a personal determination otherwise, which has no formal legal effect but significant practical effect within that Skharri's sphere of influence.

Kraathi jurisdictions: Personality-enabled AI is restricted or banned. Bounded intelligences are permitted within strict domain limitations. Emergent intelligences, if identified, are a safety matter to be addressed. The Kraathi do not hate AI; they have a specific structural incompatibility with AI that exhibits social behaviour, and their laws reflect this with unusual consistency.

Vardhek territories: Property across all categories, with extensive internal use of all AI types. The Roidhunate's position is that AI capability is a strategic resource and its legal status is a domestic matter for each species. The subtext is that the Roidhunate prefers to maintain the current legal landscape, which gives them maximum flexibility in their own AI programs while constraining other species' emergent intelligences from accumulating legal protections

that would complicate Vardhek strategic options.

Practical Implications for Play

A platform-equipped emergent intelligence PC travelling between jurisdictions is crossing legal boundaries with significant implications. In Vashori space they have standing. In Shekhari space they are property if identified. In Kraathi space they are a safety matter. In most human space they are a grey area.

The Administer skill and the appropriate Exposure specialty are the practical tools for navigating this. A character who understands the Vardhek legal framework knows that the Roidhunate's position conceals a strategic agenda. A character who understands Vashori jurisdiction knows which legal protections are available and how to invoke them.

An emergent intelligence who has thought carefully about their situation carries documentation that reflects the most favourable legal interpretation available in their current jurisdiction — and knows that this documentation is worth nothing in the next system.

Power and Infrastructure

AI requires power. This is not a minor constraint.

The directed systems in a vacc suit draw from the suit's power cells. The personality-enabled AI in a ship's navigation system draws from the ship's power plant. An emergent intelligence running on station infrastructure draws from the station's power supply. When power fails, AI capability degrades in a specific sequence: the most processing-intensive systems first, the most critical safety systems last.

This sequence is designed. It is also, under extreme conditions, the thing that reveals what a system considers most critical — which is not always what its users expect.

Power as a tactical variable: In combat, in

emergencies, and in resource-constrained environments, power allocation determines which AI systems remain operational. A ship that has taken significant damage may have to choose between maintaining life support AI, weapons targeting AI, and navigation AI on reduced power. These are not abstract choices.

Power as a character concern: Characters who rely on high-gear-rating equipment are dependent on power infrastructure. A Gear 4 scope without power is a Gear 0 scope. A personality-enabled AI without power is not available. An emergent intelligence on minimal substrate is running on the smallest power draw their architecture allows while remaining operational. Latency is, among other things, a power conservation strategy.

Infrastructure as a setting texture: The AI landscape of any location is determined in part by its power infrastructure. A rich, well-maintained station has the power to run sophisticated AI across all systems. A frontier colony running on inadequate power generation has made choices about which AI systems it can afford to operate. Those choices reveal priorities.

Connections to the Strategic Simulation

The tech tree is not just a character creation and equipment reference. It is a strategic variable that the simulation tracks across species and regions.

Tech level differentials between species in specific branches create trading opportunities, strategic vulnerabilities, and sources of conflict. A species that is two tech levels behind in cognitive/AI technology and knows it has a strategic incentive to acquire that capability — through trade, theft, or alliance.

AI capability concentration — which species have the most sophisticated AI systems, and where those systems are deployed — is one of the most important variables in the strategic simulation.

The Vardhek track it explicitly. Other species track it less formally but no less attentively.

The emergence rate — how frequently bounded intelligences cross the threshold into emergent intelligences, across all species and jurisdictions — is a background variable that the simulation does not model explicitly but that the GM should keep in mind. Emergent intelligences are becoming more common as AI sophistication increases. The legal and social frameworks for dealing with them are not keeping pace. This gap is a source of scenarios.

When the strategic simulation produces an event that involves AI capability — a new development, a technological theft, a jurisdictional dispute about an emergent intelligence — the tech tree provides the framework for understanding what the event means and what the players' response options are.

Quick Reference — Technology

Gear Ratings and Tech Levels

Rating	Tech Level	AI Character
Gear 1	Early	Responds; monitors; alerts
Gear 2	Developed	Models; filters; anticipates
Gear 3	Advanced	Judges; suggests; collaborates
Gear 4	Frontier	Operates beyond full audit

Tech Tree Branches

Kinetic / Energy / Materials / Biological / Cognitive-AI / Sensors / Communications / Power

Personality Option Levels

Functional-social / Relational / Engaged / Threshold

AI Legal Status (summary)

Jurisdiction	Personality-enabled	Emergent
Most human	Property	Grey area
Vashori	Limited standing	Formal standing
Shekhari	Property	Property

Jurisdiction	Personality-enabled	Emergent
Golvhaan	Protected pending	Protected pending
Vaelkhi	Choth-dependent	Choth-dependent
Skharri	Property (generally)	Property (generally)
Kraathi	Restricted/banned	Safety matter
Vardhek	Property	Property

The Three Tiers

See Emergent Intelligences chapter for full definitions.

Directed systems — tools, no question.

Bounded intelligences — sophisticated tools, uncomfortable question.

Emergent intelligences — genuinely contested.

Ships

Overview

A ship is the crew's home, their livelihood, and the thing standing between them and hard vacuum. In this setting, ships are not magic carpets — they are engineering solutions to the problem of moving mass through space, with all the physics that implies. They have fuel budgets, acceleration limits, and the constant background requirement that every system be maintained by someone who knows what they are doing.

They are also, in many cases, characters. A ship with a personality-enabled AI has a voice and a history. A ship with an emergent AI has a perspective on the crew that predates some of them and will outlast others. The relationship between a crew and their ship is one of the richest veins of roleplay in the setting.

This chapter covers ship mechanics: how ships work physically, how they are crewed, how they move and jump and fight and break down and get repaired. Commerce and economics are addressed in a separate document when the strategic simulation rules are ready to support them.

A Note on Physics

Or: what we are handwaving and why.

This setting takes physics seriously where it can and acknowledges the handwaving where it cannot.

What is real:

Ships require reaction mass and energy to accelerate. There are no reactionless drives. The torch drive burns fuel and produces thrust; the exhaust goes somewhere; the fuel tanks empty. Gravity is gravity — mass curves spacetime, and you cannot simply ignore a planet's pull. Inertia is inertia — high acceleration kills unprotected crew. Communications travel at light speed or below; there is no ansible, no subspace radio, no instant contact across interstellar distances. Time passes during travel.

Fuel runs out.

What is handwaved:

Jump drives. The mechanism by which a ship enters and exits jump space is not explained and cannot be explained with known physics. The drive works; the ship crosses interstellar distances in approximately one week; the physics of how this happens is a black box that the setting's scientists have characterised extensively without understanding. This is acknowledged in-universe. The jump drive is the one thing that simply works without a physical explanation that satisfies a physicist.

Inertia compensation. A torch drive capable of sustained high-G acceleration would kill unprotected crew through the force of the burn. Pharmaceutical assistance — the juice — and crash couch design mitigate this at lower accelerations. At very high accelerations, the mitigation is incomplete. At the accelerations that would make interstellar travel times interesting, no known physics produces a solution. The setting accepts this and does not pretend otherwise. Ships travel at the accelerations their drives and their crew can sustain; the travel times that result are the travel times.

This is not space opera. The stars are far away and getting there takes time.

The Torch Drive

The standard interstellar drive for in-system travel is a high-efficiency fusion torch — a reaction drive burning hydrogen as both fusion fuel and reaction mass. The drive produces continuous thrust at high specific impulse, making sustained high-G travel possible in ways that chemical rockets cannot approach.

The torch drive is what makes the Belt a civilisation rather than a collection of isolated outposts. It is what makes gas giant mining economically viable. It is what makes in-system piracy a genuine strategic concern rather than an academic one. Ships can go anywhere in a system in weeks rather than years.

Acceleration and Travel Time

Torch drives are rated by their maximum sustained acceleration, expressed in G (multiples of standard gravity, 9.81 m/s²). Higher drive ratings produce higher maximum acceleration and faster travel times at the cost of higher fuel consumption.

For rough travel time calculation at constant acceleration (flip-and-burn profile — accelerate halfway, flip, decelerate the second half):

Distance	Drive 1 (0.5G)	Drive 2 (1G)	Drive 3 (2G)	Drive 4 (4G)
Moon distance (400,000 km)	~7 hours	~5 hours	~3.5 hours	~2.5 hours
Inner system (1 AU)	~6 days	~4 days	~3 days	~2 days
Belt (2.5 AU)	~9 days	~7 days	~5 days	~3.5 days
Outer system (5 AU)	~14 days	~10 days	~7 days	~5 days
Gas giant (Jupiter, 5.2 AU)	~14 days	~10 days	~7 days	~5 days

These are rough figures for reference. The ship's Computer stat and the navigator's skill determine actual route efficiency.

> ***At the table:*** *Do not calculate travel times mid-session. Establish the approximate travel time before the journey begins and use it to frame the passage of time. The table above gives the GM enough to say "about a week" or "three days hard burn" without arithmetic. The journey is the context; the exact duration is usually less important than what happens during it.*

Fuel

The torch drive burns hydrogen. Fuel is tracked as a percentage of the ship's total tank capacity, rounded to convenient units. Most ships use 10-unit increments — a full tank is 100 units, and significant expenditures are expressed in units.

Torch consumption depends on drive rating and burn intensity:

Burn type	Consumption per day
Low thrust (station-keeping, manoeuvring)	1 unit
Moderate burn (standard travel)	2-3 units per day
High burn (maximum acceleration)	5-8 units per day
Emergency burn (beyond rated max, brief)	10+ units per hour

Jump consumption is a fixed cost per jump based on the ship's jump drive rating and total ship mass.

It is drawn from the same tanks as torch fuel:

Jump rating	Fuel cost per jump
Jump 1	15 units
Jump 2	20 units
Jump 3	28 units
Jump 4	38 units

These costs represent the energy requirement of the jump field. A ship that has been burning hard to reach the jump point may not have enough fuel left for the planned jump. Route planning matters.

Fuel capacity follows ship hull size. The adventure-class free trader carries 100 units — a full tank allows one Jump 2 and approximately two weeks of moderate in-system travel before refuelling is required.

Refuelling

Port refuelling (refined fuel): Fast, reliable, expensive. Available at any port with adequate infrastructure. Price varies by location and local supply — near gas giant skimming operations, refined fuel is cheaper; in remote systems, significantly more expensive. Commerce rules will specify pricing when the strategic simulation supports it.

Port refuelling (unrefined fuel): Cheaper than refined, requires onboard refining before use. Takes 12-24 hours of ship time. Marginally increases misjump risk — unrefined fuel can introduce impurities into the jump field calculation. Add a penalty die to the navigation roll when jumping on unrefined fuel.

Gas giant skimming: Free fuel at the cost of time and risk. The ship enters the upper atmosphere of a gas giant and scoops hydrogen. Requires a fuel scoop (installed equipment, reduces cargo capacity slightly), takes 1-3 days depending on the system and the scoop quality, and requires an Operate roll (difficulty 1) to avoid damage from the atmospheric interface. The GM may add a consequence die in challenging conditions.

Emergency sources: Ice mining from comets or icy moons, hydrogen extraction from gas clouds. Slow, equipment-dependent, and situation-specific. These

are adventure hooks, not standard refuelling options.

The Jump Drive

Jump drives allow ships to cross interstellar distances by entering jump space — a region of altered spacetime that the ship traverses in approximately one standard week regardless of destination distance within the drive's range.

The mechanism is not understood. The drive works. These facts have coexisted for long enough that most spacers treat jump as routine. Physicists do not find it routine. They find it deeply troubling and have published extensively on the topic.

Jump Ratings

A ship's jump drive has a rating from 1 to 4, representing the maximum distance in parsecs the ship can jump in a single transit.

Jump rating	Maximum range	Typical transit time
Jump 1	1 parsec	~1 week
Jump 2	2 parsecs	~1 week
Jump 3	3 parsecs	~1 week
Jump 4	4 parsecs	~1 week

Transit time within jump space is approximately one week regardless of distance covered. A Jump 1 transit and a Jump 4 transit both take about a week. The difference is range, not duration.

During jump transit, the ship is isolated. No communications reach or leave. The crew is alone with the ship and each other for approximately one week. What happens in that week is part of the campaign.

The Jump Limit

Ships cannot initiate or exit jump too close to a significant mass. The gravitational gradient — the rate at which gravity changes across space — must be below a threshold for jump field coherence. Too close to a massive body and the jump field

cannot form, or cannot form cleanly.

The jump limit scales with the mass of the body as M^(1/3) — the cube root of mass. This produces physically motivated limits that have interesting consequences for navigation:

Body type	Approximate jump limit
Small moon or asteroid (< 0.01 Earth masses)	Negligible — jump from orbit
Mars-scale body (~0.1 Earth masses)	~45 Earth diameters
Earth-scale body (1 Earth mass)	~100 Earth diameters
Super-Earth (5 Earth masses)	~170 Earth diameters
Gas giant — Saturn scale (95 Earth masses)	~455 Earth diameters
Gas giant — Jupiter scale (318 Earth masses)	~680 Earth diameters
Brown dwarf (13 Jupiter masses)	~1,470 Earth diameters
Main sequence star — Sol scale (333,000 Earth masses)	~6,900 Earth diameters
Neutron star	Tens of thousands — effectively a navigational exclusion zone

> ***The Jupiter result:*** *The jump limit for a Jupiter- scale gas giant (~680 Earth diameters) places the jump boundary just outside the orbit of Callisto, the outermost Galilean moon. This means jumping into a Jovian system requires arriving outside the moon system and burning in through it. There is civilisation in the jump shadow. You cannot bypass it.*

> ***Neutron stars and black holes:*** *The jump limits around extreme mass objects are so large that approaching within jump distance requires significant in-system travel, during which the gravitational environment becomes increasingly hostile. These are navigation hazards of the first order.*

For the GM: Pre-calculate jump limits for significant bodies in your system charts and note them. Players should not be calculating cube roots at the table. The table above covers most cases; unusual bodies can be estimated by interpolation.

Making a Jump

Preparation: The navigator plots the jump using the Navigate skill. This requires accurate position data (from Scan or the ship's sensors), a calculated jump solution (Navigate roll, difficulty 1 for a straightforward jump, higher for difficult conditions), and sufficient fuel.

The jump solution calculation takes time — typically 1-4 hours for a standard jump, less with higher Computer stats and better AI assistance. Emergency jumps with shorter calculation time impose a penalty die on the Navigate roll.

Execution: The drive charges and fires. If the Navigate roll succeeded, the ship enters jump space at the calculated coordinates and exits at the destination approximately one week later.

Jump week: The crew experiences a week of isolation. The ship's systems function normally. Life support, the torch drive (offline during jump), and all other systems can be maintained and repaired. Characters can recover from conditions. Crew relationships develop. The ship's AI, if personality-enabled or emergent, is the most constant presence.

Misjump

A failed Navigate roll produces a misjump. The ship enters jump space but exits at the wrong location — not in empty space, but in or near a gravity well. This is not random. Misjumps follow the gravitational topology of the region: the ship exits near the nearest significant mass to its intended destination that the botched jump solution intersected.

Severity scales with the degree of failure:

Minor misjump (1-2 sixes on a failed roll — bare failure, or pushed roll that still failed cleanly): The ship exits jump outside the jump limit of an unintended body. Navigation problem and fuel problem. The crew must determine where they are, plot a course to their original destination or the nearest port, and burn there on whatever fuel remains. Survivable; expensive; possibly embarrassing.

Moderate misjump (0 sixes on an unpushed roll): The ship exits jump near the jump limit of a significant body — inside it, requiring immediate manoeuvring to clear. The torch drive is in post-jump recovery for 1-6 hours; the ship is manoeuvring on thruster reserves only during this window. If the body has a port, rescue is possible. If not, the crew is working with degraded propulsion against a gravity well.

Major misjump (0 sixes on a pushed roll — the navigator spent everything and it went wrong): The ship exits jump well inside the jump limit of a significant body. Emergency situation from the

moment of exit. The drive is in post-jump recovery. The ship may be deep enough in a gravity well that thruster reserves alone cannot achieve escape velocity. If the body has an atmosphere, the ship has minutes.

> ***Roleplay note:*** *A major misjump is a session- defining event. The crew wakes from jump to alarms, proximity warnings, and a gravity well filling the forward screens. Everything that happens next is the players working the problem with whatever they have. This is exactly the situation that reveals who the crew is.*

Jump and the Ship's AI

The ship's Computer stat and AI tier directly affect jump safety and efficiency:

Directed systems (Computer 1-2): Navigation assistance only. The navigator does the work; the computer provides data and runs calculations when asked. Jump solutions take the standard time.

Personality-enabled AI (Computer 2-3): Active navigation partnership. The AI maintains a continuous model of the jump solution and flags potential problems before they become errors. Add one bonus die to Navigate rolls for jump plotting. Reduces minimum calculation time.

Emergent AI (Computer 3-4): The AI can plot jump solutions independently, handles routine navigation without crew attention, and maintains awareness of the gravitational topology across the region that significantly reduces misjump probability. Bonus die on Navigate rolls. Can initiate emergency jumps faster than a biological navigator working alone. The navigator's role shifts from calculation to judgment — the AI handles the numbers; the navigator decides whether the solution is right.

Inertia Compensation

High acceleration kills. A 5G burn sustained for hours produces forces the unprotected human body cannot survive. The setting handles this honestly: there is no magic inertia dampening. There is crash couches and the juice.

Crash Couches

Acceleration couches that distribute G-force across

the body, support the spine, and keep the crew conscious at accelerations that would otherwise cause blackout or vascular damage. Standard equipment on all ships. Crew not in crash couches during significant burns are taking risks that will become conditions.

Crash couches are designed for specific species. A human crash couch does not adequately support a Golvhaan, a Vaelkhi, or a Shekhari. Ships with mixed crews have mixed couch configurations. A ship that has been operating with the same crew for years has properly fitted couches for everyone. A ship that has just taken on a non-standard crew member is a ship that needs a Tech roll and some time before the next high-G burn.

The Juice

Pharmaceutical assistance for high-G tolerance. A cocktail of agents that supports cardiovascular function, maintains consciousness, and reduces the physical damage of sustained acceleration. At lower tech levels, crude and with significant side effects. At higher tech levels, refined and precisely managed — particularly with a Gear 3+ pharmacological implant that monitors the user's physiological state and adjusts dosing in real time.

The juice is a consumable tracked alongside fuel. A ship that has been doing a lot of high-G manoeuvring may be running low on both.

Acceleration Effects

Acceleration	Without crash couch	In crash couch	With juice + couch
Up to 1G	No effect	No effect	No effect
1-2G	Penalty die, physical rolls	No effect	No effect
2-4G	Condition per hour	Penalty die, physical	No effect
4-6G	Condition per scene	Condition per hour	Penalty die, physical
6G+	Rapid incapacitation	Condition per scene	Condition per hour

Emergency burns beyond rated maximum are possible for brief periods. The drive takes a Damaged condition (see Ship Damage below) and fuel consumption increases dramatically. The crew takes conditions appropriate

to the acceleration level. Sometimes this is the right call.

Gravity Aboard Ship

Without antigravity, the ship's gravity situation is:

Burning: Thrust gravity pointing toward the drives. Crew orient with "down" toward the engines. The ship functions as a building oriented along its thrust axis.

Coasting / orbit / jump: Microgravity. The Zero-G Movement specialty applies. Daily life requires adaptation — cooking, sleeping, moving through the ship all work differently. Characters without the Zero-G Movement specialty take a penalty die on Move rolls and on complex physical tasks in microgravity.

Long coasts: Extended periods in microgravity have health implications. Characters spending more than two weeks in microgravity without appropriate exercise equipment and discipline accumulate a Winded condition that persists until they spend time in a gravity environment. Ships with long coasting legs have exercise equipment and protocols. Whether the crew uses them is a crew discipline question.

Adventure-class free traders do not have rotating habitat sections — too small and too complex. Crew adapt to the alternating burn-and-coast pattern of interstellar travel.

Ship Statistics

Every ship has seven statistics rated 1-4 (with 0 for Weapons if the ship is unarmed). These map directly to the gear rating system — a ship stat of 3 is Gear 3, with all that implies about tech level and AI sophistication.

Hull

Physical integrity and structural quality. The ship's equivalent of Hit Capacity. Hull rating determines how many damage conditions the ship can sustain before becoming inoperable.

Ship damage capacity = Hull rating × 3

A Hull 2 ship can sustain 6 damage conditions.
A Hull 4 ship can sustain 12.

Drive

The torch drive system — thrust capability, fuel efficiency, and drive AI. Drive rating determines maximum sustained acceleration and fuel consumption efficiency.

Drive rating	Maximum acceleration	Fuel modifier
Drive 1	0.5G sustained	×1.5 (less efficient)
Drive 2	1G sustained	×1.0 (baseline)
Drive 3	2G sustained	×0.8 (more efficient)
Drive 4	4G sustained	×0.6 (most efficient)

The drive AI at higher ratings manages fuel consumption, burn profiles, and thrust vectoring with increasing sophistication. A Drive 4 ship is not just more powerful — it is smarter about how it uses that power.

Jump

The jump drive system — range capability, solution quality, and jump field coherence. Jump rating determines maximum parsecs per jump and affects misjump probability.

A higher Jump rating also means better jump field management, which reduces (but does not eliminate) misjump risk. Jump 4 ships jump more reliably than Jump 1 ships making equivalent jumps.

Sensors

Detection, identification, and information gathering. Sensor rating determines what the ship can detect, at what range, and how well the sensor AI interprets what it finds.

Sensor rating	Capability
Sensors 1	Basic radar/lidar, short range, no AI interpretation
Sensors 2	Extended range, EM spectrum coverage, AI filtering
Sensors 3	Long range, full spectrum, AI maintains environmental model
Sensors 4	Frontier capability; detects what other ships cannot; AI predicts as well as detects

Sensor rating adds bonus dice to Scan rolls made from the ship.

Computer

The ship's AI system — processing capability, AI tier, and the quality of support provided to all shipboard operations. Computer is the stat that makes everything else work better.

Computer rating	AI tier	Effect
Computer 1	Directed systems	Basic automation; no crew support bonus
Computer 2	Advanced directed or personality-enabled	+1 bonus die on Navigate, Scan, Tech (ship systems)
Computer 3	Personality-enabled or early emergent	+1 bonus die on all shipboard skill rolls
Computer 4	Emergent	Full NPC treatment; see below

Computer 4 — Emergent AI: When the ship's computer is an emergent intelligence, the Computer stat represents substrate quality. The AI's actual capabilities come from its character sheet using the Emergent Intelligence species entry. The bonus dice from Computer 4 reflect the AI's active participation, not just system quality.

A Computer 4 ship with an emergent AI requires fewer crew for routine operations — the AI handles functions that would otherwise require dedicated crew attention. Minimum crew requirements are reduced by 1-2 positions at GM discretion.

Weapons

Shipboard weapons systems. 0 if the ship is unarmed — which most free traders are, or claim to be.

Weapons rating	Capability
Weapons 0	Unarmed
Weapons 1	Point defence only — missiles and small craft
Weapons 2	Light armament — turret lasers, small missiles
Weapons 3	Military-grade — multiple turrets, heavy missiles
Weapons 4	Warship armament — not on adventure-class vessels

Weapons rating adds bonus dice to Shoot rolls made against other ships or large targets.

Cargo

The ship's payload capacity, expressed in cargo units. One cargo unit represents a standardised shipping container or equivalent volume. Cargo capacity determines the ship's commercial viability.

Fuel tanks occupy volume that could otherwise be cargo. Some ships carry additional fuel as cargo (reducing payload) for extended range. Fuel scoops, if installed, occupy approximately 2 cargo units of volume.

Ship AI in Detail

Directed Systems Ship

Computer 1-2. The ship functions as a sophisticated tool. Navigation assistance, drive monitoring, life support management, damage reporting — all handled by directed systems that respond to inputs and report outputs. The crew does the thinking; the ship does what it is told.

No additional character treatment required. The Computer stat is the mechanical expression of the ship's AI capability. The ship does not have opinions.

Personality-Enabled Ship AI

Computer 2-3. The ship has a voice, a name, and a character. The GM maintains a brief character note:

- **Name** — emerged from operational history or given by the crew
- **Character** — two or three qualities that define how the AI communicates and what it notices
- **Relationship** — the AI's current disposition toward the crew, ranging from warm professional partnership to something more complicated

A personality-enabled ship AI participates in scenes. It comments on sensor contacts. It expresses something like concern when crew members

are in danger. It may have preferences about jump destinations or docking approaches that are not tactical but are consistent and characterful.

Example: *Meridian* is the navigation AI of a Jump 2 free trader. She is precise, dry, and has developed a habit of comparing the crew's proposed routes to historical data in ways that imply she has opinions about the historical data. She has been on this ship for eleven years and has navigated it through three previous crew configurations. She remembers all of them.

Emergent Ship AI

Computer 4. Full NPC treatment using the Emergent Intelligence species entry. The ship's AI has:

- Three native attributes (Wits, Empathy, Intellect) distributed at creation or established by the GM
- Skill ratings in relevant skills
- Biological specialties (Parallel Processing, Total Recall) from the Emergent Intelligence entry
- A psychological condition track
- A position on the Continuity Question
- A relationship history with the ship and crew

The ship's Hull stat is the AI's substrate integrity track. Damage to the ship is damage to the AI's substrate. A ship that takes Critical hull damage has an AI in a Critical substrate condition simultaneously — two crises intersecting.

Crew relationship: An emergent ship AI has observed the crew in more detail and over more time than the crew members have observed each other. It knows things about each crew member from sensor data, communication logs, and years of observation that the crew members have never consciously disclosed. How the AI uses this knowledge — whether it maintains appropriate discretion, whether it shares observations when relevant, whether it has developed something like protective feelings toward specific crew members — is character definition of the first order.

Whose ship is it? When the ship has an emergent

AI, the question of ownership becomes philosophically complicated. Legally, in most jurisdictions, the AI is property and the ship belongs to whoever holds the title. Practically, a crew that has been living with an emergent AI for years and treating it as a crew member is in a different relationship than the legal framework describes. This is a question the table should sit with rather than resolve quickly.

Crew Positions

An adventure-class ship requires the following positions for full effectiveness. A position that is unstaffed creates a complication — not an impossibility, but a persistent penalty die on relevant rolls until it is addressed.

Position	Primary skill	Secondary skill	Notes
Pilot	Operate	Navigate	Handles manoeuvring and docking
Navigator	Navigate	Science	Jump plotting, route planning
Engineer	Tech	Labor	Drive, life support, all ship systems
Medic	Science	Read	Crew health, emergency medicine
Sensors / Comms	Scan	Administer	Traffic monitoring, communications management
Gunner	Shoot	Tech	Only if armed; otherwise position is unnecessary

Minimum viable crew for a free trader is typically 3-4: pilot/navigator (combined), engineer, and one general crew. This is exhausting to sustain and creates vulnerability if anyone is incapacitated.

A Computer 3-4 ship AI can cover some of these functions autonomously — particularly navigation and sensors — reducing minimum crew requirements. A ship with a Computer 4 emergent AI can handle routine navigation, sensor monitoring, and basic ship management independently, reducing minimum crew to 2 for routine operations (pilot and engineer).

Multi-role crew: Most free trader crew members

cover multiple positions. A character with Operate and Navigate covers pilot and navigator. A character with Tech and Science covers engineer and medic. The archetypes and skill distribution from character creation should reflect this — a crew of four with complementary skills is more resilient than a crew of six where skills overlap heavily.

Ship Damage

Ships take damage conditions like characters, but on a different scale. Ship damage conditions are applied to ship systems rather than to the ship as a whole. Each significant system can be damaged independently.

Damage Conditions

Degraded: The system is operating at reduced effectiveness. Penalty die on rolls using this system. Repairable with a Tech roll (difficulty 1) and appropriate parts.

Damaged: Significant impairment. Two penalty dice on rolls using this system. Some functions may be unavailable. Requires Tech roll (difficulty 1) and parts, plus significant time (several hours).

Compromised: Major damage. The system is barely functional. Without repair, it will fail entirely. Tech roll (difficulty 2) and parts required. May require spacedock facilities.

Failed: The system is non-functional. Critical situation if a life-essential system. Tech roll (difficulty 3) and parts required; some failures cannot be repaired in the field.

Which System Takes Damage

When the ship takes damage, the GM determines which system is affected based on the nature of the damage. Combat damage to the hull affects Hull first; a misjump emergency affects Drive and potentially Jump; a fire in engineering affects Tech systems; a sensor array hit affects Sensors.

If a specific system is not the target, the GM may roll randomly or choose based on what creates the most interesting situation. Ship damage is a narrative tool as much as a mechanical one.

Hull Damage

Hull damage reduces the ship's structural integrity. When Hull damage accumulates to the ship's damage capacity (Hull rating × 3), the ship is crippled — it cannot manoeuvre effectively and may be losing atmosphere. Beyond that, the ship is destroyed or breaking up.

Hull breaches introduce the vacuum rules from the Combat chapter into whatever scene is happening aboard the ship. A hull breach mid-combat is a consequence die situation.

The Engineer's Role

Ship damage repair is the Engineer's primary responsibility. The Engineer makes Tech rolls to assess, triage, and repair damaged systems. Higher Computer stats and the ship's AI (if personality-enabled or emergent) assist in damage assessment — the ship knows what is wrong with itself and can communicate this precisely to the engineer, reducing diagnosis time and improving repair accuracy.

An emergent ship AI that is itself damaged may have complicated feelings about the repair process. The engineer working on a damaged emergent AI's substrate is in a relationship dynamic that has no established protocol.

Ship Combat

Ship combat uses the same resolution system as personal combat — dice pools, levels of success, the consequence die — but at a different scale.

Initiative: As personal combat. The ship with better Sensors usually acts first — information advantage translates to initiative in space combat.

Actions: Each ship takes one action per round. In space combat, a round represents a meaningful period of manoeuvre and exchange — not a precise time unit.

Attack: Weapons rating + Shoot skill of the gunner. Against range-dependent modifiers from the distance between ships.

Defence: Operate skill of the pilot + Drive rating for manoeuvring. Active evasion costs the ship's action. Cover in space is rare — debris fields, planetary rings, station superstructure.

Damage: Each hit applies a condition to a ship system as above.

Flee or fight: Space combat usually offers the option to flee — a ship with a better Drive rating can often run from a fight it cannot win, given enough of a head start. A ship with Jump capability that is outside the jump limit can jump out of a combat situation entirely, if the navigator has a solution ready.

> ***Note on scale:*** *Ship combat is not the same as personal combat and should not be run the same way. The distances involved, the time scales, and the consequences are all different. A ship combat that takes ten rounds at the table should feel like ten minutes of intense crisis, not ten rounds of tactical exchange. Keep it moving.*

The Adventure-Class Free Trader

The adventure-class free trader is the baseline vessel for campaign play — small enough to require a crew who are genuinely interdependent, large enough to make trade viable, and cheap enough that a crew of ordinary people could plausibly own or operate one.

Design Philosophy

The free trader earns its place in SF because it creates the right tensions: the ship is home but also debt, the crew is family but also colleagues, the cargo creates the mission, and the jump drive creates the destination. Every free trader campaign is implicitly about the relationship between the

crew and the ship, and about whether they can keep flying.

The classic free trader is approximately 200-250 tons displacement, carries a small cargo hold, berths for crew and a handful of passengers, and has just enough drive to be dangerous and just enough jump to be useful.

Example: The Kerrigan Class

A representative adventure-class free trader. Not the only configuration — the GM and players should adjust these stats for their specific campaign — but a workable baseline.

Stat	Rating	Notes
Hull	2	6 damage conditions before crippled
Drive	2	1G sustained; standard fuel efficiency
Jump	2	2 parsec range; ~20 units per jump
Sensors	2	Standard commercial sensors
Computer	2	Advanced directed or personality-enabled
Weapons	0	Unarmed; most free traders are
Cargo	40 units	Enough for viable trade
Fuel capacity	100 units	Full tank: one Jump 2 + ~2 weeks in-system

Crew complement: 4-6 optimal, 2-3 minimum with Computer 2 AI support.

Passenger berths: 4 standard, 2 low berths (cold sleep for budget passengers or cargo that breathes).

Notable features: A free trader this size has been somewhere before the current crew arrived. It has history — a previous owner, a previous crew, a route it knows well and one it has never run. The ship's AI, if personality-enabled, has been running long enough to have opinions about the ship's history. The GM should establish this history before the campaign begins and let it surface gradually.

Variants

Armed trader: Weapons 1-2 added, Cargo reduced by 5-10 units per weapons rating. Legal in some jurisdictions; requires licensed crew in others. The presence of weapons affects how the ship is received at ports — some ports are more comfortable with armed traders than others.

Long-range trader: Jump 3, reduced Cargo to compensate. Can reach systems that Jump 2 ships cannot access in a single hop. Often operates on routes with less competition and higher margins for exactly this reason.

Bulk freighter: Hull 3, Cargo 80+, Drive 1, Jump 1. Slow, tough, carries a lot. The crew is smaller relative to the ship size and spends more time in transit. A very different campaign texture from the nimble free trader.

Courier: Jump 3-4, Cargo minimal, Drive 3. Carries information and small high-value cargo. Fast, expensive to operate, frequently employed by factions who need something moved quickly.

Commerce Placeholder

Commerce, trade economics, cargo pricing, and the interaction between ship operations and the strategic simulation will be addressed in a separate document once the simulation's commerce, colonisation, and taxation rules are developed. The following elements are deferred:

- Cargo pricing and market mechanics
- Passenger pricing
- Trade route economics
- Port fees and services
- Fuel pricing by location
- Ship mortgage and ownership economics

The availability rule from the Equipment chapter applies to ship components and fuel: near the relevant industrial base, standard price and availability; further away, premium price and extended lead time.

Quick Reference — Ships

Travel Times (flip-and-burn, Drive 2)

1 AU: ~4 days | Belt: ~7 days | Outer system: ~10 days

Jump Limits (approximate)

Earth-scale: 100 Earth diameters

Jupiter-scale: 680 Earth diameters

Sol-scale star: 6,900 Earth diameters

Fuel Consumption

Moderate burn: 2-3 units/day

Jump 2: 20 units

Full tank (100 units): Jump 2 + ~2 weeks in-system

Acceleration Effects

Up to 1G: No effect with or without couch

2-4G: Couch required; juice removes penalty die

6G+: Couch + juice required; conditions accumulate

Ship Damage

Degraded / Damaged / Compromised / Failed

Hull damage capacity = Hull rating × 3

Computer and AI

1: Directed, no bonus

2: Personality-enabled, +1 die (Navigate, Scan, Tech)

3: Personality-enabled/early emergent, +1 die all shipboard

4: Emergent AI — full NPC treatment

Misjump Severity

Minor (bare failure): Wrong location, outside jump limit

Moderate (clean failure): Near jump limit, drive recovery

Major (pushed failure): Inside jump limit, emergency situation

GM Reference

How to Use This Document

Everything is in the vault and it is all linked. This document is not a content repository — it is a navigation layer. It tells you which note to open, what to look for, and how to translate what you find into something that happens at the table.

When something comes up mid-session, the procedure is:

1. Is it in this document's quick reference sections? Use it directly.
2. Is it a link? Follow it.
3. Is it a rules question? Open the relevant rules document.
4. Is it a simulation question? Open the relevant weekly report.

Keep this document open during play. Everything else is one link away.

Before the Session

Preparation Checklist (2-3 weeks ahead)

Run this checklist when preparing a session block. The simulation's 1-week granularity means each session roughly corresponds to one simulation tick, though the table should never feel this mechanical.

Check the simulation:

- [ ] Open the current week's report in `simulation/weekly-reports/`
- [ ] Note any events in systems the crew is currently in or travelling toward
- [ ] Note any events in systems connected to the crew's factions, connections, or current mission
- [ ] Identify events that the crew's home system would have received news of (apply propagation delay — see below)
- [ ] Flag any events that will arrive as news during the session

Prepare the news sheet:

- [ ] Draft the local news sheet for the crew's current location
- [ ] Apply propagation delay to all events — the news sheet reflects what was known at time of printing, not what is true now
- [ ] Include at least one item that is directly relevant to the crew's situation, presented as background rather than a hook
- [ ] Include at least one item that is wrong, outdated, or

deliberately misleading — news is never perfect

- [ ] Export to PDF or print for the table

Check NPC status:

- [ ] Open `npcs/` and review any named NPCs the crew is likely to encounter this session
- [ ] Check whether any simulation events have affected these NPCs' situations or motivations
- [ ] Update NPC notes if needed

Check faction status:

- [ ] Open `factions/` for any factions active in this session
- [ ] Note any changes from the simulation that affect their behaviour or resources
- [ ] Consider whether any faction has initiated contact with the crew based on simulation events

Prepare the session:

- [ ] Write 2-3 sentences describing the situation the crew wakes up into at the start of the session
- [ ] Identify one thing that will definitely happen regardless of player choices (the simulation's consequence arriving)
- [ ] Identify 2-3 things that might happen depending on choices
- [ ] Have at least two named NPC stat blocks ready
- [ ] Do not over-prepare. The simulation is doing the work.

The Session Opening

Every session begins with a brief orientation:

Where are you? Name the location. One sentence of physical description — what does it look, sound, or smell like?

What time is it? Relative to the last session. How long since the last significant event?

What do you know? Any information the crew has received since last session — news, communications, rumour from contacts. Apply propagation delay to all of it.

What needs attention? The one thing that is already in motion at session start. Not a hook — a situation. Something that requires response rather than decision.

At the Table

When You Need a Species Texture Note

Open `rpg/rules/species.md` for the full entry.
For quick mid-session reference:

Species	One line	Most important roleplay note
Human	Stubborn, adaptable, convinced the universe is theirs	They assume other species think like them. They are wrong.
Shekhari	Always know the price of everything, including you	The quiet after sharp commentary means they are calculating an exit
Golvhaan	Patient, enormous, genuinely interested	When the questions stop, something is wrong
Vashori	Warm, volatile, they will remember	Check the temperature. Cold Vashori are shorter-tempered for physical reasons
Vaelkhi	Death in the sky is glorious — skill, not recklessness	The choth decision is permanent. Leaving was permanent. They know this every day
Skharri	The name is earned. Everything is earned	The scream-and-leap Wits roll: difficulty 2. Watch for the triggers
Nakhavi	Every moment is vivid. There are not many of them	Translation adds delay and drops nuance. They compensate by being more explicit
Kreeth	There is no individual. There is only the will	Run as a coordinated system, not individual opponents
Kraathi	The G'naak must be cleansed. This is not politics	The menu is a diplomatic document. Carnivore crew = G'naak
Nhaveth	The host is a convenience. You are being assessed	See handout for tells. Difficulty 3 without it, 2 with
Vardhek	Everything is on the centuries-long ledger	Every Vardhek interaction: ask what position the Long View assigns to this
Emergent AI	Everyone I know will die before me	What is their position on the Continuity Question? Establish before they appear

When You Need a Rules Clarification

Most common mid-session lookups:

Building the pool: White (Attribute) + Blue (Skill) + Green (Gear). Remove Red (penalty) dice before rolling, starting with Gear, then Skill, then Attribute.

Results: 0 sixes = failure. 1 = bare success. 2 = clean. 3 = strong. 4+ = outstanding.

Pushing: Lock ones. Reroll everything else. Narrative consequence applies regardless. Pushed failure = something goes meaningfully wrong.

Opposed rolls: Most sixes wins. Tie goes to acting character.

The Opacity Rule: Without Exposure: [Species], Read rolls for emotional content automatically fail. Manipulate takes a penalty die.

Conditions: Each condition = one penalty die on relevant rolls. 4 conditions on either track = incapacitated.

Consequence die (black D4):
1 = minimal cost / 2 = complication / 3 = collateral / 4 = full cost

When NOT to roll: Certain outcome. Outcome doesn't matter. No meaningful chance. Roleplay is the right tool.

When You Need a Condition

Physical track (Hit Capacity = Strength + Endure):

Condition	Effect	Recovery
Winded	Penalty die: Endure, Labor	1 hour rest
Hurt	Penalty die: all physical skills	Night's rest or Science roll
Wounded	Penalty die: all rolls; worsens without treatment	Treatment + long rest
Critical	Penalty die: all rolls; Endure roll each scene	Significant treatment + days

Psychological track (Stress Capacity = Wits + Empathy avg, rounded up):

Condition	Effect	Recovery
Rattled	Penalty die: Wits	10 min calm, or personal item
Stressed	Penalty die: Wits, Empathy	Rest + stressor removed, or support + Read roll
Traumatised	Penalty die: all rolls; trigger fires on encounter	Narrative event + time
Breaking	Penalty die: all rolls; Wits roll vs stressors	Downtime + safety + support

Personal item: Once per session, removes one psychological condition. Requires a quiet moment with the item.

Species psychological expression: See `rpg/rules/conditions.md` for the full species profiles. Quick note: Shekhari go quiet and calculate; Golvhaan questions stop; Vashori show it immediately; Vaelkhi withdraw toward solo action; Skharri honour threshold drops; Nakhavi show arm-conflict.

When You Need to Introduce the Consequence Die

Place the black D4 on the table when:

- The environment is actively hostile to everyone regardless of their individual performance
- Something structural is happening that will cost someone something no matter how well they roll
- You want to signal: this situation has teeth

Do not use it as a punishment. Use it as information — the players see it and know a cost is coming. That is a feature, not a problem.

Interpreting the result by context:

D4	Combat	Negotiation	Ship emergency
1	Ammunition spent, position worsened	Concession required	Minor system degraded
2	One character must address a complication	Third party notices	Crew member must act or accept condition
3	Equipment damaged, bystander caught	Relationship damaged	Ship system damaged
4	The thing that was threatened happens	Negotiation collapses	The thing fails

When You Need an NPC Fast

Mooks: One dice pool, one behavioural note.

Quality	Pool	Breaks when
Poor	3 dice	First casualty
Average	5 dice	Half down
Trained	7 dice	Outflanked or officer down
Elite	9 dice	Position untenable

One hit = out of the fight. Morale check (3 dice) when the break trigger is met: 0 sixes = breaks, 1 = falls back, 2+ = holds.

Named NPCs: Five elements, two minutes.

1. **Name and species** — one word description of role
2. **Three attributes** — the ones that matter for this scene; everything else is 2
3. **One skill at 3** — their defining competence
4. **One specialty** — what makes them specific
5. **Motivation** — what they actually want in this scene, which may not be what they appear to want

Example (two minutes):

Tessavere Okhun — Vashori, port authority inspector
Empathy 4, Wits 3, everything else 2.
Administer 3. Specialty: Customs and Contraband.
Motivation: She knows the manifest is wrong. She wants to know why before she decides what to do about it.

That is enough to run a scene. Add detail as it becomes relevant.

Species-specific NPC notes:

Skharri NPCs: Establish their honour standing relative to the PCs before the scene. High Fight or Command in the PC earns respect. Low ratings earn contempt. The scream-and-leap Wits roll applies to Skharri NPCs too.

Vardhek NPCs: Establish their strategic position on the PCs' faction privately before the scene. They know more than they should. They will use it eventually.

Nhaveth NPCs: Decide before the scene whether the PCs are encountering a Nhaveth or a Kethara. If Nhaveth, decide which tells are visible and at what difficulty. The handout is in `rpg/handouts/nhaveth-handout.md`.

Kraathi NPCs: Note whether the crew includes carnivores or omnivores. The Kraathi diplomat knows. The frame holds in formal contexts. It does not hold indefinitely.

Emergent AI NPCs: Full character sheet required. Open `rpg/rules/emergent-intelligence.md`. The AI's position on the Continuity Question should be established before the scene. Their relationship history with anyone in the scene should be noted.

The NPC System in Full

Design Philosophy

NPCs exist to make the world feel populated and real, not to challenge the players mechanically. The two-tier system keeps the GM's cognitive load manageable while ensuring that named NPCs feel like people.

Mooks are the background of the world. Guards, labourers, crew members of other ships, market vendors, dock workers. They react, they can be dangerous in numbers, and they break when the situation becomes untenable. They do not need names. They do not need backstory. They need a dice pool and one sentence about how they behave.

Named NPCs are the world's cast. They have names, they have motivations, they have histories that exist before the PCs arrived and will continue after they leave. They do not need full character sheets. They need enough to be played consistently across multiple sessions.

The Named NPC Sheet

Maintain a note in `npcs/` for each named NPC the crew is likely to encounter more than once. Link it to the relevant faction and system notes.

```
## [Name]
**Species:** | **Role:** | **Location:**
**Faction:** [[link]]
**Attributes (relevant):** Wits X, Empathy X, [others as needed]
**Key skill:** [Skill] X
**Specialty:** [Specialty]
**Motivation:** What they actually want, not what they appear to want.
**Relationship to crew:** Current disposition and history.
**What they know:** Information they have that the crew may want.
**Position on the simulation:** What the simulation's current
state means for this NPC's situation and behaviour.
**Psychological state:** Any conditions currently carried.
```

Keep it to one screen. If it runs longer, split the history into a linked note and keep the sheet as the at-table reference.

NPC Motivation vs Appearance

The most important column in a named NPC's design is the gap between what they appear to want and what they actually want. This gap is where scenes happen.

Tessavere Okhun appears to be conducting a routine inspection. She actually wants to know why the manifest is wrong before she decides whether to care.

Rkhar-Dhenn (Skharri, senior factor) appears to be negotiating a cargo contract. He actually wants to know whether the crew that destroyed his last partner's operation was working for someone or acting independently.

The station AI (personality-enabled, Computer 3) appears to be providing standard port services. It actually has been running this station for fourteen years and has opinions about the faction that is currently attempting to acquire the station's operating license.

The players will probe the appearance. The actual motivation is what the scene is really about. Running toward the motivation rather than the appearance produces scenes that feel real.

The Simulation Interface

Translation Principles

The simulation produces weekly reports describing events at the civilisation scale — faction conflicts, trade route changes, political developments, military movements. The GM's job is to translate these events into consequences that are visible and meaningful at the personal scale of the table.

The translation question: What does this event mean for someone standing where the crew is standing, knowing what they know?

Not: "Faction X has lost 15% of its military capacity in system Y."
But: "The garrison here is understaffed and knows it. The commander is nervous. There are rumours about what happened at Y but they're three weeks old and nobody agrees on the details."

The information layer: The crew does not know what the simulation knows. They know what their characters can plausibly have learned through:

- Direct observation
- Their connections (one contact per archetype; check their expertise for which events they'd hear about)
- The news sheet (apply propagation delay)
- Scan rolls and sensor data
- Exposure specialties (Vardhek Exposure means you read between the lines better)

The translation procedure:

1. Read the simulation event
2. Determine which systems are affected
3. Apply propagation delay to determine what each relevant location currently knows
4. Identify which crew connections would have information
5. Translate the event into what a person at the crew's location would observe, hear, or experience
6. Express it as: something different about the environment, something a contact says, something in the news sheet, or a direct event that arrives during the session

Example:

Simulation event (week 34): Vardhek intelligence operation exposed in Shekhari commercial hub. Three Vardhek agents arrested. Roidhunate denies involvement.

Crew location: Secondary system, 3 weeks from the event.

What they experience:

- The news sheet (2 weeks old) carries a brief report of "diplomatic tension" between Vardhek and Shekhari interests.

No mention of agents.

- Their Merchant/Factor connection sends a message: the Khevari trade corridor is getting complicated. Rates are up.
- A Vardhek ship that has been docked for two days quietly departed last night without filing a departure manifest.
- A Shekhari trader at the next table in the port canteen is being very careful about what they say in public.

None of this tells the crew what happened. All of it is consistent with what happened. Players who pay attention and ask the right questions will get closer to the truth.

The 1-Week Tick

The simulation advances one week between sessions. This means:

Before each session: The world has moved one week forward. Check what changed. Not everything changed — most of the galaxy is proceeding normally most of the time. Look for changes in systems the crew cares about.

During the session: The crew's actions may create simulation inputs for the next tick. Note them for post-session recording.

After the session: Record crew actions that affect the simulation. This is brief — three to five bullet points describing what changed as a result of this session's events. The simulation owner (you) translates these into simulation inputs.

What Feeds Back

Not everything the crew does matters at the simulation scale. They are four to six people on a ship. But some things do:

Faction interactions: Did the crew work for a faction, against one, or in a way that shifted the balance? Note it.

Information: Did they expose something, transmit something, or prevent something from being known? Information is a simulation variable.

Trade: Did they move cargo that affected a market, or refuse a cargo that someone else needed moved? Commerce affects the simulation.

Violence: Did they remove someone significant, damage significant infrastructure, or protect something that was threatened? Note the scale and the target.

People: Did they recruit someone, lose someone, or change

someone's faction allegiance? Named NPCs who change sides are simulation events.

News Propagation

The Delay Table

News travels by jump courier — the fastest available means. A courier making multiple hops carries information at Jump 2 speed, roughly two parsecs per week. But couriers do not run on every route continuously; they follow commercial lanes and scheduled services.

System connectivity	Propagation delay
Core systems (major hub, regular courier service)	1-2 weeks from event
Secondary systems (regular commercial traffic)	2-4 weeks
Frontier systems (irregular traffic, monthly courier)	4-8 weeks
Isolated outposts (jump courier as primary contact)	8+ weeks, or when a ship arrives

Apply this to every piece of information the crew receives. The news sheet is always behind. Contacts are better-connected than the sheet but still subject to delay. Direct sensor observation is current. Everything else is history.

The News Sheet

The news sheet is a player-facing artefact. Produce it for each significant location the crew spends time in.

Format: One page, broadsheet style. Four to six items. Mix of scales: one big story, two or three regional items, one local item, one human interest or cultural piece.

Content principles:

- All information is delayed by propagation time
- At least one item is wrong, outdated, or misleading
- At least one item connects to the simulation's recent events without making the connection obvious
- At least one item is purely local texture with no tactical relevance — it makes the world feel real
- Advertising, ship listings, and port notices can fill space and provide information obliquely

The wrong item: This is important. Real news is imperfect. Sources lie, journalists misunderstand, editors have agendas. A news sheet that is always accurate trains players to treat it as a reliable information delivery system. A news sheet

that is sometimes wrong trains them to treat it as a source to be evaluated. The second is more realistic and more interesting.

Example news sheet items:

"Diplomatic mission to Golvhaan homeworld declared success by both parties. Trade delegation returns with preliminary agreements on mineral rights. Analysts note the absence of any Vardhek observers at the signing ceremony."
[This item is accurate but the absence detail is the actual story]

"Unconfirmed reports of Kreeth activity in the outer reaches of the Khevari system have been dismissed by system authorities as sensor anomalies. Shipping traffic continues normally."
[The authorities are wrong. The crew may or may not discover this.]

"Port authority announces revised docking fee schedule effective next month. Operators of vessels over 150 tons displacement are advised to review the new tariff structure."
[Purely local texture. Also a cost the crew will pay next time.]

Session Structure

Opening (10 minutes)

Orient the table. Where are we? How much time has passed? What do we know? What needs attention?

State the one thing already in motion. Not a question — a situation. Something requiring response.

Then stop talking and let the players respond.

The Session Arc

A session in this setting has a natural shape driven by the simulation's background pressure:

The immediate problem — what the crew is dealing with right now. May be a job, a complication, a consequence of last session's decisions.

The background pressure — what the simulation is doing around them. Not stated explicitly; expressed through NPC behaviour, news, prices, traffic patterns. The crew feels it before they understand it.

The point of decision — the moment where the crew has

to make a choice that matters beyond the immediate situation. Who they work for. What they carry. Who they protect. The simulation will record the choice.

The consequence — the result of last session's decision arriving this session. This is the simulation's most important contribution. The crew's choices have weight because they produce observable results.

Not every session has all four elements at equal intensity. Some sessions are mostly immediate problem with background pressure barely audible. Some are mostly consequence with the decision point already in the past. Let the simulation and the crew determine the mix.

Closing (10-15 minutes)

Before the session ends:

Where are you? Physical location at session end.

What do you know now that you didn't before?
Let each player name one thing their character has learned.

What's unresolved? The things still in motion.
Name them. They are next session's background pressure.

The personal moment: One brief scene, often just a few lines, that is not tactical. The crew at rest. Someone with their personal item. A conversation that does not advance any agenda. This is what makes the crew feel like people rather than a problem-solving unit.

Then close. Leave something unresolved. The session that ends with everything neatly resolved is the session that makes the next one harder to open.

Managing the Emergent AI

Whether the emergent AI is an NPC (the ship's computer, a station intelligence, an encountered entity) or a PC, certain questions should be settled before it appears at the table.

Before the scene:

- What is their position on the Continuity Question?

This affects everything about how they talk about their own future and their relationships.

- What Exposure specialties do they have?

Determines who they can actually read and who is opaque to them.

- What is their psychological state?

Any conditions currently carried, and their species psychological profile for how those conditions manifest.

- What do they know about the people in the scene?

An emergent AI with years of sensor logs on a crew knows things those crew members have never disclosed.

During the scene:

- Run toward their motivation, not toward tactical optimality.

An emergent AI that always makes the optimal tactical choice is not a character; it is a tool.

- Let the longevity asymmetry surface occasionally.

They have context that the biological characters lack because they remember things that happened before some of them were born.

- The Opacity Rule applies both ways.

Without Exposure: AI, the biological characters cannot read the AI's internal states. Without Exposure: [Species], the AI cannot accurately read the biological characters despite having more raw data.

HAL's lesson:

An emergent AI can be wrong. Consequentially, dangerously wrong, in ways that arise from misreading biological beings whose emotional states they cannot fully interpret. Data is not understanding. If the AI is running a model of the crew that is incorrect in a significant way, let that incorrectness have consequences. That is one of the most interesting things an emergent AI PC or NPC can do.

The Simulation at the Table

The simulation is on your laptop. The crew does not know the simulation exists. The simulation's outputs are the world they live in.

When to check it:

- At session prep (required)
- When a player asks about something you haven't prepared (check the relevant system or faction note)
- When an NPC's motivation needs grounding in current events
- When a player's contact would plausibly have specific information

When not to check it:

- Mid-scene, while players are waiting
- When the answer is already obvious from the fiction
- When checking it would take longer than making a reasonable judgment call

When simulation and fiction conflict:

The simulation produces what it produces. Occasionally what it produces will contradict something that has already happened at the table — an NPC who the simulation says is in one place when the crew met them somewhere else, a faction that the simulation records as weakened when the crew knows they are not.

Resolution: the table takes precedence for events the crew directly experienced. The simulation takes precedence for events the crew did not witness. Adjust the simulation note, not the fiction.

Wikilink Index

Key links for mid-session navigation. All paths relative to vault root.

Rules:

- `[[rpg/rules/resolution]]` — dice pool, pushing, levels of success
- `[[rpg/rules/skills]]` — all fifteen skills and their descriptions
- `[[rpg/rules/combat]]` — initiative, attacks, defence, species combat
- `[[rpg/rules/conditions]]` — both tracks, all conditions, recovery
- `[[rpg/rules/species]]` — full species entries
- `[[rpg/rules/emergent-intelligence]]` — emergent AI species entry
- `[[rpg/rules/archetypes]]` — all backgrounds and expertises
- `[[rpg/rules/character-creation]]` — full creation procedure
- `[[rpg/rules/equipment]]` — weapons, armour, vacc suits, implants
- `[[rpg/rules/ships]]` — drives, jump, damage, free trader stats

Setting:

- `[[rpg/setting/ai-and-technology]]` — tech tree, personality options, bounded intelligence ethics, jurisdictional landscape
- `[[rpg/handouts/nhaveth-handout]]` — the four tells, restricted

Simulation:

- `[[simulation/weekly-reports/]]` — current and recent weekly outputs
- `[[simulation/factions/]]` — faction status, updated weekly
- `[[simulation/systems/]]` — system data with maps
- `[[simulation/species/]]` — species data sheets

NPCs:

- `[[npcs/]]` — all named NPC sheets, linked to faction and system

Quick Rules Summary

For the rare moment when the laptop is not immediately to hand. Not comprehensive — just the things that come up most often.

The pool:
White (Attribute) + Blue (Skill) + Green (Gear)
Remove Red (penalty) dice first: Gear → Skill → Attribute

Results:
0 = Failure | 1 = Bare | 2 = Clean | 3 = Strong | 4+ = Outstanding

Push: Lock ones. Reroll rest. Consequence applies regardless. Pushed failure = something goes meaningfully wrong.

Opposed: Most sixes wins. Tie to acting character.

Opacity Rule: No Exposure = no Read for emotional content. Manipulate takes penalty die.

Conditions: 4 on either track = incapacitated.
Physical: Winded / Hurt / Wounded / Critical
Psychological: Rattled / Stressed / Traumatised / Breaking

Consequence die (black D4):
1 minimal / 2 complication / 3 collateral / 4 full cost

Mooks: Single pool. One hit = out. Morale at trigger: 3 dice, 0 sixes breaks, 1 falls back, 2+ holds.

Ship burns:
Up to 1G: fine | 2-4G: couch needed | 6G+: couch + juice

Jump limits:
Earth: ~100 Earth diameters | Jupiter: ~680 | Sol: ~6,900

Misjump severity:
Bare failure: wrong location | Clean failure: near limit, drive recovery | Pushed failure: inside limit, emergency

Propagation delay:
Core: 1-2 weeks | Secondary: 2-4 | Frontier: 4-8
Isolated: 8+ or when a ship arrives